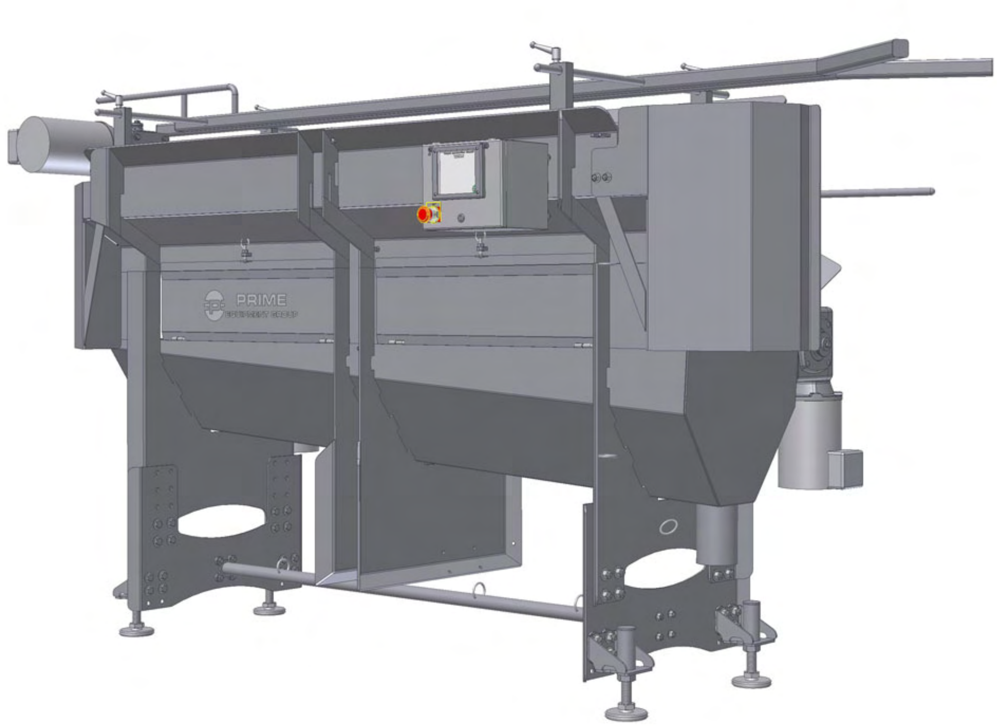


Technical Manual

Chicken Shoulder/Neck Skinner Model# CSNS-1

Wayne Farms, Union Springs, AL

Serial# SN1852

Left Hand Unit

2000 East Fulton Street

Columbus, OH 43205

USA

TEL: (614) 253-8590

FAX: (614) 253-6966

www.primeequipmentgroup.com

Revision – 8/19/2008

CONTENTS

1.0 Safety

DANGER! WARNING!

All operations, maintenance and sanitation personnel MUST read and understand this manual and become familiar with the equipment before operating, maintaining or sanitizing.

Operators **MUST REMOVE** all **RINGS, WATCHES, NECKTIES**, **LOOSE GARMENTS**, and any other items that might become entangled in the moving parts of the machine. **NO CHAIN MESH GLOVES** should be worn while operating this machine.

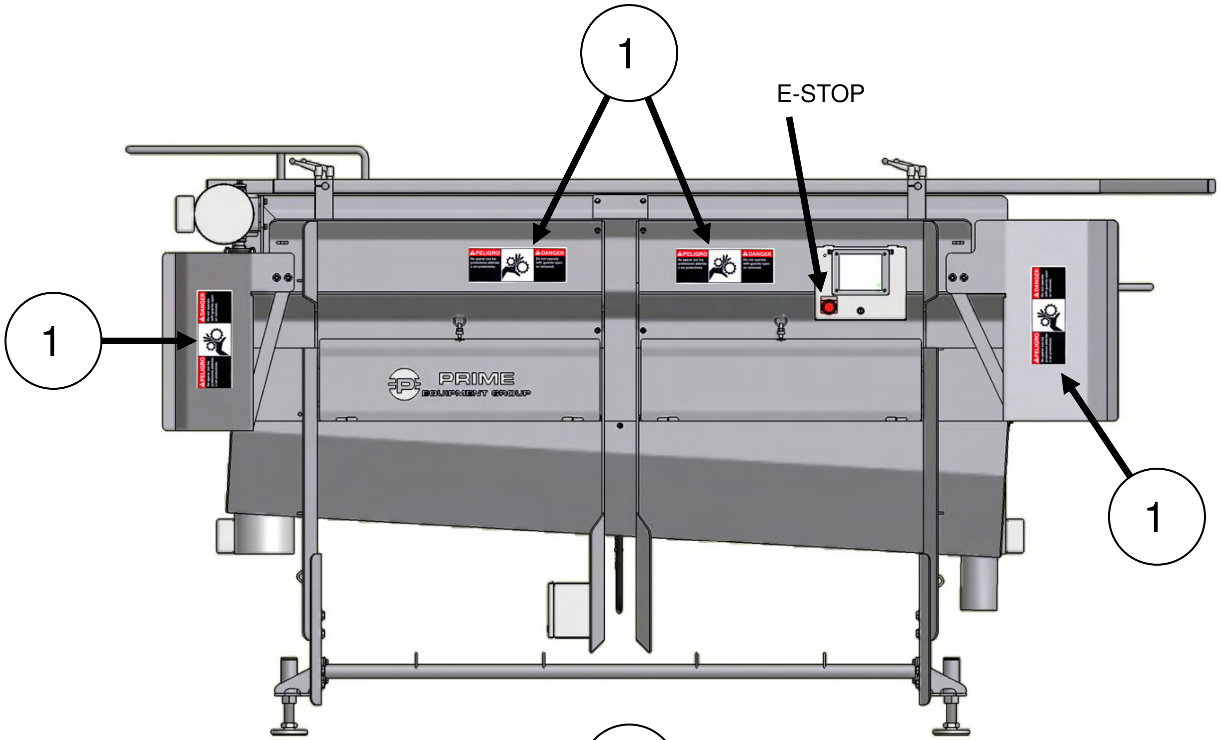
Operators **MUST KEEP HANDS AWAY** from the **CUTTING AREA** and any **ROTATING or MOVING PARTS** at all times.

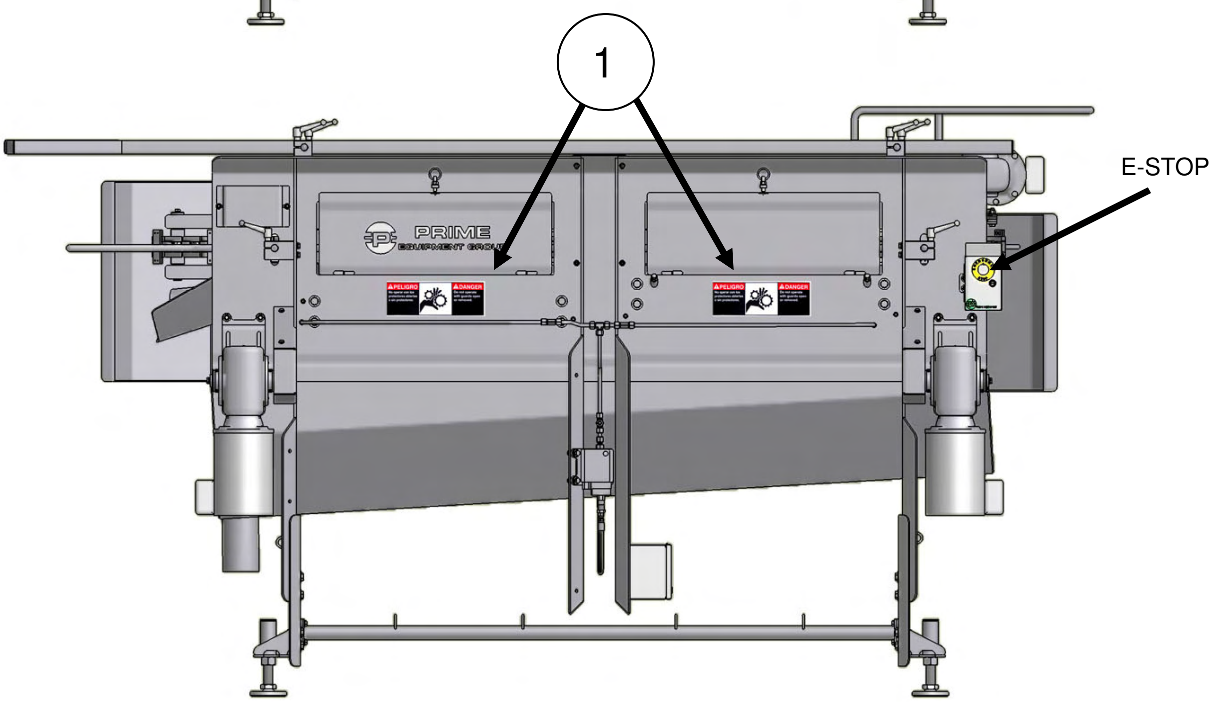
This machine is equipped with operating safety interlocks, which must be fully functional at all times. No attempt should ever be made to bypass safeties or run the machine with any guard removed.

Do not attempt to service machine unless power has been disconnected and locked out/ tagged out, and the machine has come to a complete stop.

Failure to heed these warnings and comply with the instructions for safe operation of this machine can and will cause serious bodily injury and or death.

1.2 Warning Labels and Locations





NO.	PART NO.	DESCRIPTION	REQ.
1	Sticker-3	SAFETY STICKER,GEAR	4

All individuals that will be working on or around this machine must be informed of the following procedures and safety guidelines.

All operators and any other persons working on or around the CSNS-1 must be informed of the location and operation of the safeties that are built in to the CSNS-1. All guards equipped with proximity switches must be fully operational in order to operate the machine. Safety switches should never be by-passed for any reason.

All operators and any other persons working on or around the CSNS-1 must also be informed of the location of the emergency stop buttons. Once the emergency stop button has been pressed the machine will not operate until the button has been reset and the start stop button has been reset as well.

Proper lock out tag out procedures must be followed at all times during repairs or during sanitation.

On occasion product may build up in certain areas during operation; never place your hand in to a pinch point area with out the machine being locked out.

Model : CSNS-1 Serial Number: ______________________________

I agree that I have read the above Start Up and Safety guidelines.

____

Customer (Print Name) Signature Date

This document was reviewed by the below Prime Equipment Representative and a copy was left for future reference.

____

Print Name Signature Date

Please call Prime Equipment for any service related questions;

Phone 1-614-253-8590 fax 1-614-253-6966

www.primeequipmentgroup.com

2.0 Warranty

2.1 Limited Warranty

Except original equipment components manufactured by other parties, Prime Equipment Group, Incorporated warrants the equipment described in this manual for 90 days after delivery. The warranty extended by Prime Equipment Group, Incorporated applies to the original purchaser only. This limited warranty will be void if the equipment is damaged due to abuse, negligence, and/or modification. Original equipment components including, but not limited to, motors, gearboxes, and electrical components shall be warranted by the original equipment manufacturer. A breach of warranty for an original equipment component will be the responsibility of the original equipment manufacturer, not Prime Equipment Group, Incorporated. Upon request, Prime Equipment Group, Incorporated will supply information regarding the terms and conditions of the warranties extended by original equipment manufacturers.

The exclusive and sole remedy of any breach of the limited warranty extended by Prime Equipment Group, Incorporated will be repair and replacement on nonconforming equipment or parts. Prime Equipment Group, Incorporated will not be responsible for consequential damages claimed by the purchaser as the result of any breach of this limited warranty. Consequential damages resulting from the seller's breach include:

a) any loss resulting from general or particular requirements and needs of which the seller at the time of contracting had reason to know and which could not reasonably be prevented by cover or otherwise.

b) injury to person or property proximately resulting from any breach of warranty.

There are no warranties that extend beyond the description on the face hereof. Any implied warranty of merchantability and any implied warranty of fitness for a particular purpose are excluded. The phrases "Implied Warranty of Merchantability" and "Warranty of Fitness for a Particular Purpose" shall be defined in accordance with paragraphs 2-314 and 2-315 of the Uniform Commercial Code.

3.1 Introduction

The CSNS-1 removes neck & shoulder skin from whole chickens. The unit is best placed in evisceration after the cropper and before final inspection. The CSNS is designed to work in conjunction with a shackle line. Birds hanging on the shackle line are guided through the CSNS-1 with the help of an indexer that is set to match the shackle line speed. There are two stages of patented Prime paddle wheel and pinch block assemblies. The first removes the neck skin while the neck is curled. After the first stage the neck is allowed to uncurl and hang straight. The second stage is design to catch and pull skin from the shoulders of the birds. This machine can be built as a single stage unit for neck skinning only. In this case if the customer wishes to harvest necks the line placement in line is after the cropper and before the neck breaker.

3.2 Familiarization

The component parts of this machine are identified in the Spare Parts List and Assembly drawings section of this manual. The name usually describes the function of the part.

3.3 Specifications

A. Dimensions – including standard frame. Unit approximately 38 3/4"W x 68" H x 104" L
(Unit size and height may vary due to customer's needs and location of shackle line.)

B. Production Rates

1. Depends on shackle line speed, up to 160 Birds per minute.

C. Electrical Requirements

1. Varies per customer specifications, usually 10amps at 480 volt, three phase 60hz
2. Personnel servicing this should be trained in proper Lockout/Tagout procedures and should ensure that the electrical enclosure is properly secured to prevent water from gaining access to the electrical component.

C. Water Requirements

1. At Least 400PSI, 5 GPM.

3.4 Installation

WARNING! ALL PERSONNEL WORKING ON OR AROUND THIS MACHINE MUST UNDERSTAND PROPER FUNCTION AND LOCATION OF SAFETY DEVICES BEFORE STARTING UNIT.

A. Select a site having suitable power, water, and drainage. The CSNS-1 operates its best immediately following your plant's cropper and before final inspection. Select at least nine feet of level, straight conveyor track in between the cropper the inside outside bird washer.

B. Select a location for the main control panel. (if not mounted at the machine)

C. Stub off electrical run from main panel to the machine – or – stub off main power supply to the machine.

D. Stub off high pressure water to the machine – or – stub off water supply to the machine for the supplied pressure pump.

E. Remove any existing shackle guide bars in the area in which the machine will be placed.

F. Place the machine under the overhead conveyor and shackles, generally line it up with the overhead track.

G. Center the center of the machine to the overhead conveyor.

H. Ensuring that the center remains centered, position the infeed end at least one inch to the **tail side** of the line

I. Again ensuring that the center remains centered, position the outfeed end at least one inch to **breast side** of the line .

J. Hopefully your salesman got the plant to save a few eviscerated birds, with neck on or at least has a measurement range from the bottom of the shackle to the bottom of the neck of the bird .

K. Set the infeed height so that the lowest point of the infeed neck guide is touching the bottom of the bird neck.

L. Set the outfeed height of the machine one to two inches higher than the infeed based on the size variation in the birds (more variation = more elevation change infeed to outfeed)

M. Set the breast bar at the infeed end to allow birds to enter and make contact with the indexing conveyor.

N. Set the breast bar at the outfeed end to allow the indexing conveyor to push the breast into the second stage shoulder skinning roller.

O. Set the shackle guide bars so that the birds and shackles are plumb with relation to the breast bar settings.

P. Set the shackle guide bar width far enough apart to allow the **EMPTY** shackles to run through without dragging back yet close enough for loaded shackles not to twist excessively due to machine function.

Q. Make sure all transitions between the plants shackle guide bar network and the machine's shackle guides are smooth and continuous. Test run with empty shackles.

R. Make final electrical connection.

S. Make final water connection.

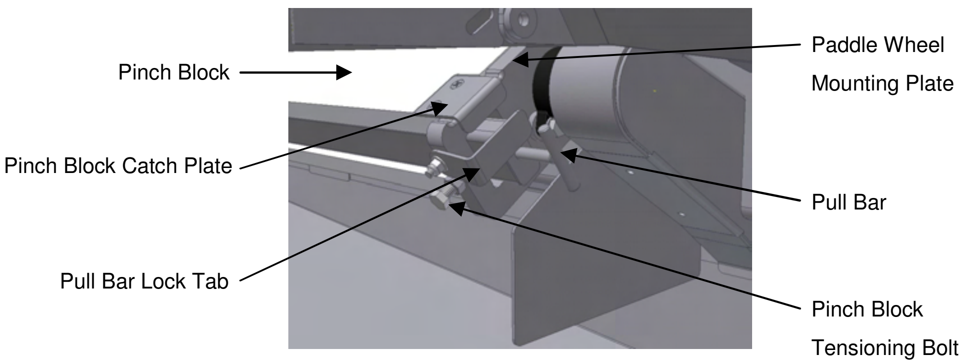
T. Test all functions.

3.5 Operation

WARNING! ALL PERSONNEL WORKING ON OR AROUND THIS MACHINE MUST UNDERSTAND PROPER FUNCTION AND LOCATION OF SAFETY DEVICES BEFORE STARTING UNIT.

A. Daily / Before Startup

1. Check E-Stop and that all proximity switches are operating properly.
2. Check all chain tensions. Tighten as necessary.
3. Inspect Pinch Block for chips, cracks, etc. Replace if necessary.
4. Reinstall Pinch Block.
    a) Place Pinch Block up against Paddle Wheel making sure that the lips on Pinch Block Catch Plates are resting over the Paddle Wheel Mounting Plates to ensure that the Pinch Blocks do not slip out of place during operation.



    b) Slide Pull Bar for Pinch Block into place.
    c) Hand-tighten pinch block tensioning bolts LIGHTLY against pull bar. **WARNING: DO NOT OVER TIGHTEN! FINGER TIGHT ONLY!** Over-tightening pinch block against star wheel will cause premature and uneven wear on those parts. Use a wrench to tighten the locking nuts only.
    d) Make sure the pinch block rides evenly against the star wheel. Use teeth on the star wheel and the edge of the pinch block as a guide. If the Pinch Block is warped or bowed, replace.
    e) Ensure Pull Bar Lock Tabs are in place.
5. Check that guards are mounted and secured properly.
6. Turn on water at ball valve (If not using the optional solenoid valve. Solenoid valve must be disabled under the maintenance screen, see 4.2 Maintenance Screen). Water will turn on automatically when unit is activated if using a solenoid valve. Verify supply and pressure. The CSNS-1 needs at least 400 psi @ 5 GPM to keep the skinning mechanism clean. Low water pressure will result in skin wrapping around the Paddle Wheel, preventing proper machine operation and causing possible damage to unit. Pressure switch is in place to prevent unit from running without adequate water pressure.
7. The CSNS-1 has a remote start feature that when used allows the unit to start and stop automatically when the shackle line is turned on/off (make sure all alarms are clear or unit will not start). This feature is enabled on the maintenance screen see 4.2 Maintenance Screen. The stop button on the touch screen display may not work in this mode use E-Stops to stop machine if needed.

3.5 Operation

B. Operation Screens

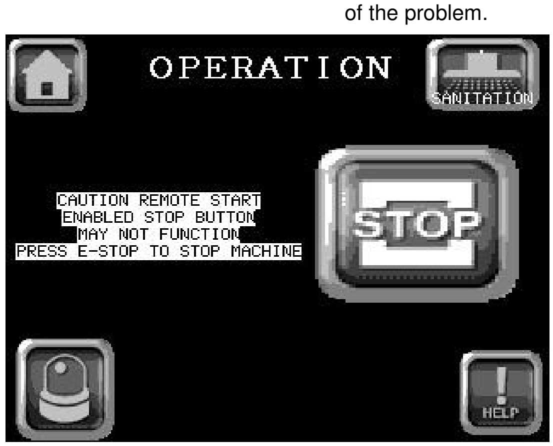
1. Select Run button from the Main screen. This will take you to the operation screen.
2. Pressing the Run Button will start the unit. (unless using the remote start feature, unit will start when shackle line is turned on if using remote start. A warning message appears on the operation screen if the remote start feature is enabled). Once running this button will turn into the Stop button, use this button to stop the unit in a non-emergency situation (unless using remote start is enabled. This button may not work if using remote start. Use E-Stops to stop machine).

This is the Main Menu, Pressing the Run Button will take you to the Operation screen shown at right.

Home button will take you back to the main menu

The Operation Screen is shown displaying an Alarm.

The unit will not start until all Alarms have been cleared. Press the alarm button to get an explanation of the problem.



Operation screen shown while machine is running with remote start enabled.

3. If machine will not start check for alarms. If there is any alarms present, lower left icon will read alarm as shown on previous page. Pressing this button will take you to the alarm screen and will explain the alarm. The alarms displayed are active. As guards are replaced and safeties reset, the alarms will reset automatically.

C. Shutdown.

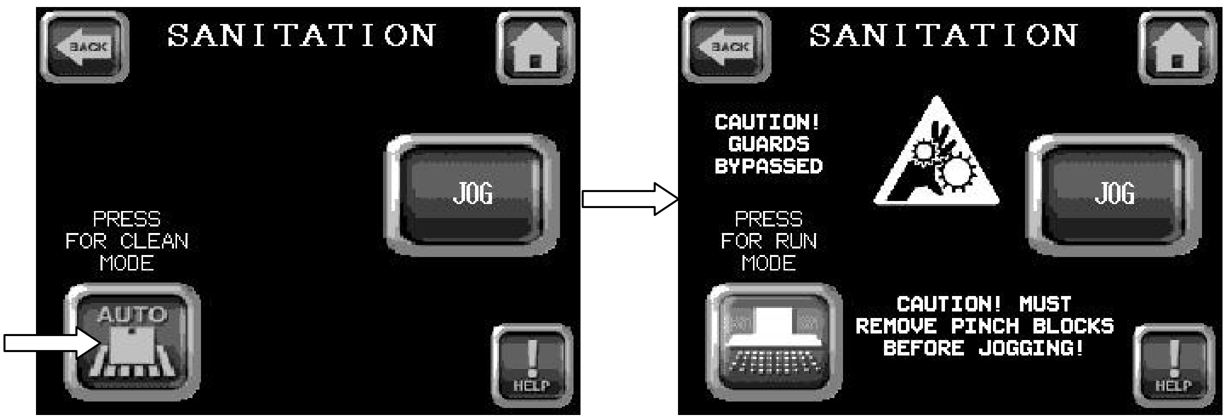
1. Turn off unit, water and electricity.
2. Remove any remaining product.

3.6 Cleanup / Sanitation

WARNING! ALL PERSONNEL WORKING ON OR AROUND THIS MACHINE MUST UNDERSTAND PROPER FUNCTION AND LOCATION OF SAFETY DEVICES BEFORE STARTING OR CLEANING UNIT. DO NOT SPRAY WATER IN MOTORS OR ELECTRICAL BOX. COVER AND WASH THESE AREAS BY HAND.

A. Sanitation Screens

1. Enter the Sanitation Mode by pressing the Sanitation button on either the main menu or in the upper right hand corner of the operation screen.



2. At the top of the sanitation menu are the home button to take you to the main menu and a back button for returning to the previous screen. A jog button is displayed but is not able to be used Until the “PRESS FOR CLEAN MODE” button Is activated.
3. Once the “PRESS FOR CLEAN MODE” button is activated the warning sign “CAUTION! GUARDS BYPASSED” appears indicating that any guard with safeties installed are now bypassed and can be opened while jogging the machine. Also the warning sign “CAUTION! MUST REMOVE PINCH BLOCKS BEFORE JOGGING” appears. This is a reminder that In sanitation mode the pressure switch is bypassed, Pinch Blocks must be removed before attempting to jog unit without Paddle Wheel nozzles on. Running the unit dry without water flow to these nozzles may cause premature wear and damage to the Pinch Block and Paddle Wheel.
4. Pressing the jog button now will activate machine. The Jog Button is maintained, meaning the machine will run when the button is pressed. Press the button again and the machine will stop. Machine will run until the jog button is released.

B. Sanitation Procedures

1. Open all doors on the sides of the CSNS-1.
2. Remove Pinch Blocks by rotating Pull Bars Lock Tabs to the side. While holding the Pinch Block, remove Pinch Block Pull Bar by sliding it out of the Pinch Block. Set loose parts aside for cleaning.
3. Jog machine as needed. In sanitation mode the pressure switch is by passed, Pinch Blocks must be removed before attempting to jog unit without Paddle Wheel nozzles on.
4. Rinse with water. (Disassemble as needed.)
5. Wash and sanitize with USDA accepted agent.
6. Rinse with water. (Reassemble.)

4.0 Maintenance

WARNING! ALL PERSONNEL WORKING ON OR AROUND THIS MACHINE MUST UNDERSTAND PROPER FUNCTION AND LOCATION OF SAFETY DEVICES. PROPER LOCKOUT/TAGOUT PROCEDURES SHOULD BE FOLLOWED AT ALL TIMES WHILE WORKING ON MACHINE

4.1 Scheduled Maintenance

A. Daily.

1. Check all safeties, switches and E-stops for proper function.
2. Verify that all electrical enclosures, leads are closed and sealed.
3. Check Pinch Block for cracks or chips. Replace if necessary.

B. Weekly (all above +):

1. Check Paddle Wheel for unusual wear, chips, and cracks. If excessive wear is observed and/or skinner performance is deteriorating, replace.
2. Check all bearings for smooth movement.
3. Apply grease to all bearings.
4. Check for any damage or unusual or excessive wear, replace, repair parts as needed.
5. Check all roller chain tension. Tighten when necessary.
6. Check Water Filter and Filter Screen for debris and blockage.
7. Gearbox is factory filled with food grade synthetic lube for universal mounting. The gearbox is lubed for life and does not require oil changes. To ensure that the system operates properly, DO NOT REMOVE THE VENT PLUG. (refer to gearbox manual for further information.)

C. Monthly (all above +):

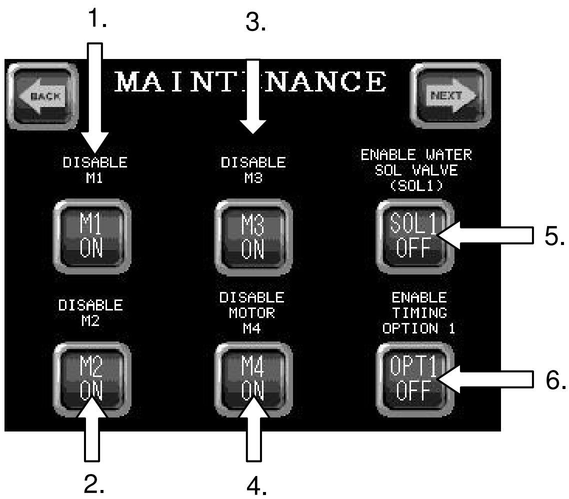
1. Check all fasteners and set screws and verify that they are snug. Tighten as necessary

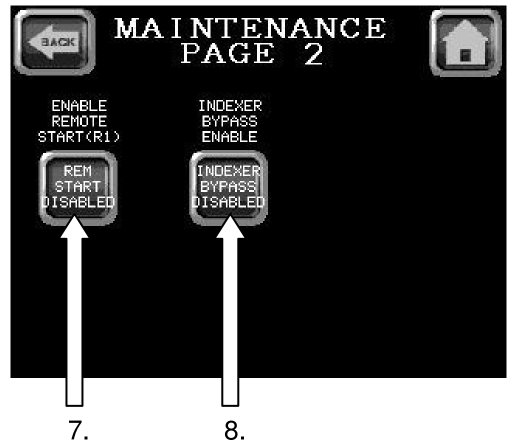
D. Six months (all above +):

1. Check Chain and sprocket wear (replace as needed).

4.2 Maintenance Screen

A. The maintenance screen can be accessed by the lower right hand button on the main menu. There you can turn off and on the optional controls inputs and outputs.



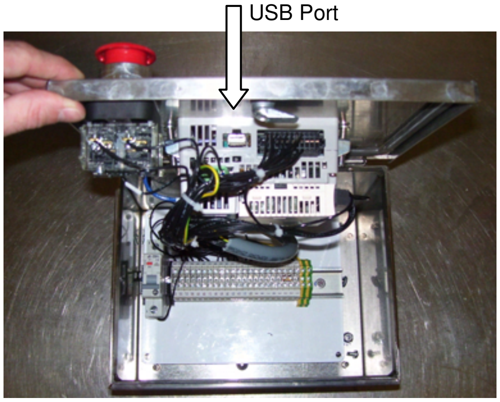


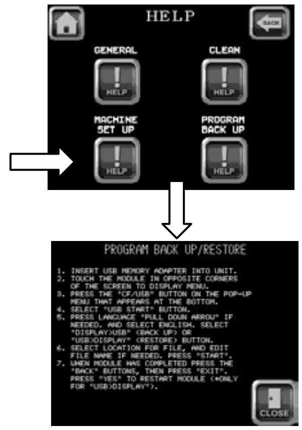
1. Disable/Enable Motor1 (Stage 1 Neck Skinning Paddle Wheel Assembly)
2. Disable/Enable Motor2 (Stage 2 Shoulder Skinning Paddle Wheel Assembly)
3. Disable/Enable Motor3 (Indexer Assembly Drive Motor)
4. Disable/Enable Motor4 (Optional Water Pump Drive Motor)
5. Disable/Enable optional solenoid valve for water inlet. This valve will automatically turn on Paddle Wheel Nozzles when unit is activated and will shut them off when unit is stopped.
6. Disable/Enable Timing Option1 (this is for future indexer timing developments. As of now CSNS1 Indexer speed is to be tuned to match that of the shackle line speed using the speed pot located on VFD3)
7. Disable/Enable Remote Start. If enabled unit is to be wired by means of a relay to the shackle line drive system. Unit will start/stop automatically as the shackle line is activated and deactivated.
8. Indexer Bypass Disable/Enable. This option allows the CSNS Indexer to continue to run if CSNS unit has stopped due to a water pressure fault. (Both Paddle Wheels will come to a stop to prevent damage from running dry but indexer will keep running)

4.3 Program Back Up/ Restore

A. The CSK12 unit’s programming can be backed up and restored by up/downloading to and from a USB drive. The USB port can be found by opening the enclosure which houses the touch screen display, port is on the bottom of the HMI touch screen unit.

USB Port





Instructions for this can also be found through the main help screen under Backup.

1. Insert USB memory adapter into unit.
2. Touch the Module in opposite corners of the screen to display menu.
3. Press the “CF/USB” button on the pop-up menu that appears at the bottom.
4. Select “USB START” button
5. Press the language “pull down arrow” if needed and select English. Select “DISPLAY>USB” (to backup) or “USB>DISPLAY” (to restore) button.
6. Select location for file, and edit file name if needed. Press “START”.
7. When module has completed press the “BACK” buttons, then press”EXIT”. Press “YES” to restart module (only for “USB>DISPLAY”).

4.4 Troubleshooting

A. Skinner Will Not Start

1. Check Touch screen for any alarms that may be present. If there is any alarms present, lower left icon will read alarm . Pressing this button will take you to the alarm screen and will explain the alarm. The alarms displayed are active. As guards are replaced and safeties reset, the alarms will reset automatically.
2. Check all power supplies, relays and disconnects.

B. Product Jams

1. Check all clearances between Shackle Guide Bars and shackles. Shackle Guide Bars are to prevent the shackles from twisting or turning as they run through the unit. Guide Bars are adjustable and should be spaced so the shackles passed freely through them and do not bind.
2. Check CSNS Indexer speed. Indexer speed must match that of the shackle line, adjustments can be made by using the speed pot knob on VFD3.

C. Removing Excess Meat with Skin

1. Adjust clearance between Star Wheel and Pinch Block. Adjust Pinch Block Take-ups evenly on both sides so that the Pinch Block is gently pushed against the Star Wheel. You should hear a slight chatter.

D. Skin wrapping on Paddlewheel.

1. Check for proper water pressure. Skin ejection nozzles should be operating at around 400 PSI. Low water pressure will result in skin wrapping around the star wheel. Skin wrap can result in damage to unit.
2. Check Paddlewheel nozzles for clogs. High water pressure may be a sign of blockage.
3. Check Paddlewheel nozzle alignment. Nozzles are positioned so that the water stream is aimed straight down the “valley” of each flight on the Paddle Wheel cleaning and removing any skin or fat from the Paddle Wheel. Nozzle aim may need to be adjusted.

E. Skin remaining on some product.

1. Check Paddle Wheel rotation. When viewed from the top Paddle Wheel should be turning into the Pinch Block, not away.
2. Excessive Paddle Wheel wear. Replace Paddle Wheel.
3. Uneven Paddle Wheel wear: Pinch block bolts tightened with wrenches and or tightened unevenly. Replace Paddle Wheel.
4. Product loading issues.
    a) Birds must enter machine end where the Shackle Guide Bars and Neck Guide Plate extend out past unit.
    b) Birds should be orientated to where the breasts are facing away from the Indexer.
    c) Product must not be frozen.
    d) Product must not overlap other product.
    e) Chipped pinch block. This will cause a sharp or jagged edge that will rip skin before it is completely pulled off.

Quick Reference / Spare Parts List

PRIME EQUIPMENT GROUP

Model No. CSNS-1
Serial No. SN1852

614-253-8590 FAX 614-253-6966 www.primeequipmentgroup.com

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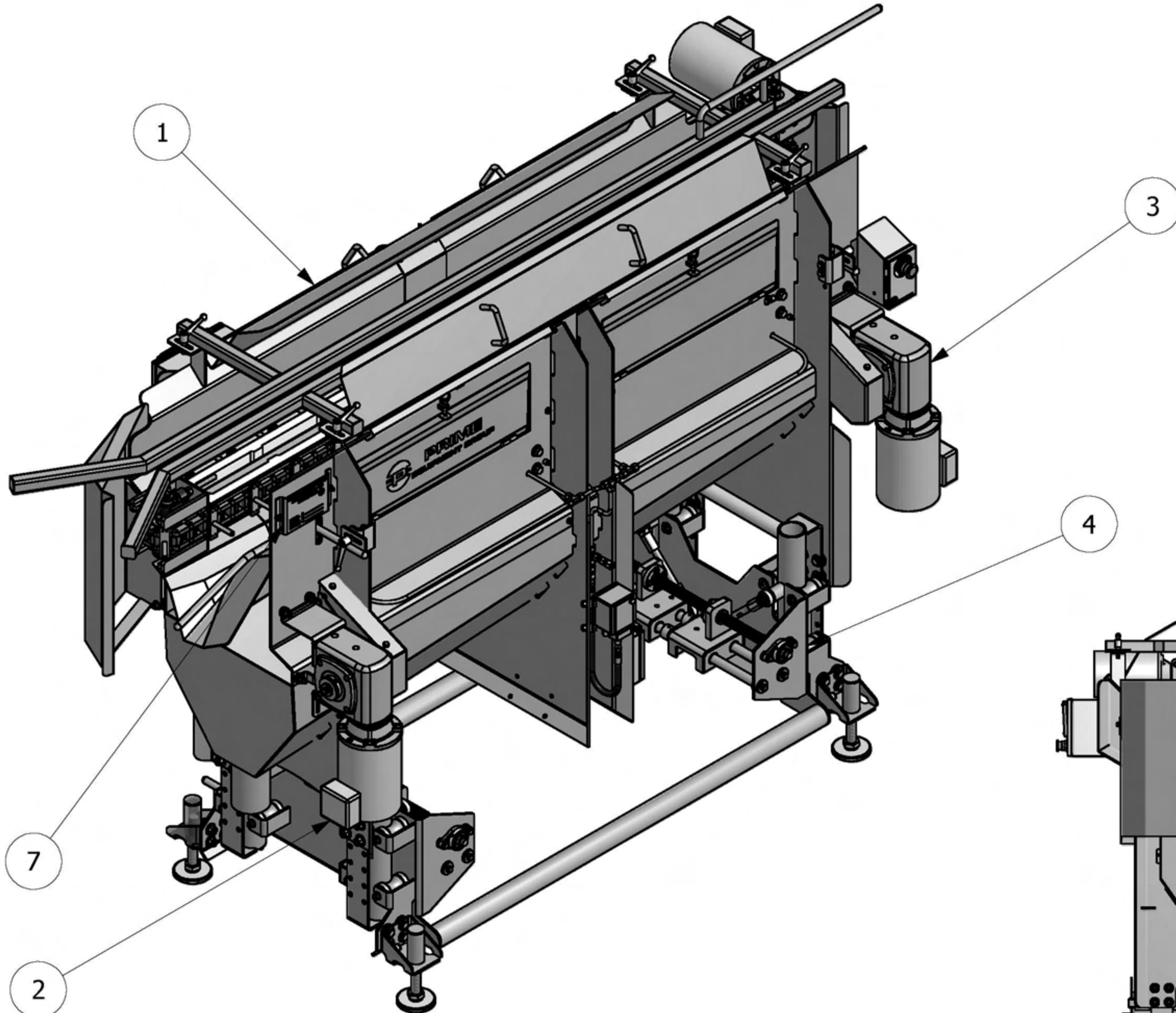
Paddle Wheel Assembly Stage 1		
Part #	**Description**	**Qty. per Unit**
14156	PADDLE WHEEL,20PT,24",STAGE 1	1
14162	PINCH BLOCK ASSY,24",STAGE 1 (COMES WITH CATCH PLATE AND HARDWARE)	1
14161	PINCK BLOCK,24",STAGE 1,CSNS-1 (PINCH BLOCK WITHOUT CATCH PLATE AND FASTENERS)	
13906	PULL BAR WLDMT,1X2X30.21L	1
13517	BEARING,FLG,2 BOLT 1"BORE	4
577B	SPRKT,40B14SS,1"BR,1/4"KW (PADDLE WHEEL DRIVE SPROCKET)	2
SP45-2-B	NOZZLE ASY,0º,0.125OD,0.055ID	4
761F	MOTOR,1 HP,1750 RPM,3 PH,60 HZ	3
13683	GEARBOX,726,40:1,BOSTON	2
782	SPROCKET,40B20SS,1"B,0.25"KW	3
533	CHAIN,ROLLER,40-1,SS,ASA	
525B	CONNECTING,LINK#40,S.S.	3
HTBS037C0175	3/8-16X1.75,CAP BOLT,FULL THRD (STAGE 1 PINCH BLOCK TENSIONING BOLT)	2
FHNS037C	0.38-16,SS,FIN,HEX,NUT (JAM NUT FOR PINCH BLOCK TENSIONING BOLTS)	4

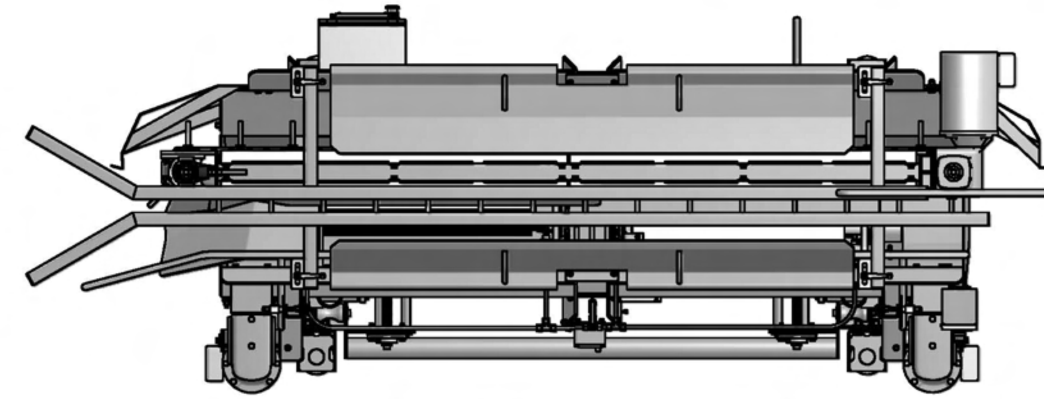
Paddle Wheel Assembly Stage 2		
Part #	**Description**	**Qty. per Unit**
14157	PADDLE WHEEL,20PT,24",STAGE 2	1
14163	PINCH BLOCK ASSY,24",STAGE 2 (COMES WITH CATCH PLATE AND HARDWARE)	1
14180	PINCH BLOCK,24",STAGE 2,CSNS-1 (PINCH BLOCK WITHOUT CATCH PLATE AND FASTENERS)	
13906	PULL BAR WLDMT,1X2X30.21L	1
13517	BEARING,FLG,2 BOLT 1"BORE	4
577B	SPRKT,40B14SS,1"BR,1/4"KW (PADDLE WHEEL DRIVE SPROCKET)	2
SP45-2-B	NOZZLE ASY,0º,0.125OD,0.055ID	4
761F	MOTOR,1 HP,1750 RPM,3 PH,60 HZ	3
13683	GEARBOX,726,40:1,BOSTON	2
782	SPROCKET,40B20SS,1"B,0.25"KW	3
533	CHAIN,ROLLER,40-1,SS,ASA	
525B	CONNECTING,LINK#40,S.S.	3
HTBS037C0275	3/8-16X2.75,CAP BOLT,FULL THRD (STAGE 2 PINCH BLOCK TENSIONING BOLT)	2
FHNS037C	0.38-16,SS,FIN,HEX,NUT (JAM NUT FOR PINCH BLOCK TENSIONING BOLTS)	4

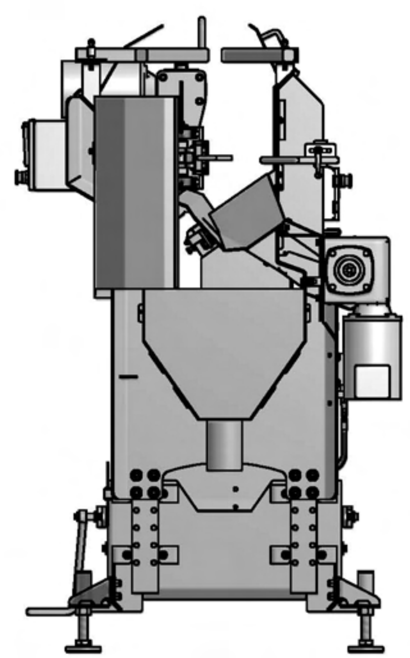
Indexer Assembly		
Part #	**Description**	**Qty. per Unit**
761F	MOTOR,1 HP,1750 RPM,3 PH,60 HZ	3
13684	GEARBOX,718,15:1,BOSTON	1
14444	SHAFT,DRIVE,INDEXER,CSNS-1	1
14443	SHAFT,IDLER,INDEXER,CSNS-1	1
550B	BEARING,TAKE-UP,1" BORE (CONVEYOR FLIP UP END IDLE SHAFT BEARING)	4
575	BEARING,FLANGE,4BOLT,Ø1"BORE (CONVEYOR DRIVE SHAFT BEARING)	6
871	CHAIN,KNUCKLE,3000,GRY W/TAB	5.3 FT (32 pcs)
871A	CHAIN,KNUCKLE,3000,GRY,W/O TAB	10.6 FT (64 pcs)
872	SPROCKET,SERIES,3000,5.20	2
16000	WEARSTRIP ASSY,CHAIN,INDEXER	4
12601	WLDMT,INDEXER,PUSH,PIN,CSNS-1 (INDEXER PUSH PIN)	32
SNAPRINGSH212SS	SNAPRING,2.12",EXTERNAL	4

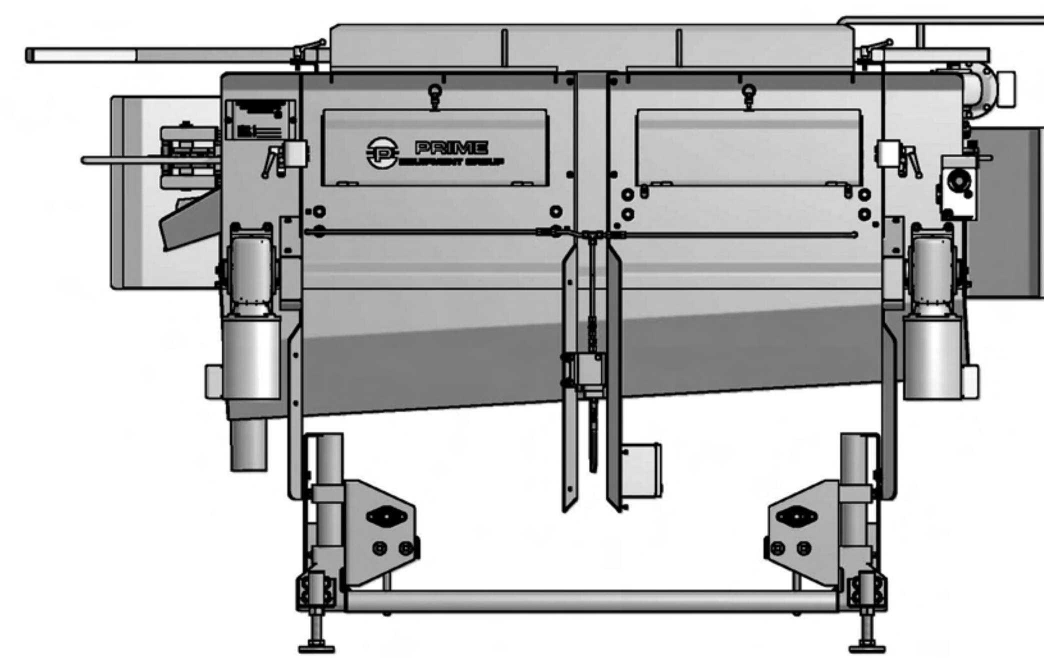
Electrical Parts List		
Part #	**Description**	**Qty. per Unit**
13098	PLC,HMI,COMPLETE, (TOUCH SCREEN INTERFACE)	1
800T-FXT6A5	E-STOP, PUSH BUTTON, 30MM	2
11589	PROX SWITCH,24-220VDC/AC,	4
13011	SWITCH,PRESS,ADJ DIFF,170-560	1
13732	SWITCH,D-CONNECT,FUSIBLE,30A	1
13797	FUSE,15A,TIME DELAY,	3
11815	HANDLE,DISCONNECT SWITCH,30A	1
11825	POWERSUPPLY,170-550VAC,24VDC	1
11822	CONTACTOR,5HP,24VDC	2
11829	CIRCUIT BREAKER,2POLE,2AMP	1
11828	CIRCUIT BREAKER,1 POLE,2AMP	1
11823	CIRCUIT BREAKER,1.6-2.5A	3
12975	RELAY,DPDT,24VDC,	1
13734	VFD,1HP,3Ø,480VAC,TELE,	1

Part #	Description	Qty. per Unit
892	ADJUSTABLE HANDLE 3/8-16 X1.18 (FOR SHACKLE LINE GUIDE RODS)	6
560B	3/8-16,SS,PULLPIN,W.233,END (DOOR CATCH PINS)	4
511	HOSE,4500,20",1/4"(#4JIC),SWIV	1









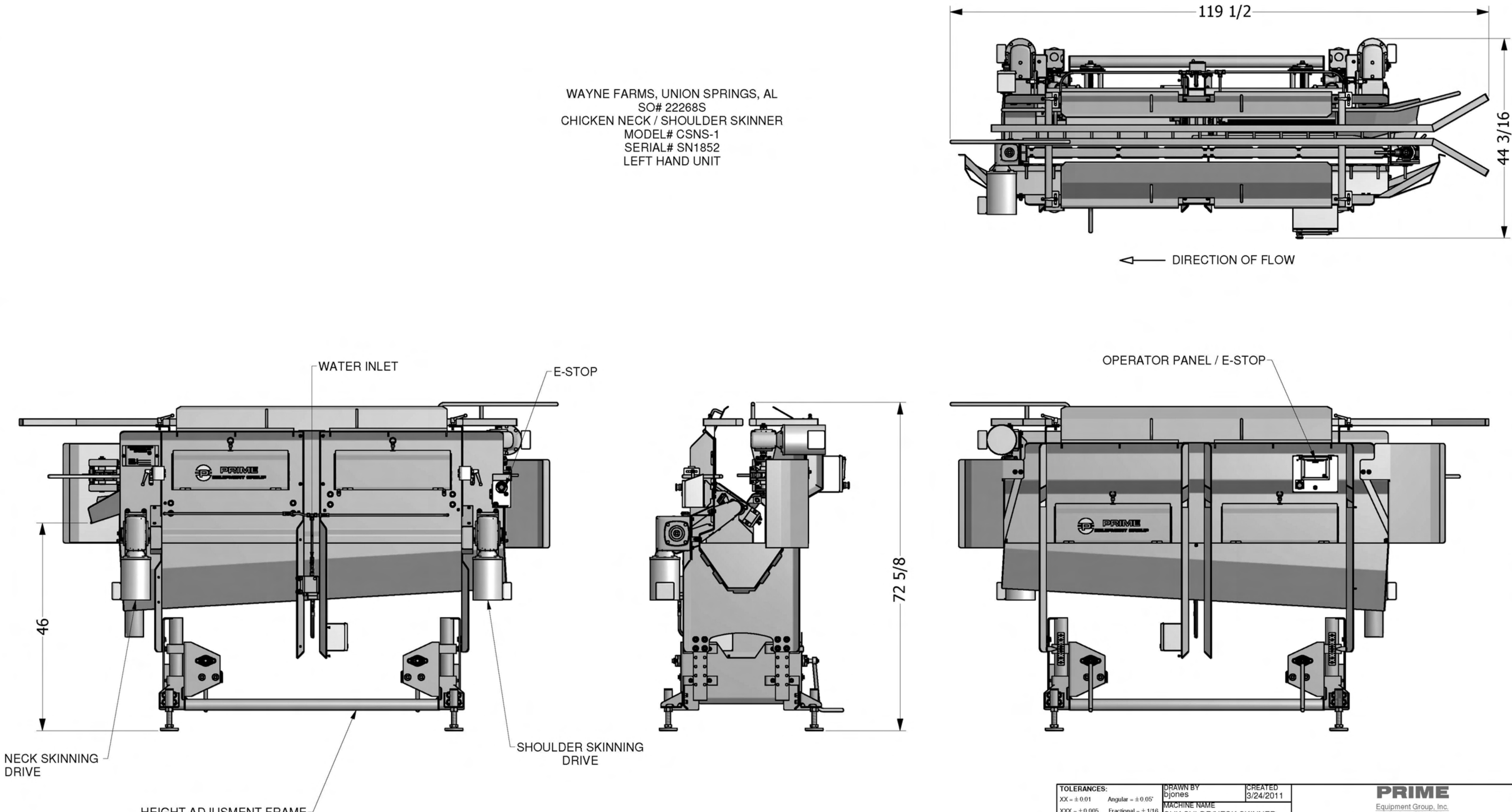
WAYNE FARMS, UNION SPRINGS, AL
SO# 22268S
CHICKEN NECK / SHOULDER SKINNER
MODEL# CSNS-1
SERIAL# SN1852
LEFT HAND UNIT

Parts List

Item	Drawing No.	Part No.	Description	Qty.
1	XX-CS00-17027	17027	BASE ASSY,CSNS,LH	1
2	CS00-14221	14221	DRIVE ASSY,STAGE 1,CSNS-1	1
3	CS00-14222	14222	DRIVE ASSY,STAGE 2,CSNS-1	1
4	16616	16616	FRAME ASSY,ADJ,HEIGHT,CSNS	1
5		14210	ELECTRICAL,MCP,ASSY,CSNS-1,	1
6	XX-CS00-16004	16004	INDEXER ASSY,CHAIN,CSNS-1	1
7	17315	17315	PLT,ID NAME TAG	1

TOLERANCES: XX = ± 0.01 Angular = ± 0.05' XXX = ± 0.005 Fractional = ± 1/16	DRAWN BY bjones / CREATED 3/24/2011 MACHINE NAME CHK SHLDR/NECK SKINNER	PRIME Equipment Group, Inc. Columbus, Ohio USA
	MODEL # CSNS-1 MATERIAL BLANK FILE NAME QUANTITY PER MACHINE 1 CLIENT/APPLICATION	TITLE CHICKEN NECK & SHOULDER SKINNER DWG NO SN1852 / PART NO SN1852 SIZE D SHEET 2 OF 4

WAYNE FARMS, UNION SPRINGS, AL
SO# 22268S
CHICKEN NECK / SHOULDER SKINNER
MODEL# CSNS-1
SERIAL# SN1852
LEFT HAND UNIT



TOLERANCES: XX = ± 0.01 Angular = ± 0.05˚ XXX = ± 0.005 Fractional = ± 1/16	DRAWN BY bjones / CREATED 3/24/2011 MACHINE NAME CHK SHLDR/NECK SKINNER	PRIME Equipment Group, Inc. Columbus, Ohio USA
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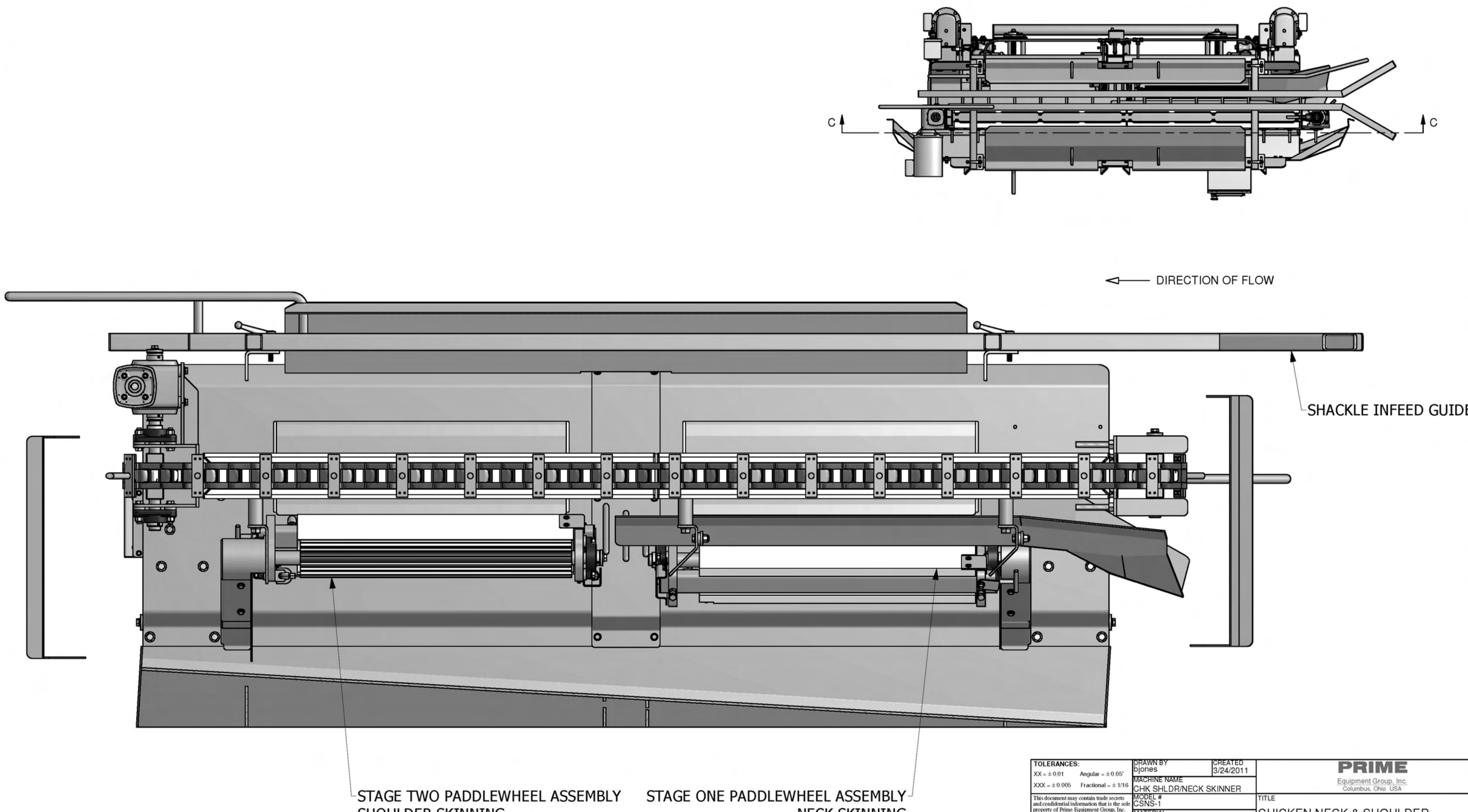
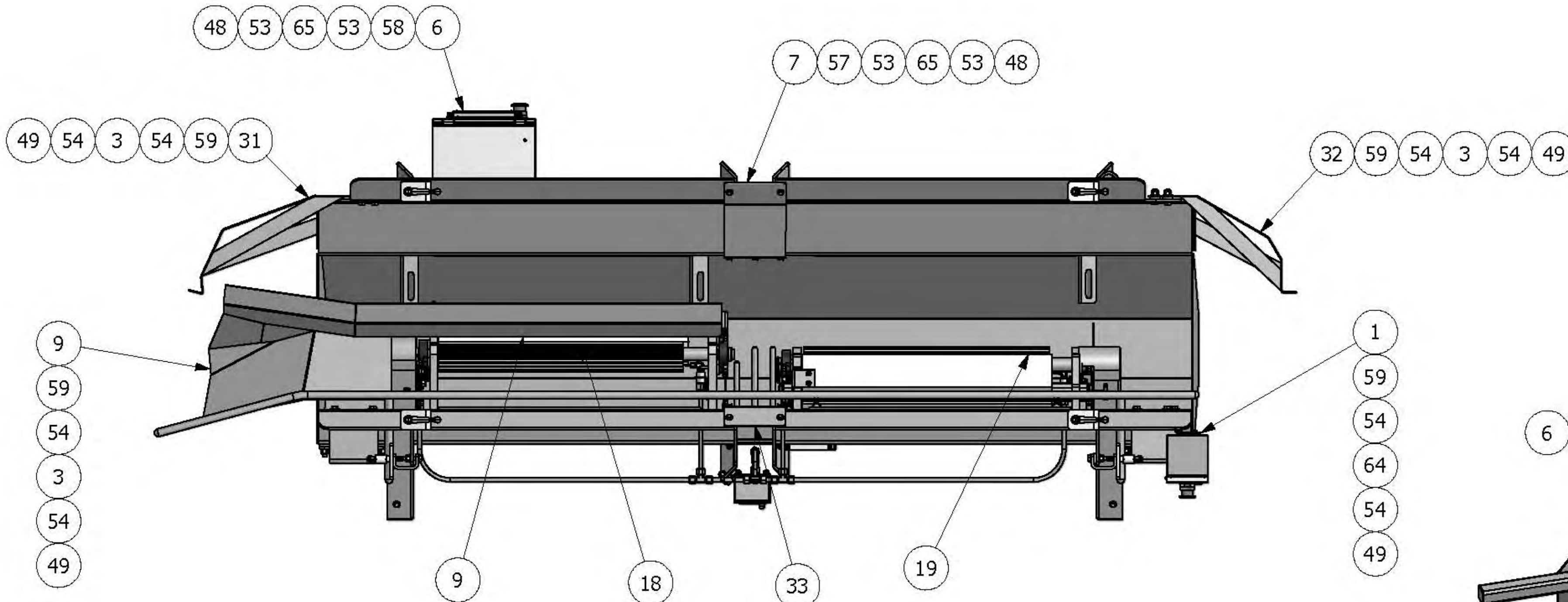
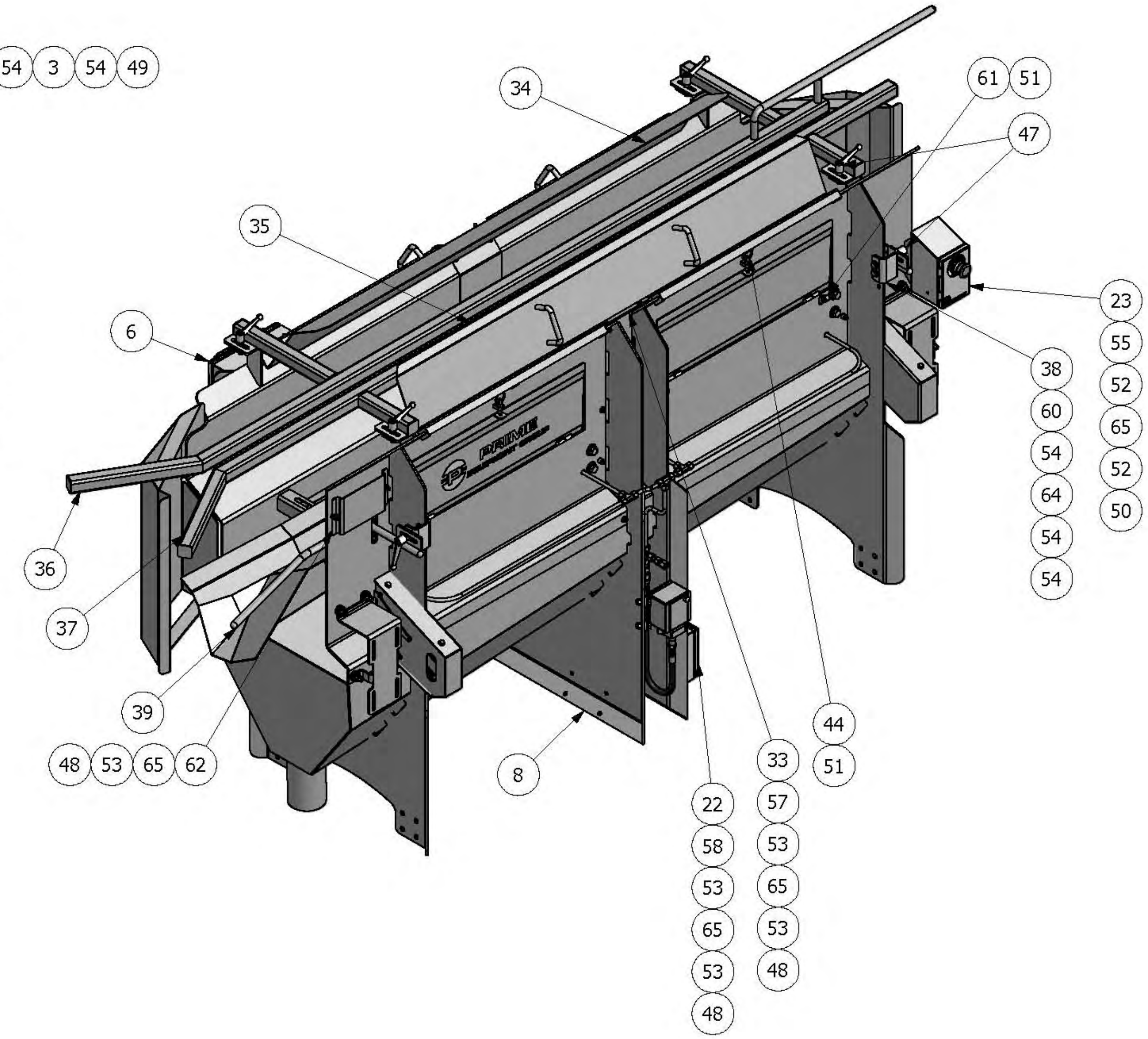


SECTION C-C

TOLERANCES: XX = ± 0.01 Angular = ± 0.05' XXX = ± 0.005 Fractional = ± 1/16	DRAWN BY bjones	CREATED 3/24/2011	PRIME Equipment Group, Inc. Columbus, Ohio USA
	MACHINE NAME CHK SHLDR/NECK SKINNER		
	MODEL # CSNS-1		TITLE CHICKEN NECK & SHOULDER SKINNER
	MATERIAL		
	BLANK FILE NAME		
	QUANTITY PER MACHINE 1		DWG NO SN1852 — PART NO SN1852
	CLIENT/APPLICATION		SIZE D — SHEET 3 OF 4

NOTE: TOP SHIELDS REMOVED





Parts List					Parts List				
Item	Drawing No.	Part No.	Description	Qty.	Item	Drawing No.	Part No.	Description	Qty.
1	KS03-11593	11593	PLT,FRMD,BRKT,E-STOP	1	38	XX-CS05-16034	16034	BRACKET,GUIDE,CHICKEN	2
2	13011	13011	SWITCH,PRESS,ADJ DIFF,170-560	1	39	XX-CS05-16035	16035	GUIDE WLDMT,CHICKEN,CSNS-1	1
3		13516	SPACER,0.25X1ODX0.38ID,SS	17	40		511	HOSE,4500,20",1/4"(#4JIC),SWIV	1
4	CS04-13871	13871	PLT,FRMD,MTG,GEARMOTOR,726	2	41		525B	CONNECTING,LINK#40,S.S	2
5	CS04-13907	13907	PLT,FRMD,MTG,CONNECTION,H2O	1	42		533	CHAIN,ROLLER,40-1,SS,ASA	76
6	13934	13934	CONTROL,HMI,ASSY,COMPLETE,SM	1	43		533	CHAIN,ROLLER,40-1,SS,ASA	80
7	CS07-13968	13968	PLT,FRMD,COVER,SIDE,LH	1	44		560B	3/8-16 SS LOCKING PULLPIN	4
8	CS900-14014	14014	FRAME WLDMT,LH,CSNS-1	1	45		621	1/4" CLOSE NIPPLE	2
9	CS09-14048	14048	GUIDE WLDMT,NECK,CSNS-1	1	46		648B	CONNECTOR,MALE,#4JIC-3/8"NPT	2
10	CS10-14088	14088	GRD WLDMT,BACK,CHAIN,DRIVE,LH	1	47		892	ADJUSTABLE HANDLE 3/8-16 X1.18	6
11	CS10-14089	14089	PLT,FRMD,GRD,FRNT,CHAIN,DRV,LH	1	48		ESNS025C	1/4-20 SS NYLON INSERT LOCKNUT	35
12	CS10-14090	14090	PLT,FRMD,GRD,FRNT,CHAIN,DRV,RH	1	49		ESNS037C	3/8-16 SS NYLON INSERT LOCKNUT	23
13	CS10-14091	14091	GRD WLDMT,BACK,CHAIN,DRIVE,RH	1	50		ESNS10C	LOCKNUT,NYL.INSERT,SS,#10	4
14	CS10-14105	14105	GRD WLDMT,BACK,CHAIN,STAGE 1	1	51		FHNS037C	0.38-16,SS,FIN,HEX,NUT	6
15	CS10-14106	14106	GRD WLDMT,BACK,CHAIN,STAGE 2	1	52		FWCS10	WASHER,FLAT,SS,#10	8
16	CS10-14111	14111	GRD WLDMT,FRONT,CHAIN,STAGE 1	1	53		FWUS025	1/4 SS FLAT WASHER, 5/8 OD	66
17	CS10-14112	14112	GRD WLDMT,FRONT,CHAIN,STAGE 2	1	54		FWUS037	3/8 SS FLAT WASHER, 7/8 OD	46
18	CS00-14198	14198	PADDLE WHEEL ASSY,STAGE 1	1	55		HCSS010C0075	10-24X3/4 SS HEX CAP SCREW	4
19	CS00-14199	14199	PADDLE WHEEL ASSY,STAGE 2	1	56		HCSS025C0050	1/4-20X1/2 SS HEX CAP SCREW	6
20		14207	FTG,STRT,1/4FNPT,3/8 TUBE	6	57		HCSS025C0100	1/4-20X1 SS HEX CAP SCREW	11
21		14208	FTG,COMPRESSION,TEE,3/8OD	4	58		HCSS025C075	HEX CAP SCREW,SS,1/4-20X.75	14
22	14231	14231	ELECTRICAL,ASSY,J-BOX,CSNS-1,	1	59		HCSS037C0125	3/8-16X1-1/4 SS HEX CAP SCREW	11
23	14233	14233	ELECTRICAL,E-STOP,ASSY,	1	60		HCSS037C0150	3/8-16 X 1 1/2" SS HEX CAP SCREW	12
24		14321	FTG,GREASE,1/4-28,1/8"Ø,STRGT	1	61		HTBS037C0275	3/8-16X2.75,CAP BOLT,FULL THRD	2
25		14322	FTG,GREASE,1/4-28,1/8"Ø,90°	3	62		ID PLATE	PLT,ID NAME TAG	1
26		14323	SSTB0.12X0.049 WALLX2.56	1	63		SHCS025C0225	1/4-20X2 1/4 SS SOC. CAP SCREW	2
27		14323	SSTB0.12X0.049 WALLX1.69	1	64		SPACER13SRS100555	SPACER, 3/8 ID X 1 OD X 1/4"	4
28		14323	SSTB0.12X0.049 WALLX1.38	1	65		SPACER16FW250125	SPACER, .5OD,.25IDX.125 THICK	39
29		14323	SSTB0.12X0.049 WALLX1.50	1	66		SSTB0.38X0.02WALL	SSTB0.38X0.02WALLX11.5	1
30		14400	FTG,TUBE,1/8",STRT,1/8" MNPT	8	67		SSTB0.38X0.02WALL	SSTB0.38X0.02WALLX8.5	1
31	CS04-14509	14509	GRD WLDMT,INDEXER,INLET	1	68		SSTB0.38X0.02WALL	SSTB0.38X0.02WALLX8	1
32	CS04-14510	14510	GRD,WLDMT,INDEXER,EXIT	1	69		SSTB0.38X0.02WALL	SSTB0.38X0.02WALLX4.25	1
33	CS07-14572	14572	COVER WLDMT,SIDE,SPLASH,RH	1	70		SSTB0.38X0.02WALL	SSTB0.38X0.02WALLX1.625	3
34	CS04-14695	14695	PLT,FRMD,GRD,SPLASH,TOP,RH	1	71		SSTB0.38X0.02WALL	SSTB0.38X0.02WALLX34.88	1
35	CS04-14696	14696	GRD WLDMT,SPLASH,TOP,RH	1	72		SSTB0.38X0.02WALL	SSTB0.38X0.02WALLX33	1
36	XX-CS05-16013	16013	GUIDE WLDMT,SHACKLE,LH,CSNS-1	1	73		ZERK-1966S	FTG,ZERK,1/4-28,STRAIGHT,SS	4
37	XX-CS05-16016	16016	GUIDE WLDMT,SHACKLE,RH,CSNS-1	1					

TOLERANCES:
XX = ± 0.01 Angular = ± 0.05°
XXX = ± 0.005 Fractional = ± 1/16

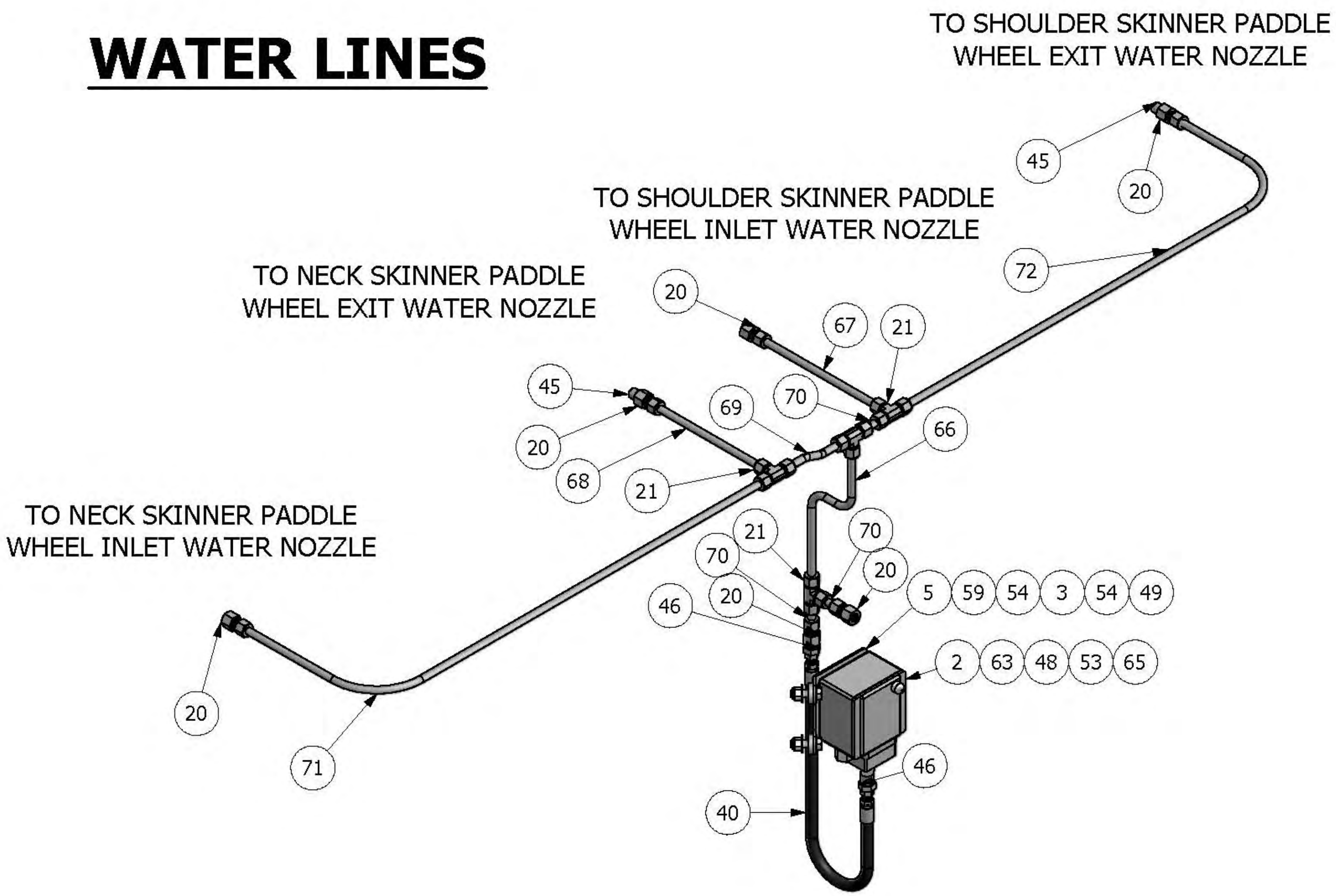
DRAWN BY	bsnyder
CREATED	3/3/2010
MACHINE NAME	CHK SHLDR/NECK SKINNER
MODEL #	CSNS-1
MATERIAL	
BLANK FILE NAME	
QUANTITY PER MACHINE	1
CLIENT/APPLICATION	

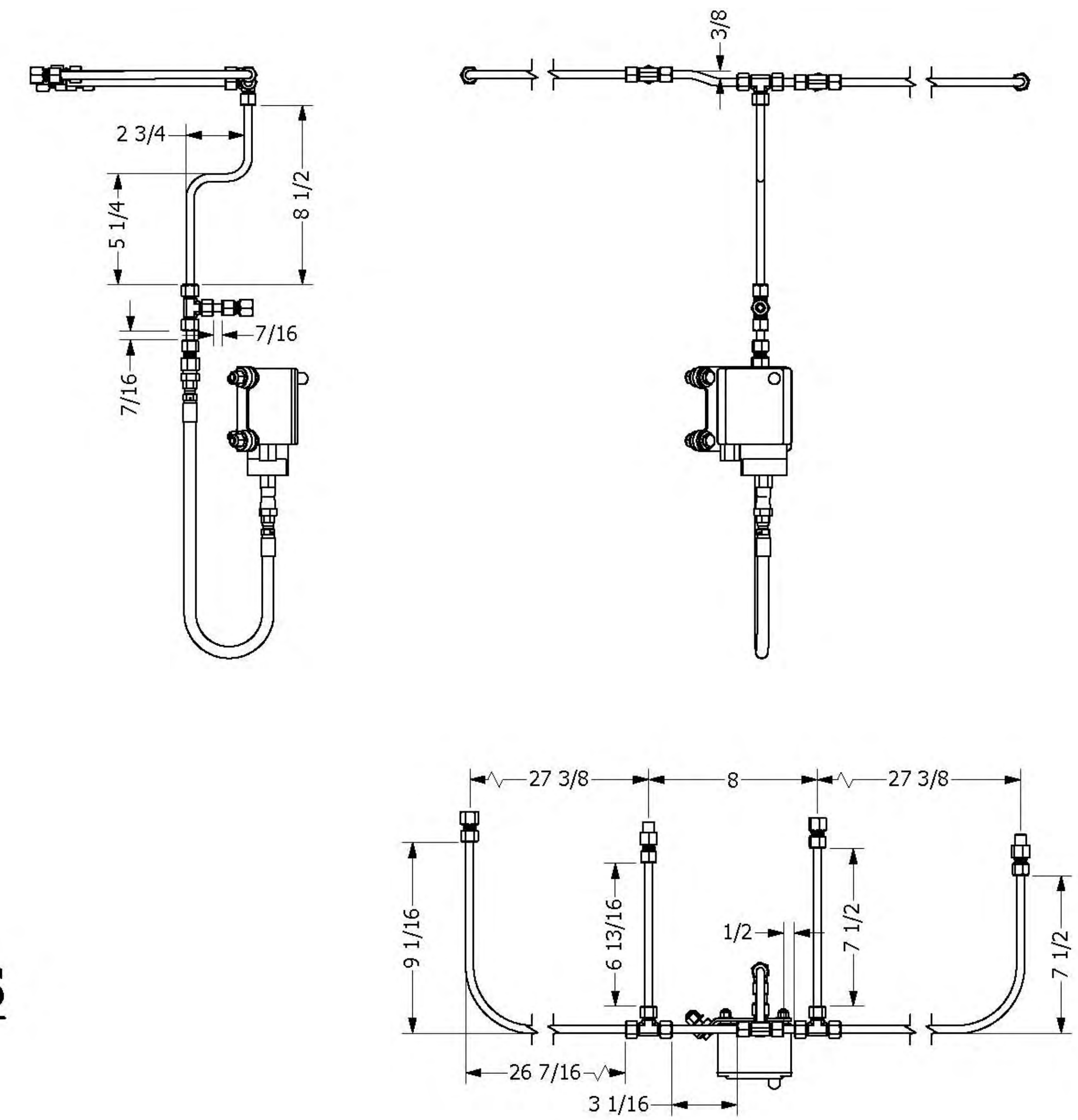
PRIME
Equipment Group, Inc.
Columbus, Ohio USA

TITLE
BASE ASSY,CSNS,LH

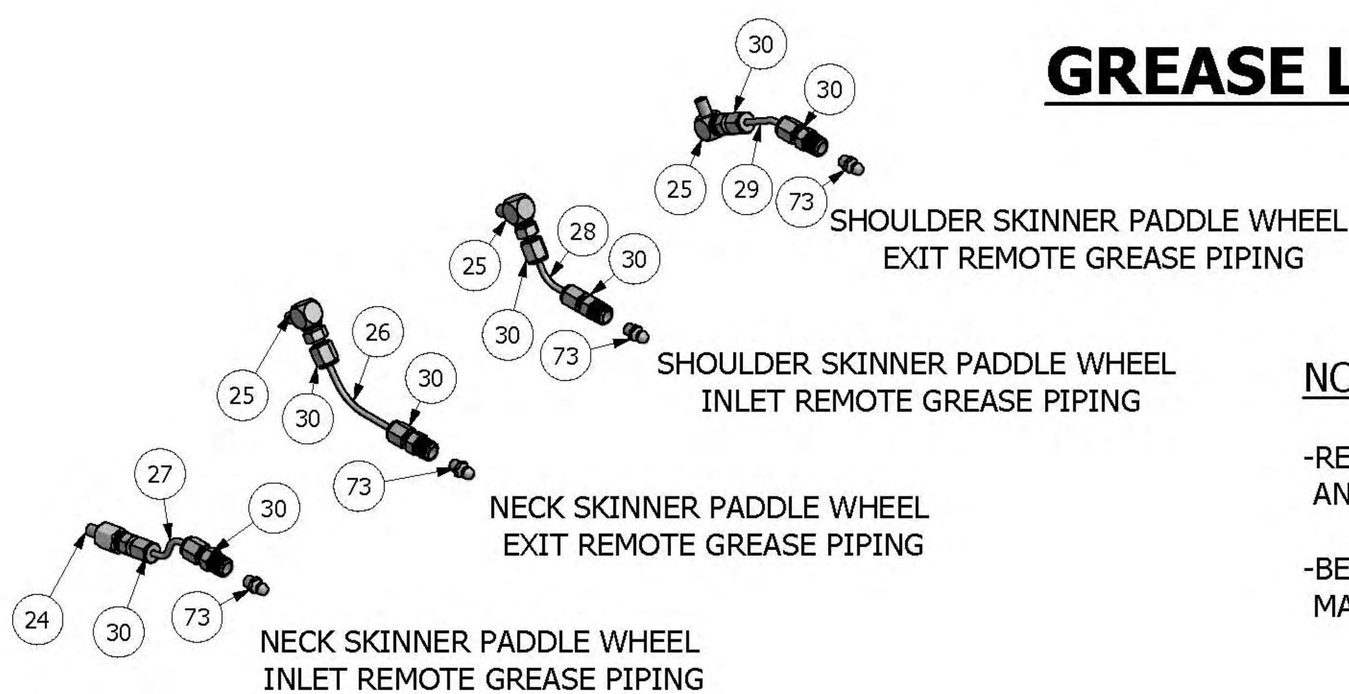
DWG NO XX-CS00-17027
PART NO 17027
SIZE D SHEET 1 OF 3

WATER LINES





GREASE LINES



NOTES:

-REMOVE BEARING GREASE ZERKS AND INSTALL ITEMS 24 & 25

-BEND TUBING AS REQUIRED TO MAKE CONNECTIONS

TOLERANCES: XX = ± 0.01 Angular = ± 0.05° XXX = ± 0.005 Fractional = ± 1/16	DRAWN BY bsnyder	CREATED 3/3/2010	PRIME Equipment Group, Inc. Columbus, Ohio USA
	MACHINE NAME CHK SHLDR/NECK SKINNER		
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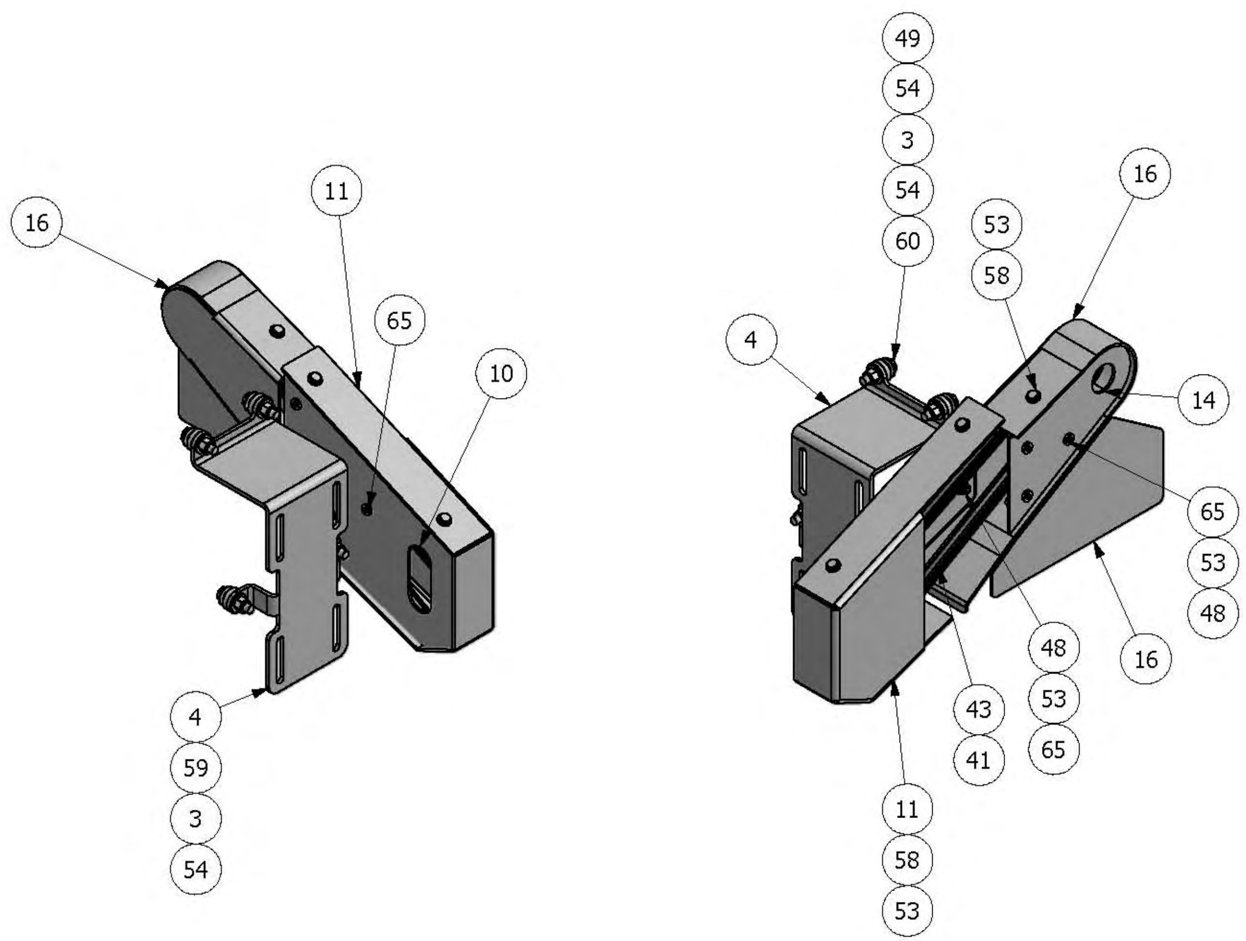
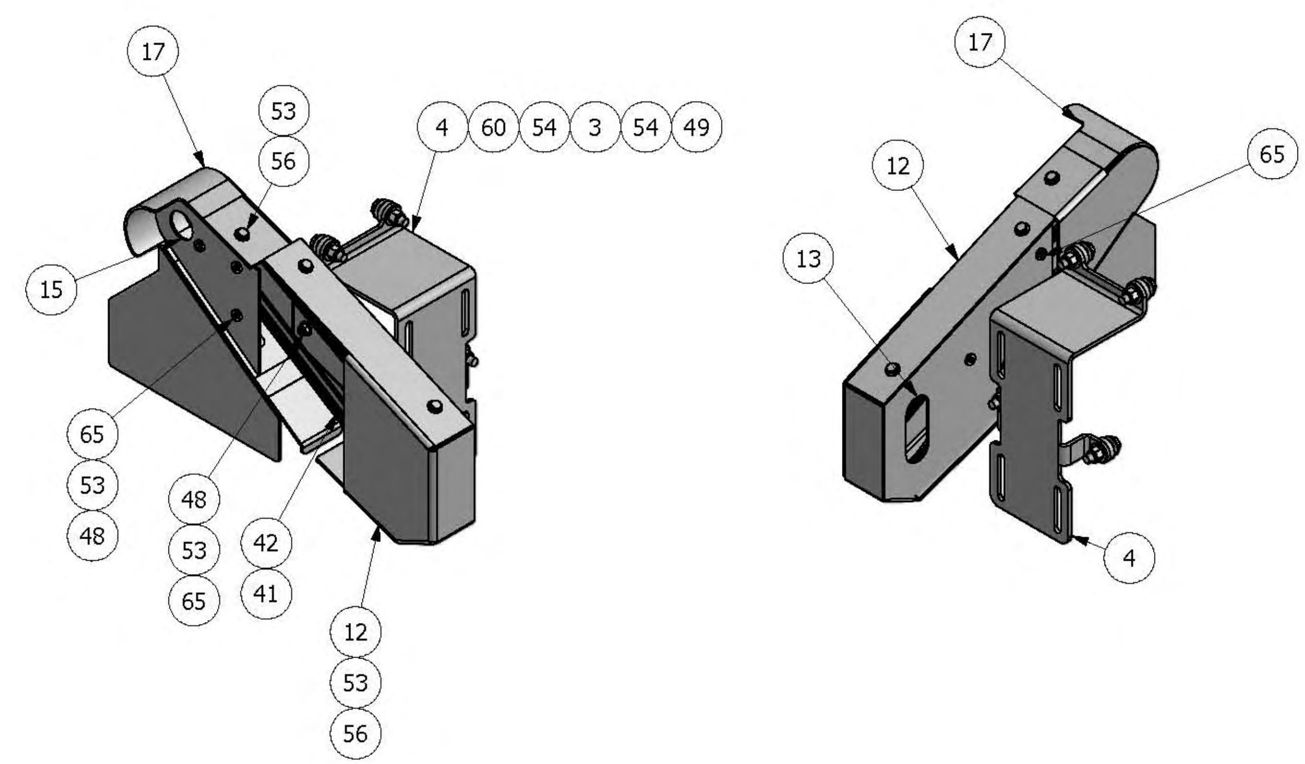
	MATERIAL		
	BLANK FILE NAME		
	QUANTITY PER MACHINE 1		DWG NO XX-CS00-17027 — PART NO 17027
	CLIENT/APPLICATION		SIZE D — SHEET 2 OF 3

NECK SKINNER DRIVE ASSY



SHOULDER SKINNER DRIVE ASSY



NOTE: INSTALL MOTOR ASSY'S BEFORE ASSEMBLING

TOLERANCES: XX = ± 0.01 Angular = ± 0.05° XXX = ± 0.005 Fractional = ± 1/16	DRAWN BY bsnyder	CREATED 3/3/2010	PRIME Equipment Group, Inc. Columbus, Ohio USA
	MACHINE NAME CHK SHLDR/NECK SKINNER		
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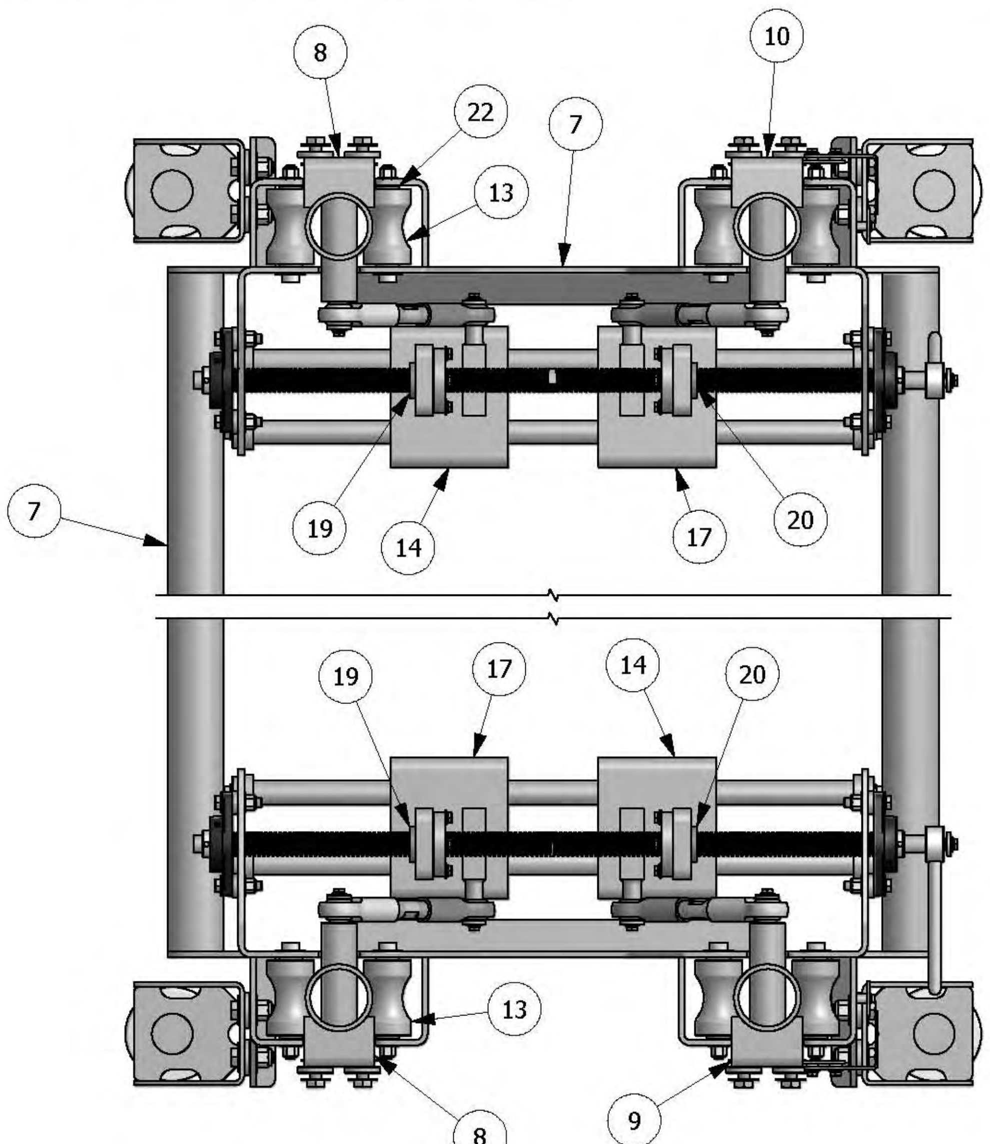
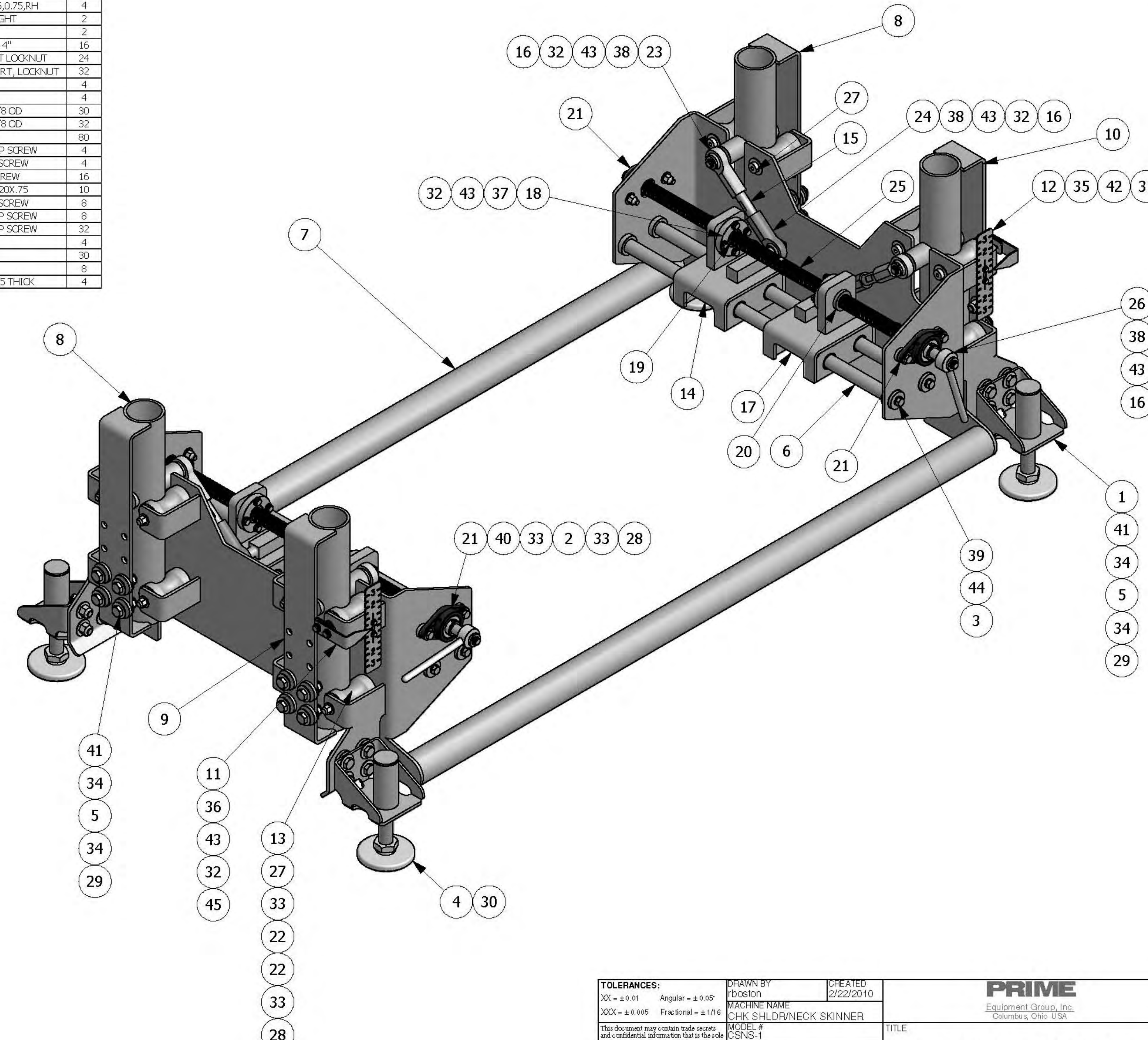
	MATERIAL		
	BLANK FILE NAME		
	QUANTITY PER MACHINE 1		DWG NO XX-CS00-17027 / PART NO 17027
	CLIENT/APPLICATION		SIZE D / SHEET 3 OF 3

Parts List				
Item	Drawing No.	Part No.	Description	Qty.
1	KS07-10988	10988	ADJUSTER WLDMT,HEIGHT	4
2	13516	13516	SPACER,0.25X1ODX0.38ID,SS	8
3	14197	14197	SPACER,1.38OD,0.44ID,0.25 THK	8
4	14224	14224	SHOE,MACHINE,4"Ø,1"-8,6"LONG	4
5	14554	14554	SPACER,0.25X1.50ODX0.56ID,SS	32
6	16573	16573	ROD,SLIDE,SSRD1.0X26.75,3/8-16	4
7	16576	16576	FRAME WLDMT,HEIGHT,ADJ,CSNS	1
8	16581	16581	LEG WLDMT,ADJUSTMENT,HEIGHT	2
9	16582	16582	LEG WLDMT,ADJUSTMENT,HEIGHT,LH	1
10	16583	16583	LEG WLDMT,ADJUSTMENT,HEIGHT,RH	1
11	16595	16595	PLT,FRMD,INDICATOR,HEIGHT	2
12	16596	16596	PLT,SCALE,INDICATOR,HEIGHT	2
13	16597	16597	ROLLER,LEG,ADJUSTMENT,HEIGHT	16
14	16609	16609	SLIDE ASSY,ADJ,HEIGHT,LH	2
15	16614	16614	ROD,CONNECTING,END,ROD	4
16	16615	16615	SPACER,0.188X0.313ODX1.0OD,SS	10
17	16626	16626	SLIDE ASSY,ADJ,HEIGHT,RH	2
18	16645	16645	FLG,MTG,NUT,ACME,1-5,SS	4
19	16646	16646	NUT,ACME,1-5,BRONZE,LH	2
20	16647	16647	NUT,ACME,1-5,BRONZE,RH	2
21	16650	16650	BRG,FLG,2 BOLT,0.75"	4
22	16651	16651	BSHG,FLG,BRNZ,0.5ID,1.0OD,1.0L	32
23	16652	16652	END,ROD,FEMALE,3/4-16,0.75,LH	4

Parts List				
Item	Drawing No.	Part No.	Description	Qty.
24	16653	16653	END,ROD,FEMALE,3/4-16,0.75,RH	4
25	16663	16663	SCREW, ACME, ADJ, HEIGHT	2
26	17154	17154	CRANK WLDMT,SSRD	2
27		897A	SHOULDER BOLT, 1/2" X 4"	16
28		ESNS037C	3/8-16 SS NYLON INSERT LOCKNUT	24
29		ESNS050C	1/2-13, SS, NYLON, INSERT, LOCKNUT	32
30		FHNS100C	1-8,S.S.FIN,HEX,NUT	4
31		FWCS10	WASHER,FLAT,SS,#10	4
32		FWUS025	1/4 SS FLAT WASHER, 5/8 OD	30
33		FWUS037	3/8 SS FLAT WASHER, 7/8 OD	32
34		FWUS050	1/2 SS FLAT WASHER	80
35		HCSS010C0050	#10-24X1/2 S.S. HEX CAP SCREW	4
36		HCSS025C0050	1/4-20X1/2 SS HEX CAP SCREW	4
37		HCSS025C0100	1/4-20X1 SS HEX CAP SCREW	16
38		HCSS025C075	HEX CAP SCREW,SS,1/4-20X.75	10
39		HCSS037C0075	3/8-16X3/4 SS HEX CAP SCREW	8
40		HCSS037C0175	3/8-16X1-3/4 SS HEX CAP SCREW	8
41		HCSS050C0150	1/2-13X1-1/2 SS HEX CAP SCREW	32
42		LWSS01	# 10 LOCK WASHER	4
43		LWSS025	1/4 SS LOCK WASHER	30
44		LWSS037	3/8 SS LOCK WASHER	8
45		SPACER16FW250125	SPACER, .5OD,.25IDX.125 THICK	4

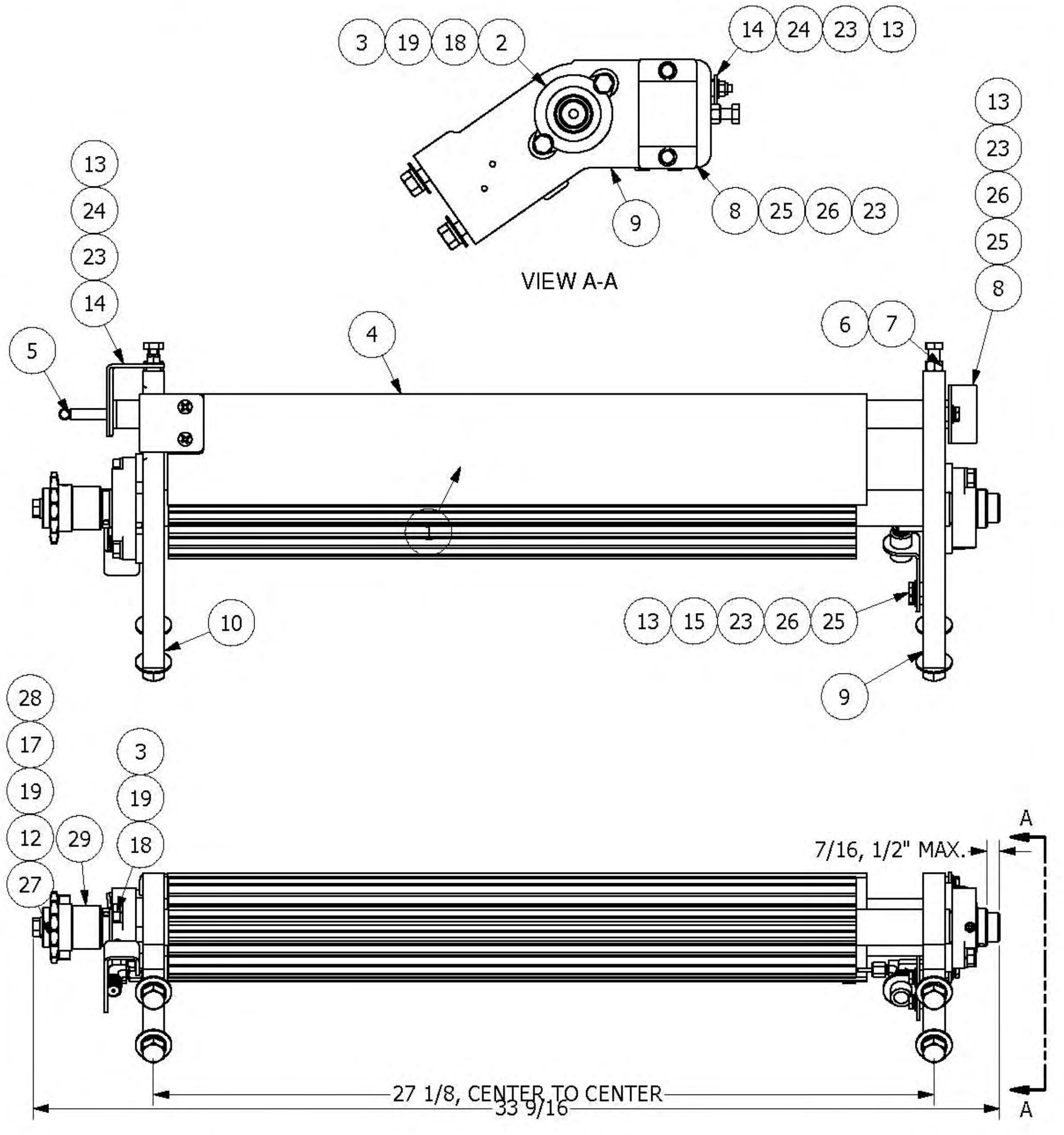


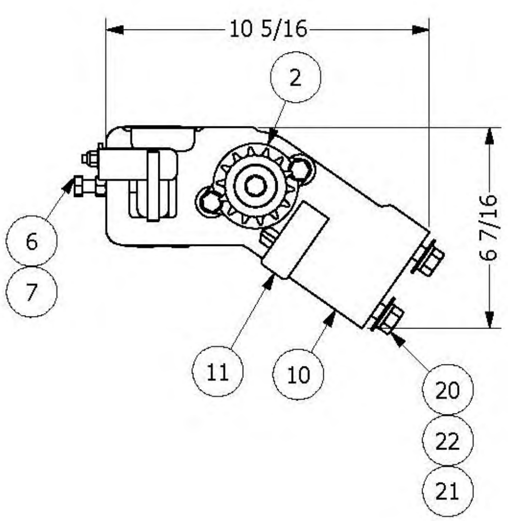


TOLERANCES: XX = ± 0.01 Angular = ± 0.05° XXX = ± 0.005 Fractional = ± 1/16	DRAWN BY rboston	CREATED 2/22/2010	PRIME Equipment Group, Inc. Columbus, Ohio USA
	MACHINE NAME CHK SHLDR/NECK SKINNER		
	MODEL # CSNS-1		TITLE FRAME ASSY,ADJ,HEIGHT,CSNS
	MATERIAL		
	BLANK FILE NAME		
	QUANTITY PER MACHINE 1		DWG NO 16616 — PART NO 16616
	CLIENT/APPLICATION		SIZE D SHEET 1 OF 1

Parts List

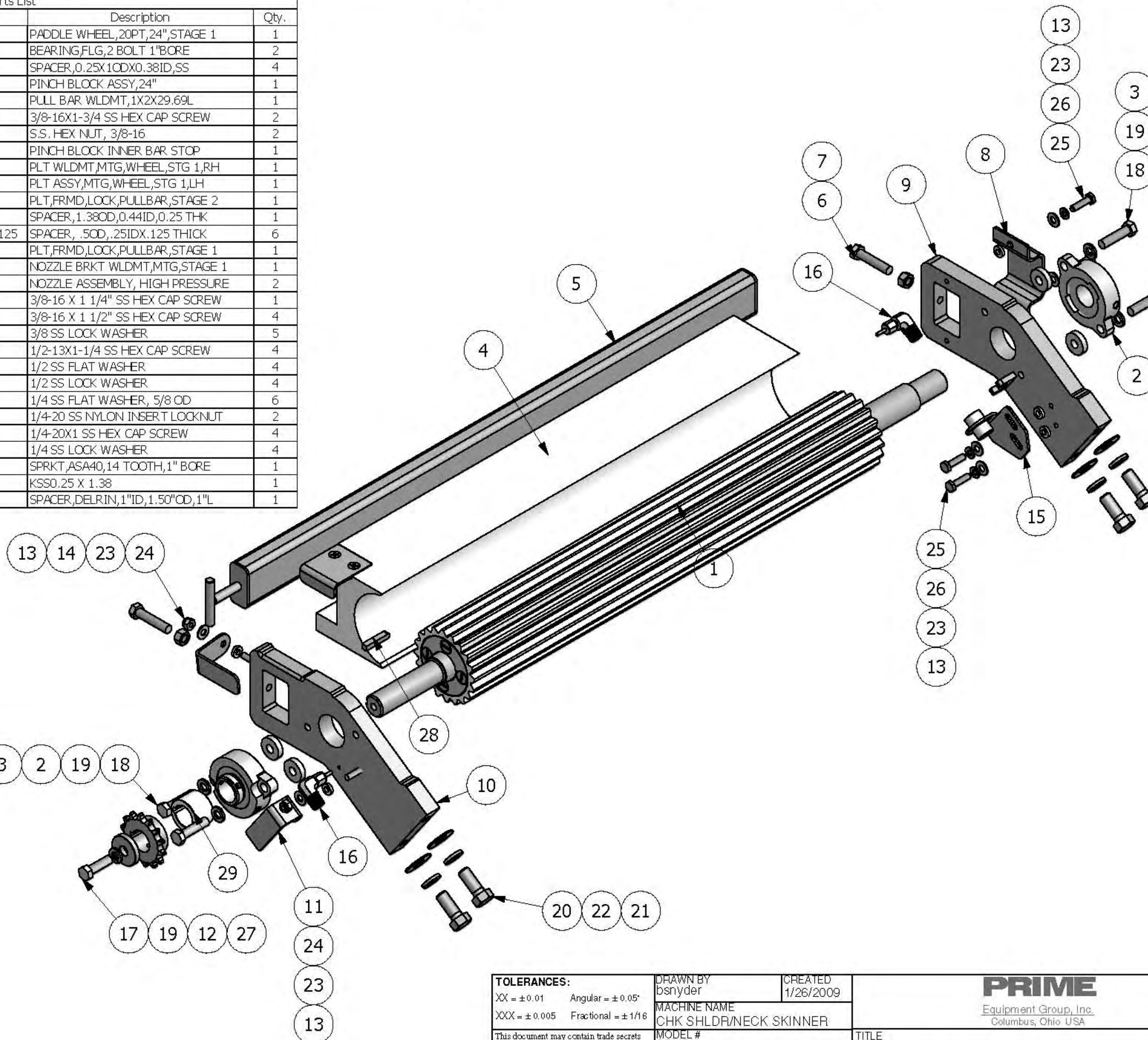
Item	Drawing No.	Part No.	Description	Qty.
1	CS02-14156	14156	PADDLE WHEEL,20PT,24",STAGE 1	1
2	13517	13517	BEARING,FLG,2 BOLT 1"BORE	2
3	13516	13516	SPACER,0.25X1ODX0.38ID,SS	4
4	CS03-14162	14162	PINCH BLOCK ASSY,24"	1
5	CS03-13906	13906	PULL BAR WLDMT,1X2X29.69L	1
6	HCSS037C0175	HCSS037C0175	3/8-16X1-3/4 SS HEX CAP SCREW	2
7	FHNS038C	FHNS038C	S.S. HEX NUT, 3/8-16	2
8	SP03-6-A	SP03-6-A	PINCH BLOCK INNER BAR STOP	1
9	CS01-14181	14181	PLT WLDMT,MTG,WHEEL,STG 1,RH	1
10	CS01-14194	14194	PLT ASSY,MTG,WHEEL,STG 1,LH	1
11	CS12-14186	14186	PLT,FRMD,LOCK,PULLBAR,STAGE 2	1
12	14197	14197	SPACER,1.38OD,0.44ID,0.25 THK	1
13	SPACER16FW250125	SPACER16FW250125	SPACER, .5OD,.25IDX.125 THICK	6
14	CS12-14185	14185	PLT,FRMD,LOCK,PULLBAR,STAGE 1	1
15	CS12-14205	14205	NOZZLE BRKT WLDMT,MTG,STAGE 1	1
16	XX-SP45-2-B	SP45-2-B	NOZZLE ASSEMBLY, HIGH PRESSURE	2
17	HCSS037C0125	HCSS037C0125	3/8-16 X 1 1/4" SS HEX CAP SCREW	1
18	HCSS037C0150	HCSS037C0150	3/8-16 X 1 1/2" SS HEX CAP SCREW	4
19	LWSS037	LWSS037	3/8 SS LOCK WASHER	5
20	HCSS050C0125	HCSS050C0125	1/2-13X1-1/4 SS HEX CAP SCREW	4
21	FWUS050	FWUS050	1/2 SS FLAT WASHER	4
22	LWSS050	LWSS050	1/2 SS LOCK WASHER	4
23	FWUS025	FWUS025	1/4 SS FLAT WASHER, 5/8 OD	6
24	ESNS025C	ESNS025C	1/4-20 SS NYLON INSERT LOCKNUT	2
25	HCSS025C0100	HCSS025C0100	1/4-20X1 SS HEX CAP SCREW	4
26	LWSS025	LWSS025	1/4 SS LOCK WASHER	4
27		577B	SPRKT,ASA40,14 TOOTH,1" BORE	1
28	KSS0.25	KSS0.25	KSS0.25 X 1.38	1
29	CS11-14179	14179	SPACER,DELRIN,1"ID,1.50"OD,1"L	1





TOLERANCES:			
XX = ±0.01 Angular = ±0.05°	DRAWN BY bsnyder	CREATED 1/26/2009	PRIME Equipment Group, Inc. Columbus, Ohio USA
XXX = ±0.005 Fractional = ±1/16	MACHINE NAME CHK SHLDR/NECK SKINNER		
	MODEL # CSNS-1		TITLE PADDLE WHEEL ASSY,STAGE 1
	MATERIAL		
	BLANK FILE NAME		
	QUANTITY PER MACHINE 1		DWG NO XX-CS00-14198 — PART NO 14198
	CLIENT/APPLICATION		SIZE C SHEET 1 OF 2

Parts List				
Item	Drawing No.	Part No.	Description	Qty.
1	CS02-14156	14156	PADDLE WHEEL,20PT,24",STAGE 1	1
2	13517	13517	BEARING,FLG,2 BOLT 1"BORE	2
3	13516	13516	SPACER,0.25X1ODX0.38ID,SS	4
4	CS03-14162	14162	PINCH BLOCK ASSY,24"	1
5	CS03-13906	13906	PULL BAR WLDMT,1X2X29.69L	1
6	HCSS037C0175	HCSS037C0175	3/8-16X1-3/4 SS HEX CAP SCREW	2
7	FHNS038C	FHNS038C	S.S. HEX NUT, 3/8-16	2
8	SP03-6-A	SP03-6-A	PINCH BLOCK INNER BAR STOP	1
9	CS01-14181	14181	PLT WLDMT,MTG,WHEEL,STG 1,RH	1
10	CS01-14194	14194	PLT ASSY,MTG,WHEEL,STG 1,LH	1
11	CS12-14186	14186	PLT,FRMD,LOCK,PULLBAR,STAGE 2	1
12	14197	14197	SPACER,1.38OD,0.44ID,0.25 THK	1
13	SPACER16FW250125	SPACER16FW250125	SPACER, .5OD,.25IDX.125 THICK	6
14	CS12-14185	14185	PLT,FRMD,LOCK,PULLBAR,STAGE 1	1
15	CS12-14205	14205	NOZZLE BRKT WLDMT,MTG,STAGE 1	1
16	XX-SP45-2-B	SP45-2-B	NOZZLE ASSEMBLY, HIGH PRESSURE	2
17	HCSS037C0125	HCSS037C0125	3/8-16 X 1 1/4" SS HEX CAP SCREW	1
18	HCSS037C0150	HCSS037C0150	3/8-16 X 1 1/2" SS HEX CAP SCREW	4
19	LWSS037	LWSS037	3/8 SS LOCK WASHER	5
20	HCSS050C0125	HCSS050C0125	1/2-13X1-1/4 SS HEX CAP SCREW	4
21	FWUS050	FWUS050	1/2 SS FLAT WASHER	4
22	LWSS050	LWSS050	1/2 SS LOCK WASHER	4
23	FWUS025	FWUS025	1/4 SS FLAT WASHER, 5/8 OD	6
24	ESNS025C	ESNS025C	1/4-20 SS NYLON INSERT LOCKNUT	2
25	HCSS025C0100	HCSS025C0100	1/4-20X1 SS HEX CAP SCREW	4
26	LWSS025	LWSS025	1/4 SS LOCK WASHER	4
27		577B	SPRKT,ASA40,14 TOOTH,1" BORE	1
28	KSS0.25	KSS0.25	KSS0.25 X 1.38	1
29	CS11-14179	14179	SPACER,DELRIN,1"ID,1.50"OD,1"L	1



TOLERANCES:
XX = ± 0.01 Angular = ± 0.05°
XXX = ± 0.005 Fractional = ± 1/16

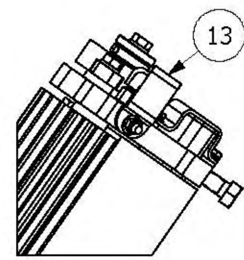
DRAWN BY: bsnyder
CREATED: 1/26/2009
MACHINE NAME: CHK SHLDR/NECK SKINNER
MODEL #: CSNS-1
MATERIAL:
BLANK FILE NAME:
QUANTITY PER MACHINE: 1
CLIENT/APPLICATION:

PRIME
Equipment Group, Inc.
Columbus, Ohio USA

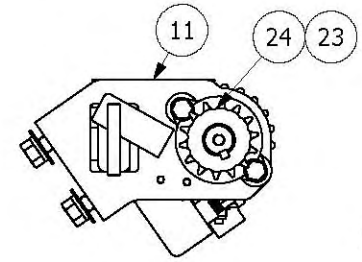
TITLE: PADDLE WHEEL ASSY,STAGE 1

DWG NO: XX-CS00-14198
PART NO: 14198
SIZE C SHEET 2 OF 2

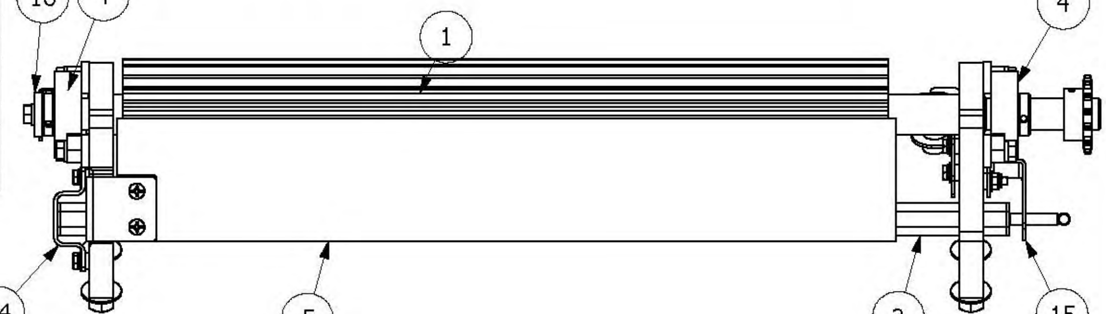
Parts List				
Item	Drawing No.	Part No.	Description	Qty.
1	CS02-14157	14157	PADDLE WHEEL,20PT,24",STAGE 2	1
2	CS03-13906	13906	PULL BAR WLDMT,1X2X29.69L	1
3	13516	13516	SPACER,0.25X1ODX0.38ID,SS	4
4	13517	13517	BEARING,FLG,2 BOLT 1"BORE	2
5	CS03-14163	14163	PINCH BLOCK ASSY,24",STAGE 2	1
6	CS01-14196	14196	PLT,ASSY,MTG,WHEEL,STAGE 2,LH	1
7	SPACER16FW250125	SPACER16FW250125	SPACER, .5OD,.25IDX.125 THICK	6
8	HCSS025C0100	HCSS025C0100	1/4-20X1 SS HEX CAP SCREW	4
9	FWUS025	FWUS025	1/4 SS FLAT WASHER, 5/8 OD	6
10	LWSS025	LWSS025	1/4 SS LOCK WASHER	4
11	XX-CS01-14182	14182	PLT WLDMT,MTG,WHEEL,STG 2,RH	1
12	CS12-14206	14206	NOZZLE BRKT WLDMT,MTG,STAGE 2	1
13	CS12-14188	14188	PLT,FRMD,LOCK,NOZZLE,STAGE 2	1
14	CS12-14187	14187	PLT,FRMD,STOP,PULLBAR,STAGE 2	1
15	CS12-14186	14186	PLT,FRMD,LOCK,PULLBAR,STAGE 2	1
16	14197	14197	SPACER,1.38OD,0.44ID,0.25 THK	1
17	HCSS037C0150	HCSS037C0150	3/8-16 X 1 1/2" SS HEX CAP SCREW	4
18	LWSS037	LWSS037	3/8 SS LOCK WASHER	5
19	ESNS025C	ESNS025C	1/4-20 SS NYLON INSERT LOCKNUT	2
20	HCSS050C0125	HCSS050C0125	1/2-13X1-1/4 SS HEX CAP SCREW	4
21	LWSS050	LWSS050	1/2 SS LOCK WASHER	4
22	FWUS050	FWUS050	1/2 SS FLAT WASHER	4
23	11505	11505	KEY,0.25X0.25X1.25,SS	1
24		577B	SPRKT,ASA40,14 TOOTH,1" BORE	1
25	HCSS037C0125	HCSS037C0125	3/8-16 X 1 1/4" SS HEX CAP SCREW	1
26	XX-SP45-2-B	SP45-2-B	NOZZLE ASSEMBLY, HIGH PRESSURE	2

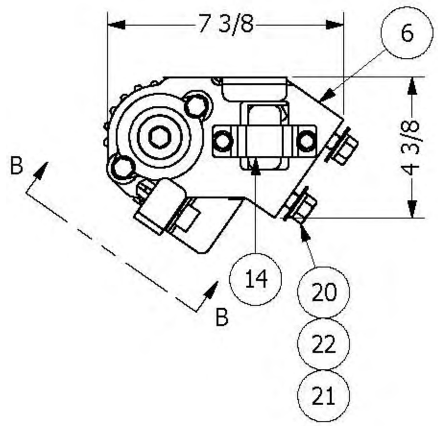


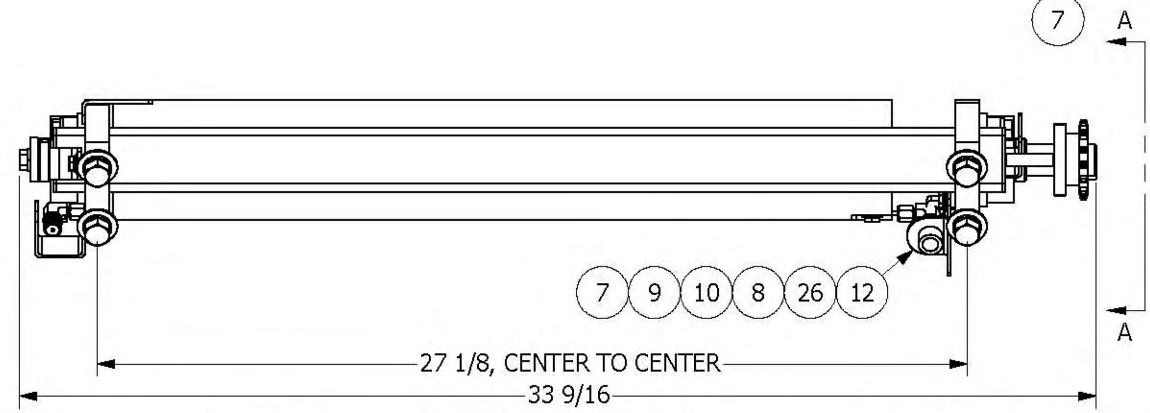
SECTION B-B



VIEW A-A

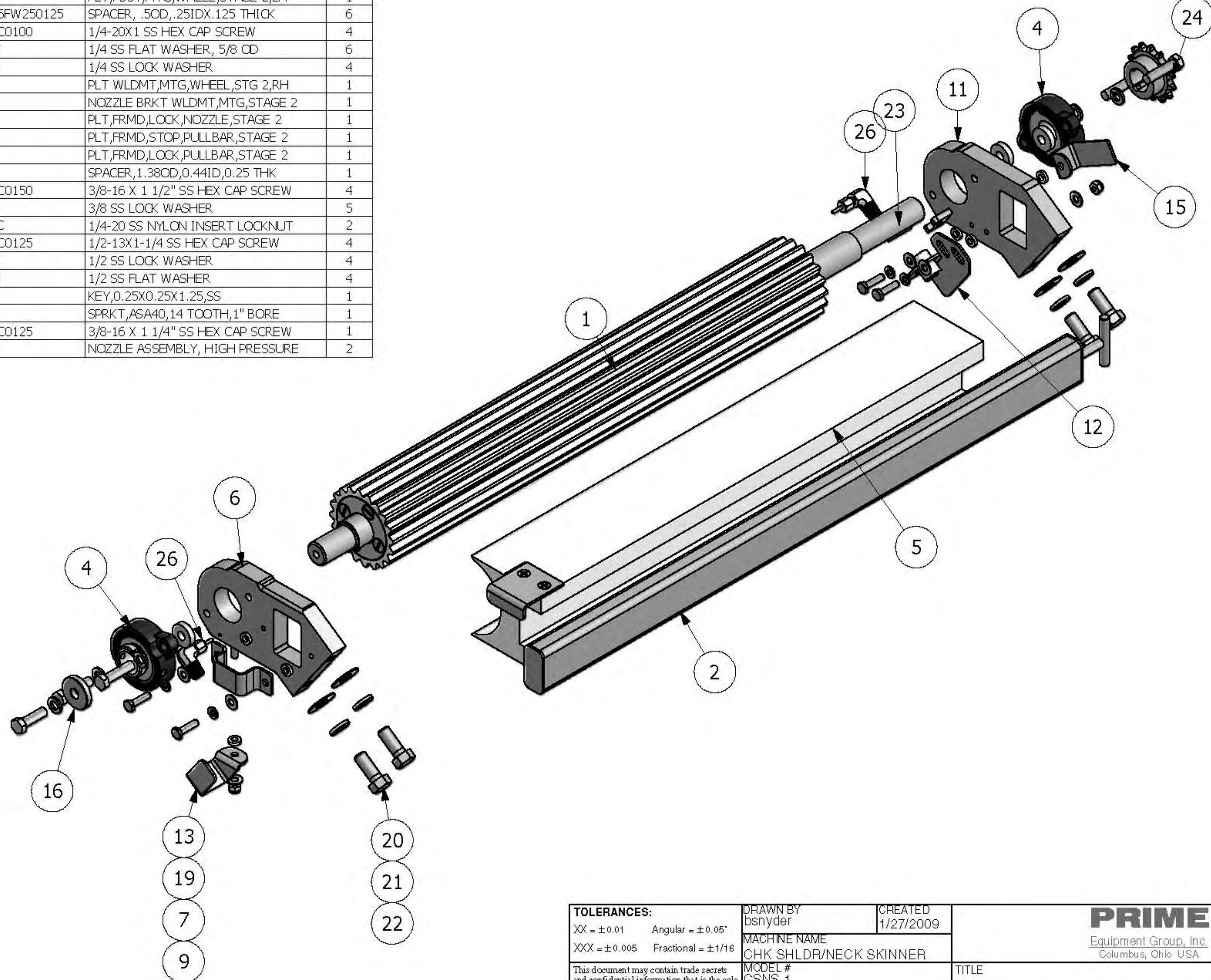






TOLERANCES: XX = ±0.01 Angular = ±0.05° XXX = ±0.005 Fractional = ±1/16	DRAWN BY bsnyder	CREATED 1/27/2009	PRIME Equipment Group, Inc. Columbus, Ohio USA
	MACHINE NAME CHK SHLDR/NECK SKINNER		
	MODEL # CSNS-1		TITLE PADDLE WHEEL ASSY,STAGE 2
	MATERIAL		
	BLANK FILE NAME		
	QUANTITY PER MACHINE 1		DWG NO XX-CS00-14199 / PART NO 14199
	CLIENT/APPLICATION		SIZE C SHEET 1 OF 2

Parts List				
Item	Drawing No.	Part No.	Description	Qty.
1	CS02-14157	14157	PADDLE WHEEL,20PT,24",STAGE 2	1
2	CS03-13906	13906	PULL BAR WLDMT,1X2X29.69L	1
3	13516	13516	SPACER,0.25X1ODX0.38ID,SS	4
4	13517	13517	BEARING,FLG,2 BOLT 1"BORE	2
5	CS03-14163	14163	PINCH BLOCK ASSY,24",STAGE 2	1
6	CS01-14196	14196	PLT,ASSY,MTG,WHEEL,STAGE 2,LH	1
7	SPACER16FW250125	SPACER16FW250125	SPACER, .5OD,.25IDX.125 THICK	6
8	HCSS025C0100	HCSS025C0100	1/4-20X1 SS HEX CAP SCREW	4
9	FWUS025	FWUS025	1/4 SS FLAT WASHER, 5/8 OD	6
10	LWSS025	LWSS025	1/4 SS LOCK WASHER	4
11	XX-CS01-14182	14182	PLT WLDMT,MTG,WHEEL,STG 2,RH	1
12	CS12-14206	14206	NOZZLE BRKT WLDMT,MTG,STAGE 2	1
13	CS12-14188	14188	PLT,FRMD,LOCK,NOZZLE,STAGE 2	1
14	CS12-14187	14187	PLT,FRMD,STOP,PULLBAR,STAGE 2	1
15	CS12-14186	14186	PLT,FRMD,LOCK,PULLBAR,STAGE 2	1
16	14197	14197	SPACER,1.38OD,0.44ID,0.25 THK	1
17	HCSS037C0150	HCSS037C0150	3/8-16 X 1 1/2" SS HEX CAP SCREW	4
18	LWSS037	LWSS037	3/8 SS LOCK WASHER	5
19	ESNS025C	ESNS025C	1/4-20 SS NYLON INSERT LOCKNUT	2
20	HCSS050C0125	HCSS050C0125	1/2-13X1-1/4 SS HEX CAP SCREW	4
21	LWSS050	LWSS050	1/2 SS LOCK WASHER	4
22	FWUS050	FWUS050	1/2 SS FLAT WASHER	4
23	11505	11505	KEY,0.25X0.25X1.25,SS	1
24		577B	SPRKT,ASA40,14 TOOTH,1" BORE	1
25	HCSS037C0125	HCSS037C0125	3/8-16 X 1 1/4" SS HEX CAP SCREW	1
26	XX-SP45-2-B	SP45-2-B	NOZZLE ASSEMBLY, HIGH PRESSURE	2



TOLERANCES: XX = ±0.01 Angular = ±0.05° XXX = ±0.005 Fractional = ±1/16	DRAWN BY bsnyder	CREATED 1/27/2009	**PRIME** Equipment Group, Inc. Columbus, Ohio USA
	MACHINE NAME CHK SHLDR/NECK SKINNER		
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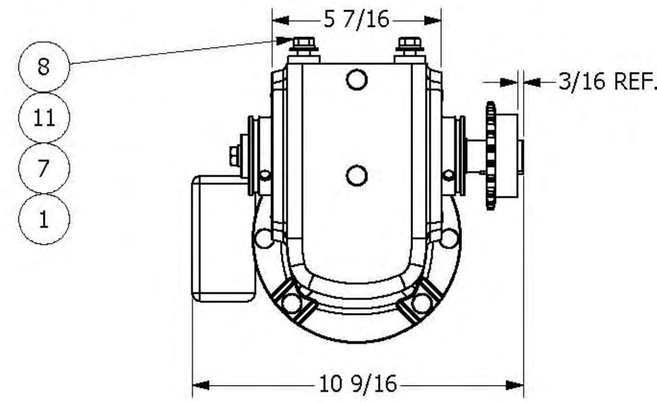
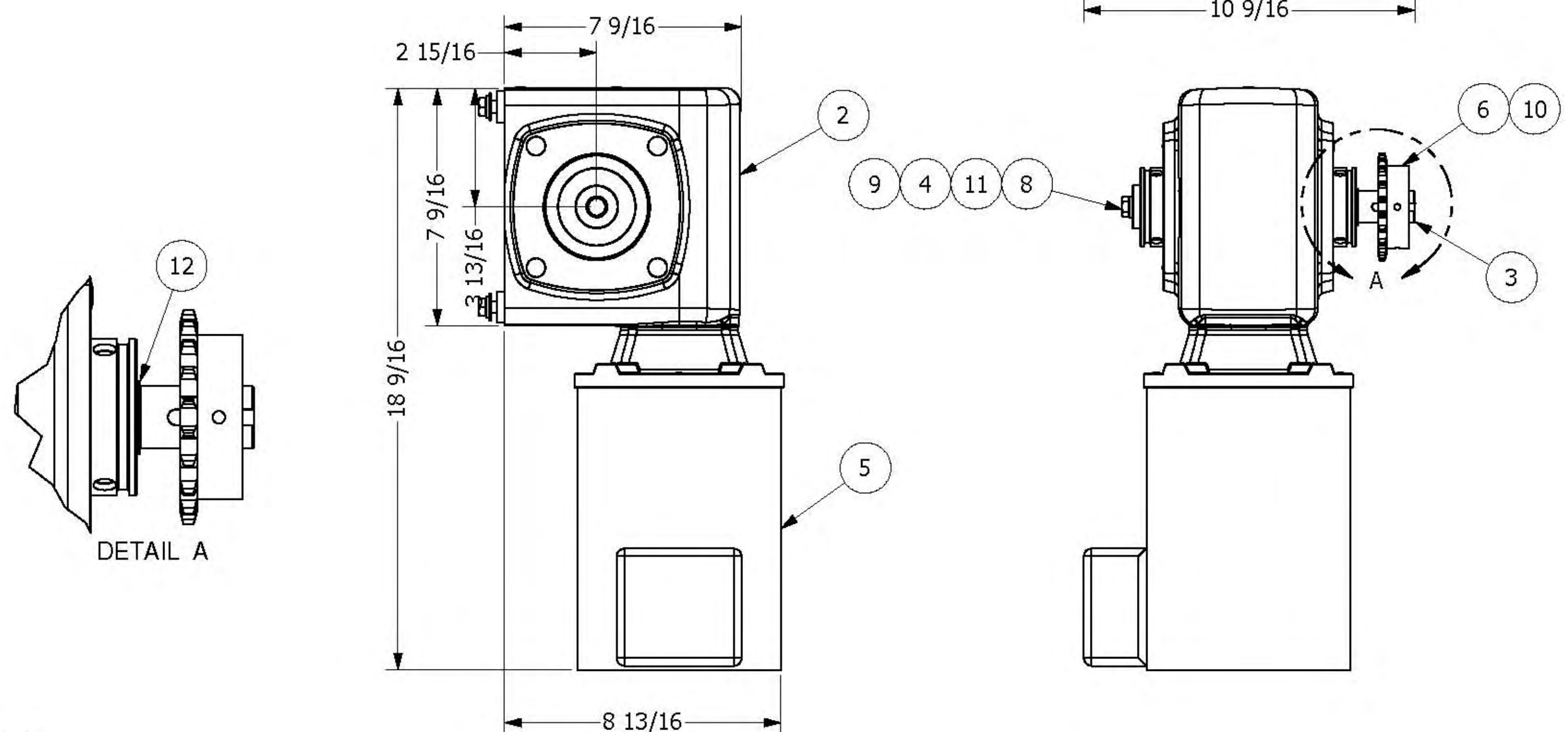
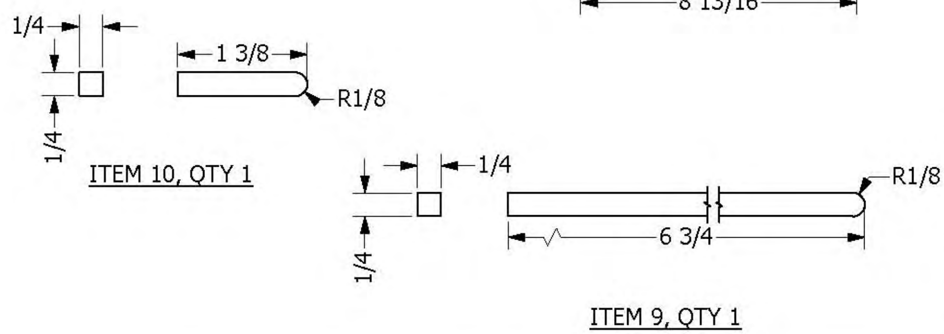
Parts List				
Item	Drawing No.	Part No.	Description	Qty.
1		13516	SPACER,0.25X1ODX0.38ID,SS	4
2		13683	GEARBOX,726,40:1,BOSTON	1
3	CS11-14159	14159	SHAFT,DRIVE,1"ØX8.75"L	1
4		14197	SPACER,1.38OD,0.44ID,0.25 THK	1
5		761F	MOTOR,1 HP,1750 RPM,3 PH,60 HZ	1
6		782	SPROCKET,40B20SS,1"B,0.25"KW	1
7		FWUS037	3/8 SS FLAT WASHER, 7/8 OD	4
8		HCSS037C0125	3/8-16X1-1/4 SS HEX CAP SCREW	5
9		KSS0.25	KSS0.25 X 6.75	1
10		KSS0.25	KSS0.25 X 1.38	1
11		LWSS037	3/8 SS LOCK WASHER	5
12		SNAPRING-SH100SS	SNAP RING, 1"	1







TOLERANCES:
XX = ±0.01 Angular = ±0.05°
XXX = ±0.005 Fractional = ±1/16

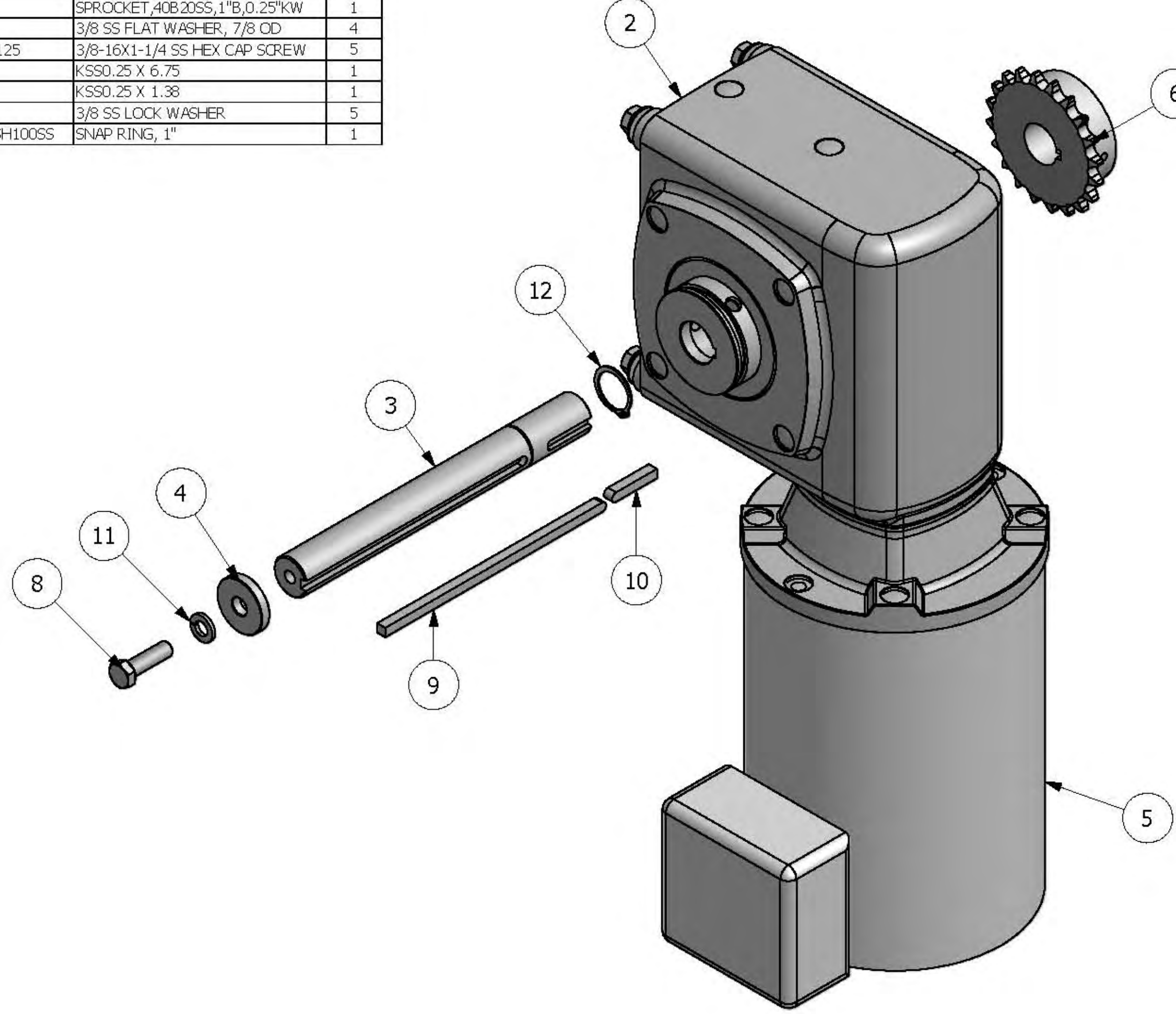
DRAWN BY: bsnyder
CREATED: 1/26/2009
MACHINE NAME: CHK SHLDR/NECK SKINNER
MODEL #: CSNS-1
MATERIAL:
BLANK FILE NAME:
QUANTITY PER MACHINE: 0
CLIENT/APPLICATION:

PRIME
Equipment Group, Inc.
Columbus, Ohio USA

TITLE: DRIVE ASSY,STAGE 1,CSNS-1

DWG NO: XX-CS00-14221
PART NO: 14221
SIZE: C SHEET 1 OF 2

Parts List				
Item	Drawing No.	Part No.	Description	Qty.
1		13516	SPACER,0.25X1ODX0.38ID,SS	4
2		13683	GEARBOX,726,40:1,BOSTON	1
3	CS11-14159	14159	SHAFT,DRIVE,1"ØX8.75"L	1
4		14197	SPACER,1.38OD,0.44ID,0.25 THK	1
5		761F	MOTOR,1 HP,1750 RPM,3 PH,60 HZ	1
6		782	SPROCKET,40B20SS,1"B,0.25"KW	1
7		FWUS037	3/8 SS FLAT WASHER, 7/8 OD	4
8		HCSS037C0125	3/8-16X1-1/4 SS HEX CAP SCREW	5
9		KSS0.25	KSS0.25 X 6.75	1
10		KSS0.25	KSS0.25 X 1.38	1
11		LWSS037	3/8 SS LOCK WASHER	5
12		SNAPRING-SH100SS	SNAP RING, 1"	1



TOLERANCES:
XX = ± 0.01 Angular = ± 0.05°
XXX = ± 0.005 Fractional = ± 1/16

DRAWN BY: bsnyder
CREATED: 1/26/2009
MACHINE NAME: CHK SHLDR/NECK SKINNER
MODEL #: CSNS-1
MATERIAL:
BLANK FILE NAME:
QUANTITY PER MACHINE: 0
CLIENT/APPLICATION:

PRIME
Equipment Group, Inc.
Columbus, Ohio USA

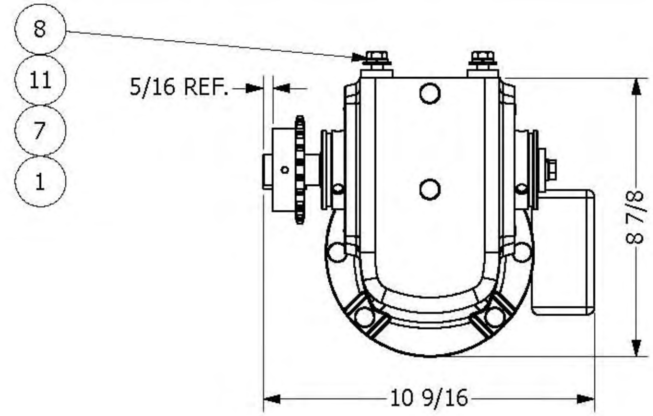
TITLE: DRIVE ASSY,STAGE 1,CSNS-1

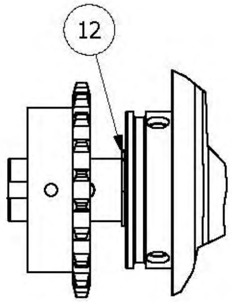
DWG NO: XX-CS00-14221
PART NO: 14221

SIZE C SHEET 2 OF 2

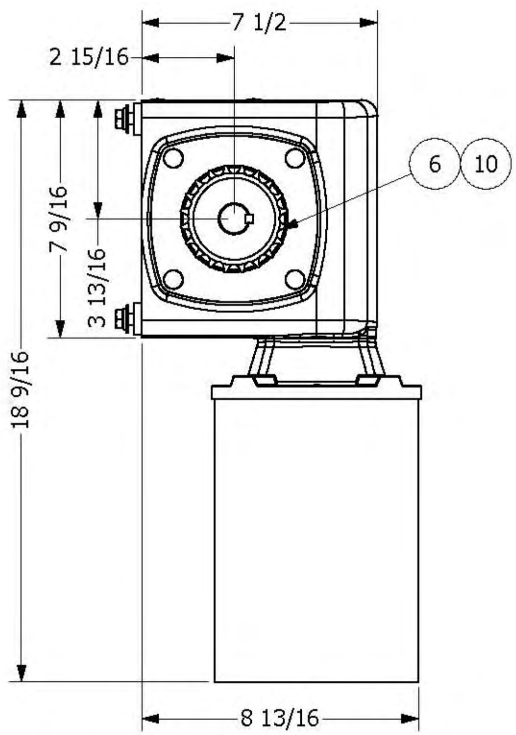
Parts List

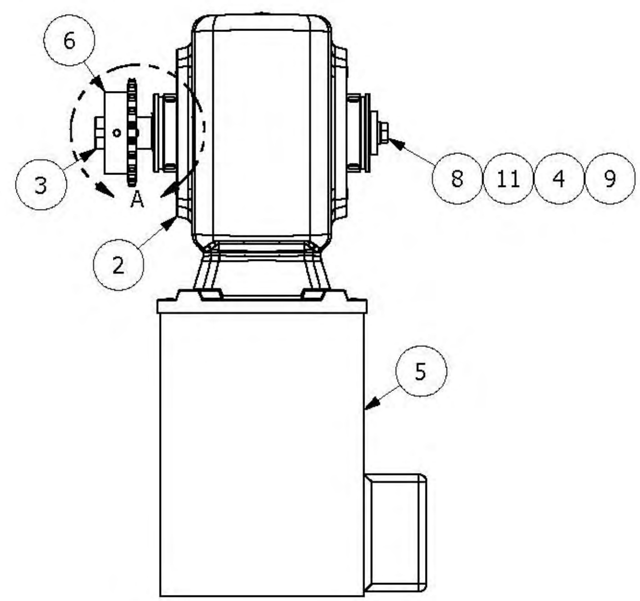
Item	Drawing No.	Part No.	Description	Qty.
1		13516	SPACER,0.25X1ODX0.38ID,SS	4
2		13683	GEARBOX,726,40:1,BOSTON	1
3	CS11-14159	14159	SHAFT,DRIVE,1"ØX8.75"L	1
4		14197	SPACER,1.38OD,0.44ID,0.25 THK	1
5		761F	MOTOR,1 HP,1750 RPM,3 PH,60 HZ	1
6		782	SPROCKET,40B20SS,1"B,0.25"KW	1
7		FWUS037	3/8 SS FLAT WASHER, 7/8 OD	4
8		HCSS037C0125	3/8-16X1-1/4 SS HEX CAP SCREW	5
9		KSS0.25	KSS0.25 X 6.75	1
10		KSS0.25	KSS0.25 X 1.38	1
11		LWSS037	3/8 SS LOCK WASHER	5
12		SNAPRING-SH100SS	SNAP RING, 1"	1

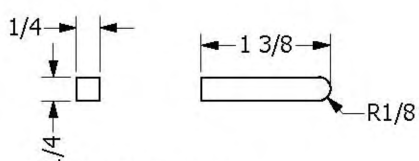




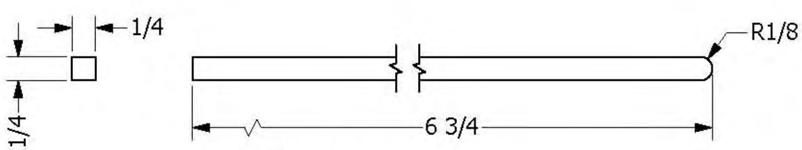
DETAIL A







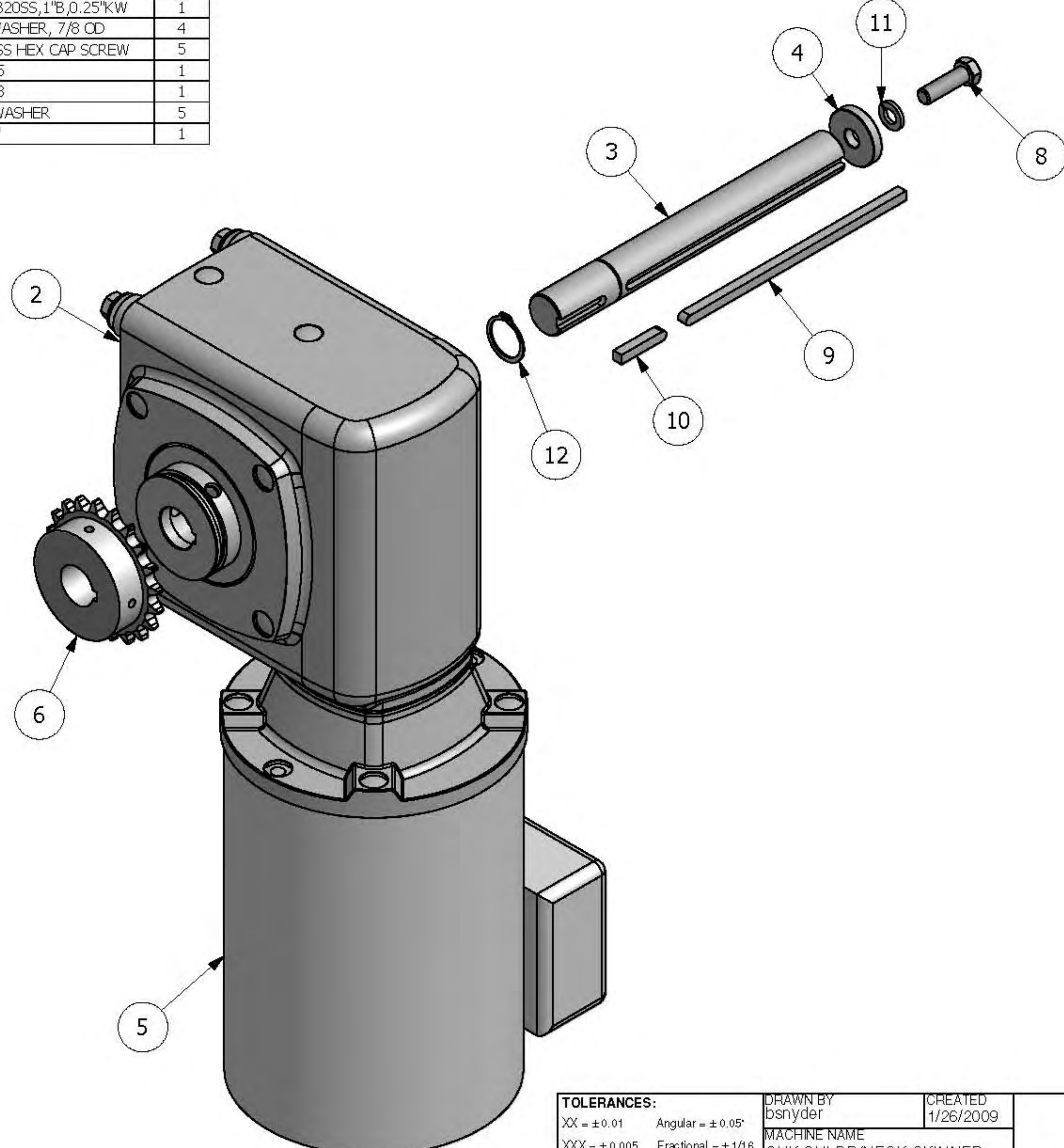
ITEM 10, QTY 1



ITEM 9, QTY 1

TOLERANCES: XX = ±0.01 Angular = ±0.05° XXX = ±0.005 Fractional = ±1/16	DRAWN BY bsnyder	CREATED 1/26/2009	PRIME Equipment Group, Inc. Columbus, Ohio USA
	MACHINE NAME CHK SHLDR/NECK SKINNER		
	MODEL # CSNS-1		TITLE DRIVE ASSY,STAGE 2,CSNS-1
	MATERIAL		
	BLANK FILE NAME		
	QUANTITY PER MACHINE 1		DWG NO XX-CS00-14222 / PART NO 14222
	CLIENT/APPLICATION		SIZE C SHEET 1 OF 2

Parts List				
Item	Drawing No.	Part No.	Description	Qty.
1		13516	SPACER,0.25X1ODX0.38ID,SS	4
2		13683	GEARBOX,726,40:1,BOSTON	1
3	CS11-14159	14159	SHAFT,DRIVE,1"ØX8.75"L	1
4		14197	SPACER,1.38OD,0.44ID,0.25 THK	1
5		761F	MOTOR,1 HP,1750 RPM,3 PH,60 HZ	1
6		782	SPROCKET,40B20SS,1"B,0.25"KW	1
7		FWUS037	3/8 SS FLAT WASHER, 7/8 OD	4
8		HCSS037C0125	3/8-16X1-1/4 SS HEX CAP SCREW	5
9		KSS0.25	KSS0.25 X 6.75	1
10		KSS0.25	KSS0.25 X 1.38	1
11		LWSS037	3/8 SS LOCK WASHER	5
12		SNAPRING-SH100SS	SNAP RING, 1"	1



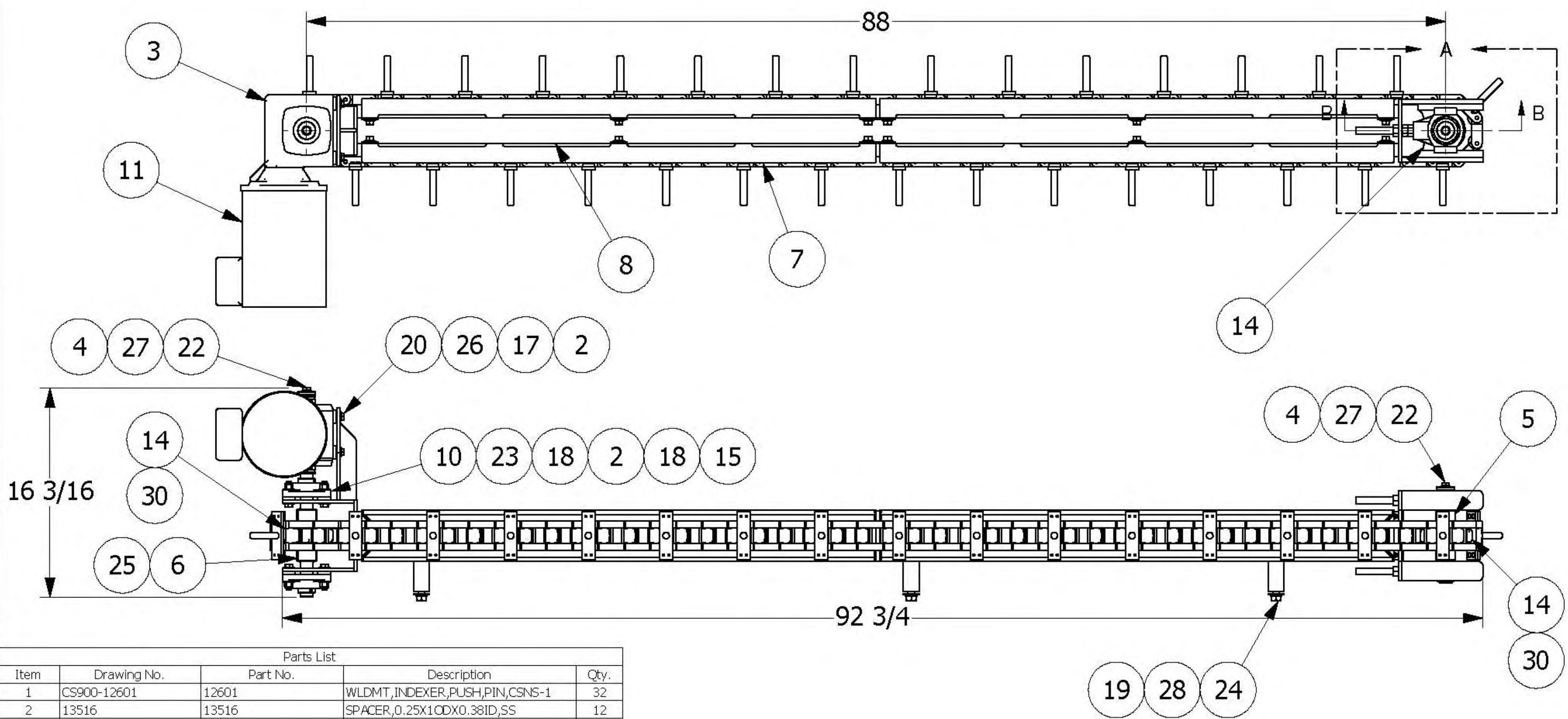
TOLERANCES:
XX = ± 0.01 Angular = ± 0.05°
XXX = ± 0.005 Fractional = ± 1/16

DRAWN BY: bsnyder
CREATED: 1/26/2009
MACHINE NAME: CHK SHLDR/NECK SKINNER
MODEL #: CSNS-1
MATERIAL:
BLANK FILE NAME:
QUANTITY PER MACHINE: 1
CLIENT/APPLICATION:

PRIME
Equipment Group, Inc.
Columbus, Ohio USA

TITLE
DRIVE ASSY,STAGE 2,CSNS-1

DWG NO: XX-CS00-14222
PART NO: 14222
SIZE C SHEET 2 OF 2



Parts List				
Item	Drawing No.	Part No.	Description	Qty.
1	CS900-12601	12601	WLDMT,INDEXER,PUSH,PIN,CSNS-1	32
2	13516	13516	SPACER,0.25X1ODX0.38ID,SS	12
3	13684	13684	GEARBOX,718,15:1,BOSTON	1
4	14197	14197	SPACER,1.38OD,0.44ID,0.25 THK	2
5	CS01-14443	14443	SHAFT,IDLER,INDEXER,CSNS-1	1
6	CS01-14444	14444	SHAFT,DRIVE,INDEXER,CSNS-1	1
7	XX-CS01-16000	16000	WEARSTRIP ASSY,CHAIN,INDEXER	4
8	XX-CS900-16002	16002	FRAME WLDMT,INDEXER,6",CSNS-1	1
9	550B	550B	BRG,TAKE-UP,1" BORE	2
10	575	575	4-BOLT,FLANGE,BEARING	2
11	761F	761F	MOTOR,1 HP,1750 RPM,3 PH,60 HZ	1
12	871	871	KNUCKLE CHAIN, SERIES 3000	32
13	871A	871A	KNUCKLE CHAIN, SERIES 3000	64
14	872	872	SPRKT,3000,8 TOOTH,1.5SQ BORE	2
15	ESNS037C	ESNS037C	3/8-16 SS NYLON INSERT LOCKNUT	8
16	FHNS050C	FHNS050C	0.50-13,S.S.FIN,HEX,NUT	6
17	FWUS031	FWUS031	5/16 SS FLAT WASHER, 3/4OD	4
18	FWUS037	FWUS037	3/8 SS FLAT WASHER, 7/8 OD	40
19	FWUS050	FWUS050	1/2 SS FLAT WASHER	3
20		HCSS031C0125	5/16-18X1-1/4 SS HEX CAP SCREW	4
21	HCSS037C0075	HCSS037C0075	3/8-16X3/4 SS HEX CAP SCREW	24
22	HCSS037C0125	HCSS037C0125	3/8-16X1-1/4 SS HEX CAP SCREW	2
23	HCSS037C0175	HCSS037C0175	3/8-16X1-3/4 SS HEX CAP SCREW	8
24	HCSS050C0125	HCSS050C0125	1/2-13X1-1/4 SS HEX CAP SCREW	3
25	KSS0.25	KSS0.25	KSS0.25 X 5.875	1
26	LWSS031	LWSS031	5/16 SS LOCK WASHER	4
27	LWSS037	LWSS037	3/8 SS LOCK WASHER	26
28	LWSS050	LWSS050	1/2 SS LOCK WASHER	3
29	MSRS10C050	MSRS10C050	10-24X1/2 MACHINE SCREW ROUND	128
30	SNAPRING-SH212SS	SNAPRING-SH212SS	SNAP,RING,2.12"	4
31	SSAT0.50-13X5.0	SSAT0.50-13	SS ALLTHREAD, 1/2-13 X 5" LONG	2

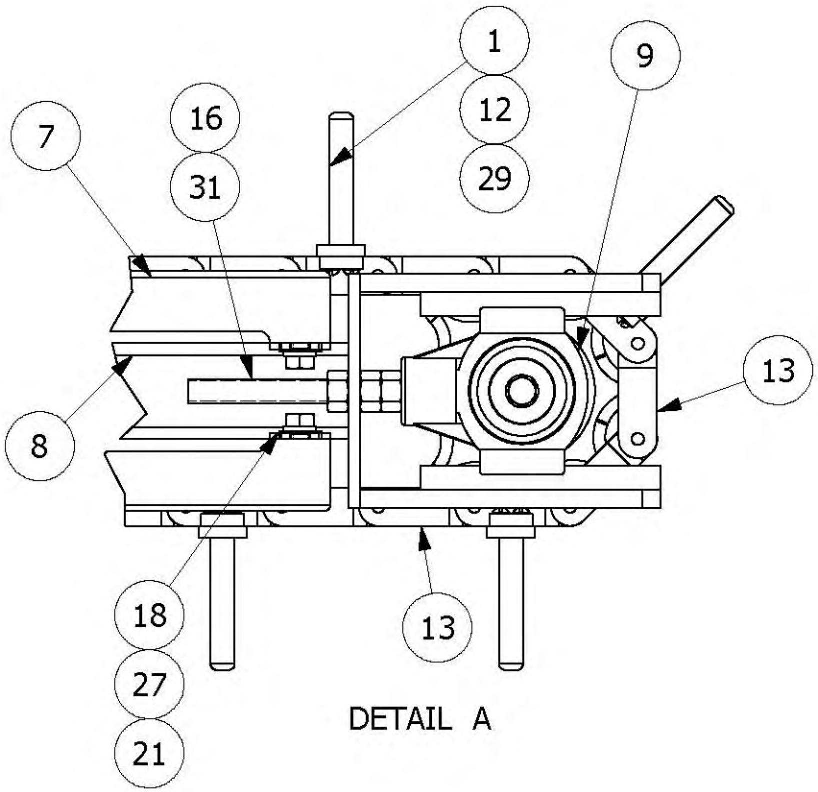
TOLERANCES:
XX = ±0.01 Angular = ±0.05°
XXX = ±0.005 Fractional = ±1/16

DRAWN BY: scribb
CREATED: 10/15/2009
MACHINE NAME: CHK SHLDR/NECK SKINNER
MODEL #: CSNS-1
MATERIAL:
BLANK FILE NAME:
QUANTITY PER MACHINE: 1
CLIENT/APPLICATION:

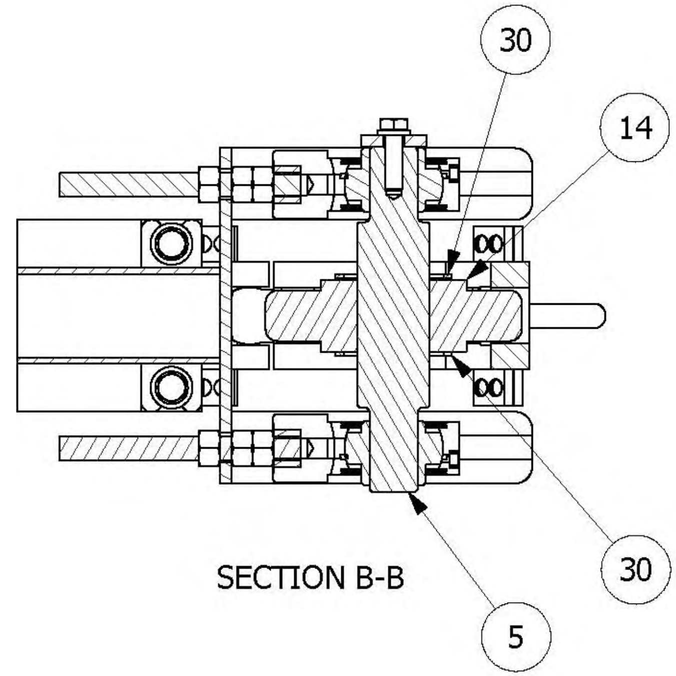
PRIME
Equipment Group, Inc.
Columbus, Ohio USA

TITLE: INDEXER ASSY,CHAIN,CSNS-1

DWG NO: XX-CS00-16004
PART NO: 16004
SIZE C SHEET 1 OF 2



DETAIL A



SECTION B-B

TOLERANCES: XX = ±0.01 Angular = ±0.05° XXX = ±0.005 Fractional = ±1/16	DRAWN BY scribb	CREATED 10/15/2009	**PRIME** Equipment Group, Inc. Columbus, Ohio USA
	MACHINE NAME CHK SHLDR/NECK SKINNER		
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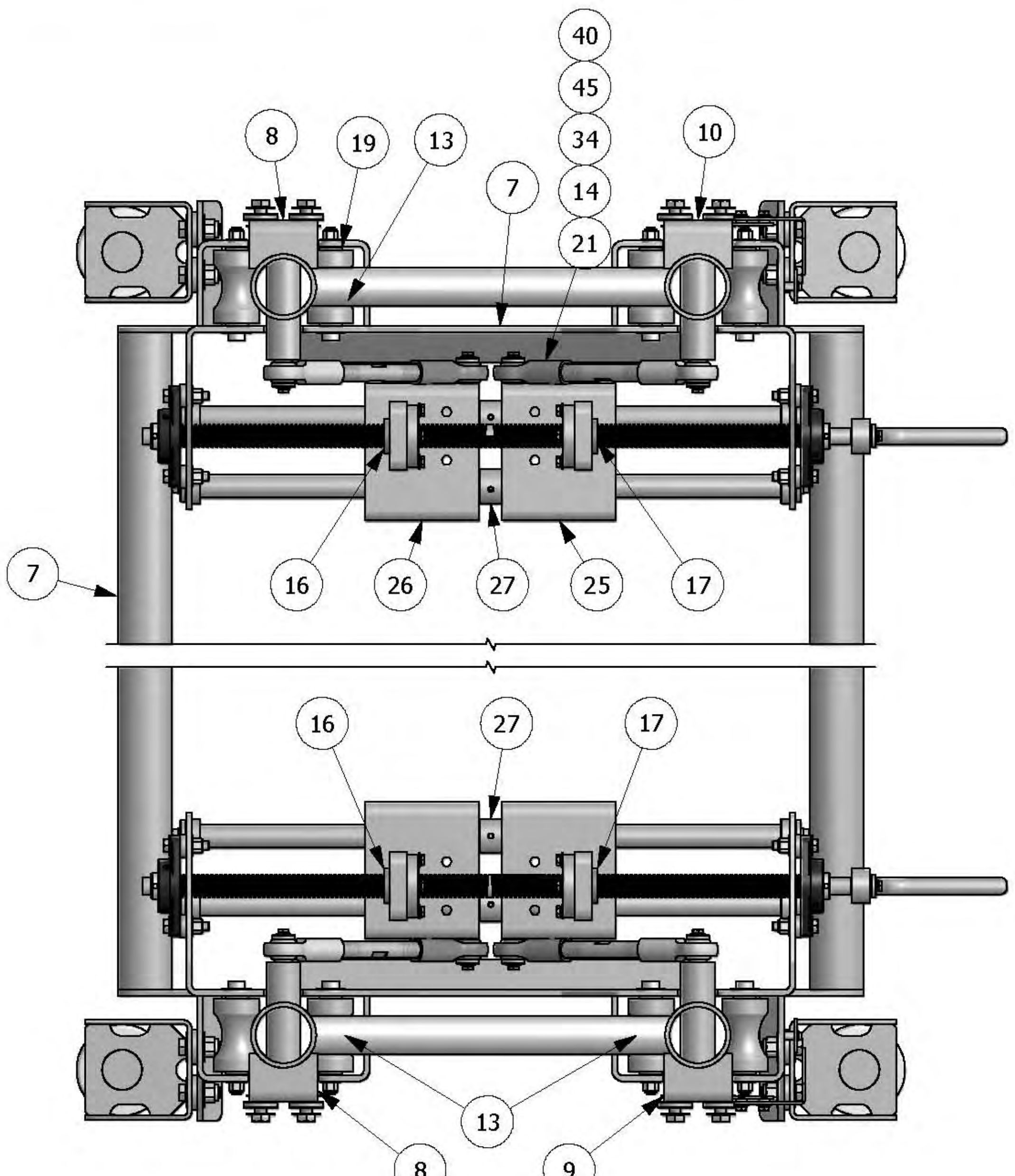
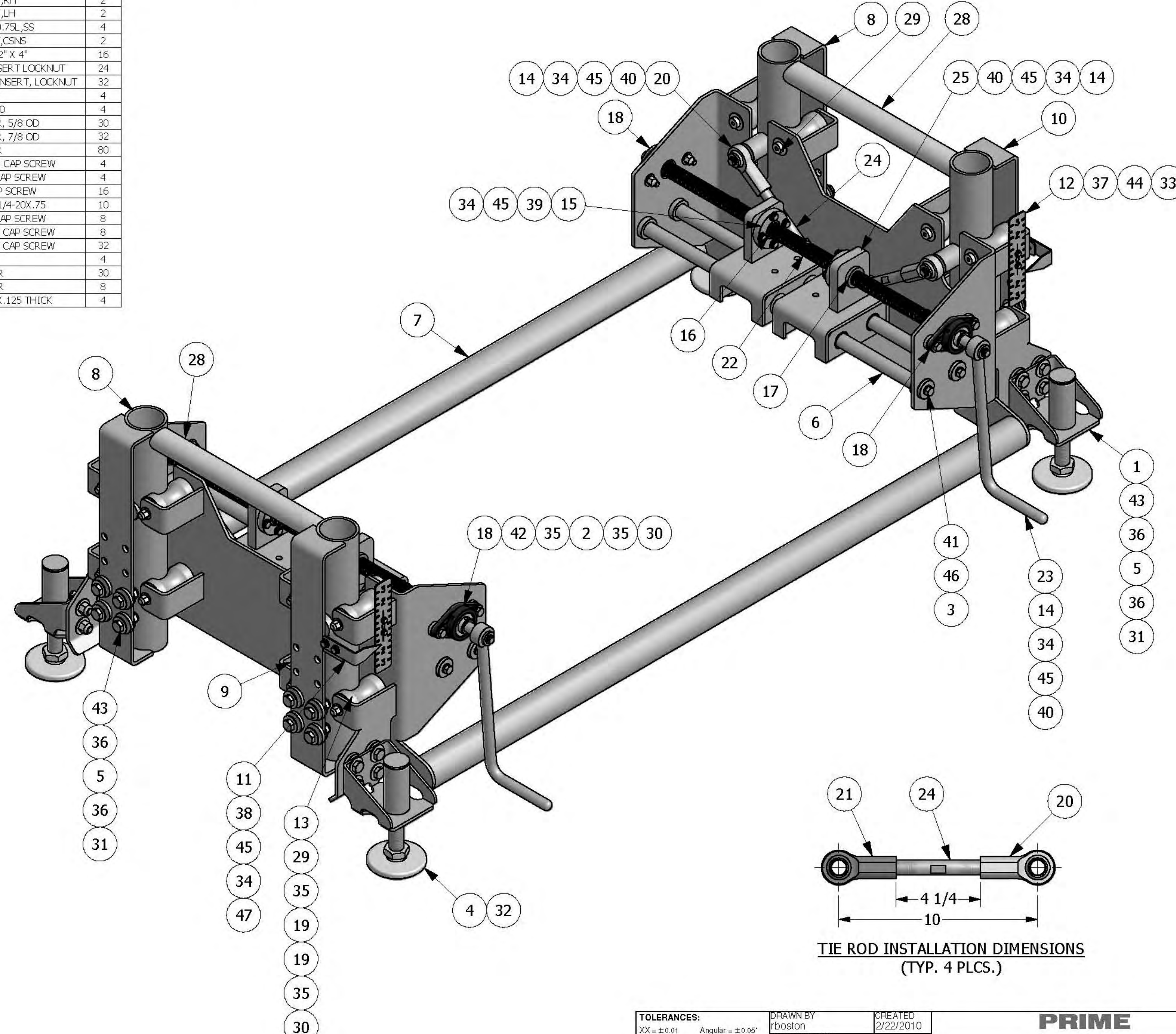
	MATERIAL		
	BLANK FILE NAME		
	QUANTITY PER MACHINE 1		DWG NO XX-CS00-16004 — PART NO 16004
	CLIENT/APPLICATION		SIZE C SHEET 2 OF 2

Parts List				
Item	Drawing No.	Part No.	Description	Qty.
1	KS07-10988	10988	ADJUSTER WLDMT,HEIGHT	4
2	13516	13516	SPACER,0.25X1ODX0.38ID,SS	8
3	14197	14197	SPACER,1.38OD,0.44ID,0.25 THK	8
4	14224	14224	SHOE,MACHINE,4"Ø,1"-8,6"LONG	4
5	14554	14554	SPACER,0.25X1.50ODX0.56ID,SS	32
6	16573	16573	ROD,SLIDE,SSRD1.0X26.75,3/8-16	4
7	16576	16576	FRAME WLDMT,HEIGHT,ADJ,CSNS	1
8	16581	16581	LEG WLDMT,ADJUSTMENT,HEIGHT	2
9	16582	16582	LEG WLDMT,ADJUSTMENT,HEIGHT,LH	1
10	16583	16583	LEG WLDMT,ADJUSTMENT,HEIGHT,RH	1
11	16595	16595	PLT,FRMD,INDICATOR,HEIGHT	2
12	16596	16596	PLT,SCALE,INDICATOR,HEIGHT	2
13	16597	16597	ROLLER,LEG,ADJUSTMENT,HEIGHT	16
14	16615	16615	SPACER,0.188X0.313ODX1.0OD,SS	10
15	16645	16645	FLG,MTG,NUT,ACME,1-5,SS	4
16	16646	16646	NUT,ACME,1-5,BRONZE,LH	2
17	16647	16647	NUT,ACME,1-5,BRONZE,RH	2
18	16650	16650	BRG,FLG,2 BOLT,0.75"	4
19	16651	16651	BSHG,FLG,BRNZ,0.5ID,1.0OD,1.0L	32
20	16652	16652	END,ROD,FEMALE,3/4-16,0.75,LH	4
21	16653	16653	END,ROD,FEMALE,3/4-16,0.75,RH	4
22	16663	16663	SCREW, ACME, ADJ, HEIGHT	2
23	18236	18236	HANDLE WLDMT,CRANK,CSNS-1	2
24	18241	18241	ROD,CONNECTING,3/4-16,6.75L	4

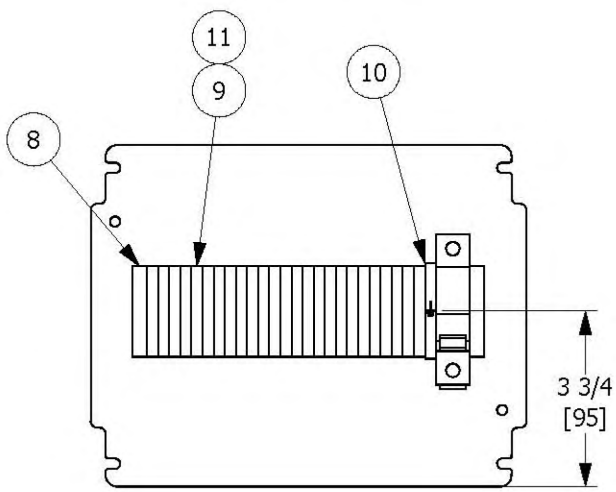
Parts List				
Item	Drawing No.	Part No.	Description	Qty.
25	18252	18252	SLIDE ASSY,ADJ,HGT,RH	2
26	18253	18253	SLIDE ASSY,ADJ,HGT,LH	2
27	18268	18268	LUG,1.25OD,0.77ID,0.75L,SS	4
28	20198	20198	BAR,TIE,ADJ,HEIGHT,CSNS	2
29		897A	SHOULDER BOLT, 1/2" X 4"	16
30		ESNS037C	3/8-16 SS NYLON INSERT LOCKNUT	24
31		ESNS050C	1/2-13, SS, NYLON, INSERT, LOCKNUT	32
32		FHNS100C	1-8,S.S.FIN,HEX,NUT	4
33		FWCS10	WASHER,FLAT,SS,#10	4
34		FWUS025	1/4 SS FLAT WASHER, 5/8 OD	30
35		FWUS037	3/8 SS FLAT WASHER, 7/8 OD	32
36		FWUS050	1/2 SS FLAT WASHER	80
37		HCSS010C0050	#10-24X1/2 S.S. HEX CAP SCREW	4
38		HCSS025C0050	1/4-20X1/2 SS HEX CAP SCREW	4
39		HCSS025C0100	1/4-20X1 SS HEX CAP SCREW	16
40		HCSS025C075	HEX CAP SCREW,SS,1/4-20X.75	10
41		HCSS037C0075	3/8-16X3/4 SS HEX CAP SCREW	8
42		HCSS037C0175	3/8-16X1-3/4 SS HEX CAP SCREW	8
43		HCSS050C0150	1/2-13X1-1/2 SS HEX CAP SCREW	32
44		LWSS01	# 10 LOCK WASHER	4
45		LWSS025	1/4 SS LOCK WASHER	30
46		LWSS037	3/8 SS LOCK WASHER	8
47		SPACER16FW250125	SPACER, .5OD,.25IDX.125 THICK	4

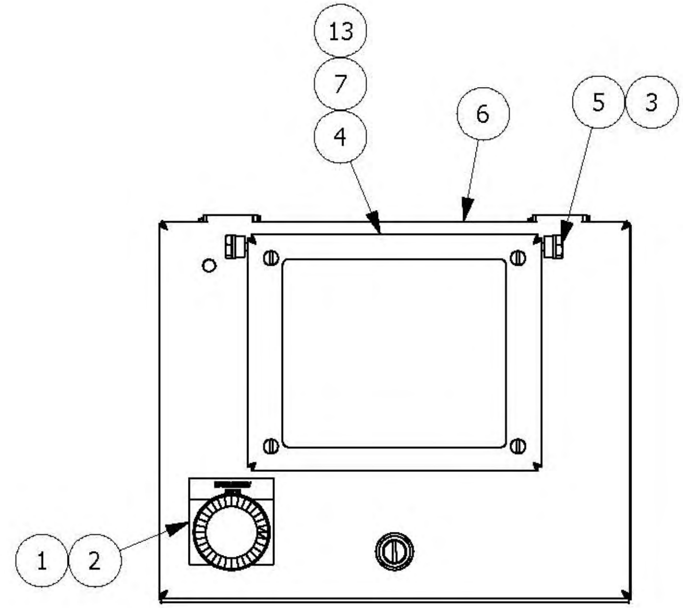




TIE ROD INSTALLATION DIMENSIONS
(TYP. 4 PLCS.)

TOLERANCES: XX = ±0.01 Angular = ±0.05° XXX = ±0.005 Fractional = ±1/16	DRAWN BY rboston	CREATED 2/22/2010	PRIME Equipment Group, Inc. Columbus, Ohio USA
	MACHINE NAME CHK SHLDR/NECK SKINNER		
	MODEL # CSNS-1		TITLE FRAME ASSY,ADJ,HEIGHT,CSNS
	MATERIAL		
	BLANK FILE NAME		
	QUANTITY PER MACHINE 1		DWG NO 16616 / PART NO 16616
	CLIENT/APPLICATION		SIZE D / SHEET 1 OF 1

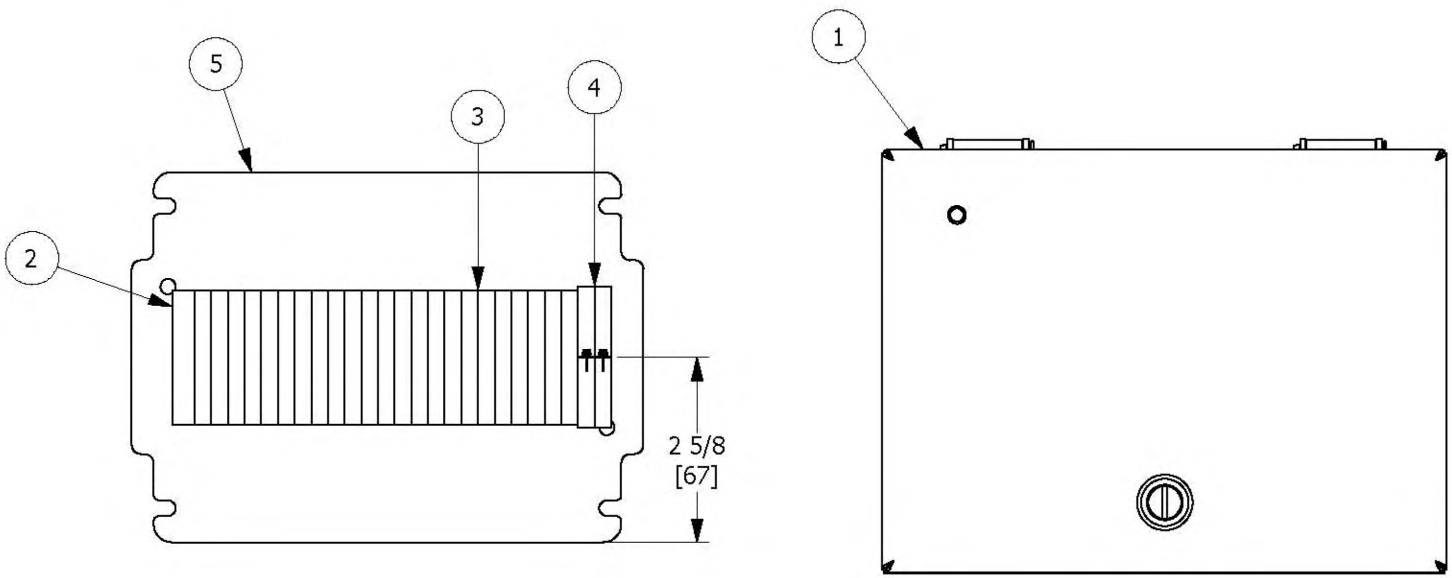




Parts List

Item	Part No.	Description	Qty.
1	800T-FXT6A5	E-STOP, PUSH BUTTON, 30MM	1
2	LEGEND-800T-X648	E-STOP LEGEND PLATE	1
3	LWSS025	1/4 SS LOCK WASHER	2
4	13068	ASSY,COVER,PROTECT,PROFACE	1
5	13095	BOLT,HINGE,1/4-20X1/2,MACHINED	2
6	13935	ENCLOSURE,10X8,PROFACE CUT,	1
7	13098	PLC,HMI,COMPLETE,	1
8	11820	END CLAMP,TERMINAL,4MM	2
9	11817	TERMINAL,4MM,GRAY	25
10	11819	TERMINAL,4MM,GROUNDING	1
11	11818	END PLATE,TERMINAL,4MM	1
12	11828	CIRCUIT BREAKER,1 POLE,2AMP	1
13	13931	MODULE,OUTPUT,RELAY,8PT,PROFAC	1

TOLERANCES: XX = ±0.01 Angular = ±0.05° XXX = ±0.005 Fractional = ±1/16	DRAWN BY TChristian	CREATED 1/8/2009	PRIME Equipment Group, Inc. Columbus, Ohio USA
	MACHINE NAME		
	MODEL #		TITLE
	MATERIAL		CONTROL,HMI,ASSY,COMPLETE,SM
	BLANK FILE NAME		
	QUANTITY PER MACHINE 0		DWG NO 13934 / PART NO 13934
	CLIENT/APPLICATION		SIZE C / SHEET 1 OF 1



Parts List				
Item	Drawing No.	Part No.	Description	Qty.
1	11852	11852	ENCLOSURE,8HX6WX4DP,SS	1
2	11820	11820	END CLAMP,TERMINAL,4MM	1
3	11817	11817	TERMINAL,4MM,GRAY	23
4	11819	11819	TERMINAL,4MM,GROUNDING	2
5	A8P6	A8P6	PANEL,BACK,F/A8064CMSS/11852	1

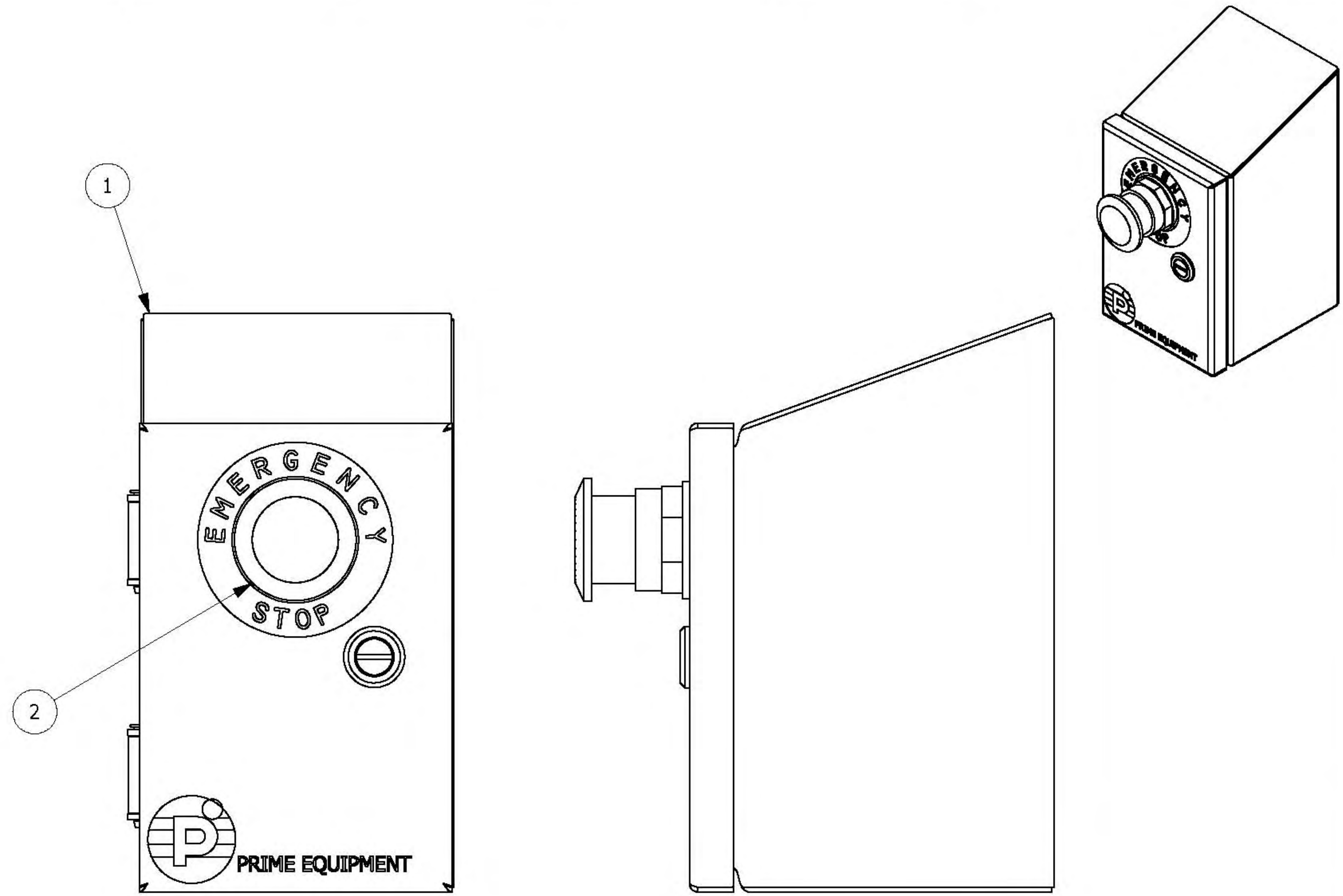
TOLERANCES:
XX = ± 0.01 Angular = ± 0.05°
XXX = ± 0.005 Fractional = ± 1/16

DRAWN BY: TChristian
CREATED: 1/26/2009
MACHINE NAME
MODEL #
MATERIAL
BLANK FILE NAME
QUANTITY PER MACHINE: 0
CLIENT/APPLICATION

PRIME
Equipment Group, Inc.
Columbus, Ohio USA

TITLE: ELECTRICAL,ASSY,J-BOX,CSNS-1,

DWG NO: 14231
PART NO: 14231
SIZE: C
SHEET 1 OF 1



Parts List			
Item	Part No.	Description	Qty.
1	12075	ENCLOSURE,ETCHED,E-STOP,4X6	1
2	800T-FXT6A5	E-STOP, PUSH BUTTON, 30MM	1
3	A6P4	PANEL, BACK,6X4 ENCLOSURE,	1
4	11820	END CLAMP,TERMINAL,4MM	1
5	11817	TERMINAL,4MM,GRAY	4
6	11819	TERMINAL,4MM,GROUNDING	1

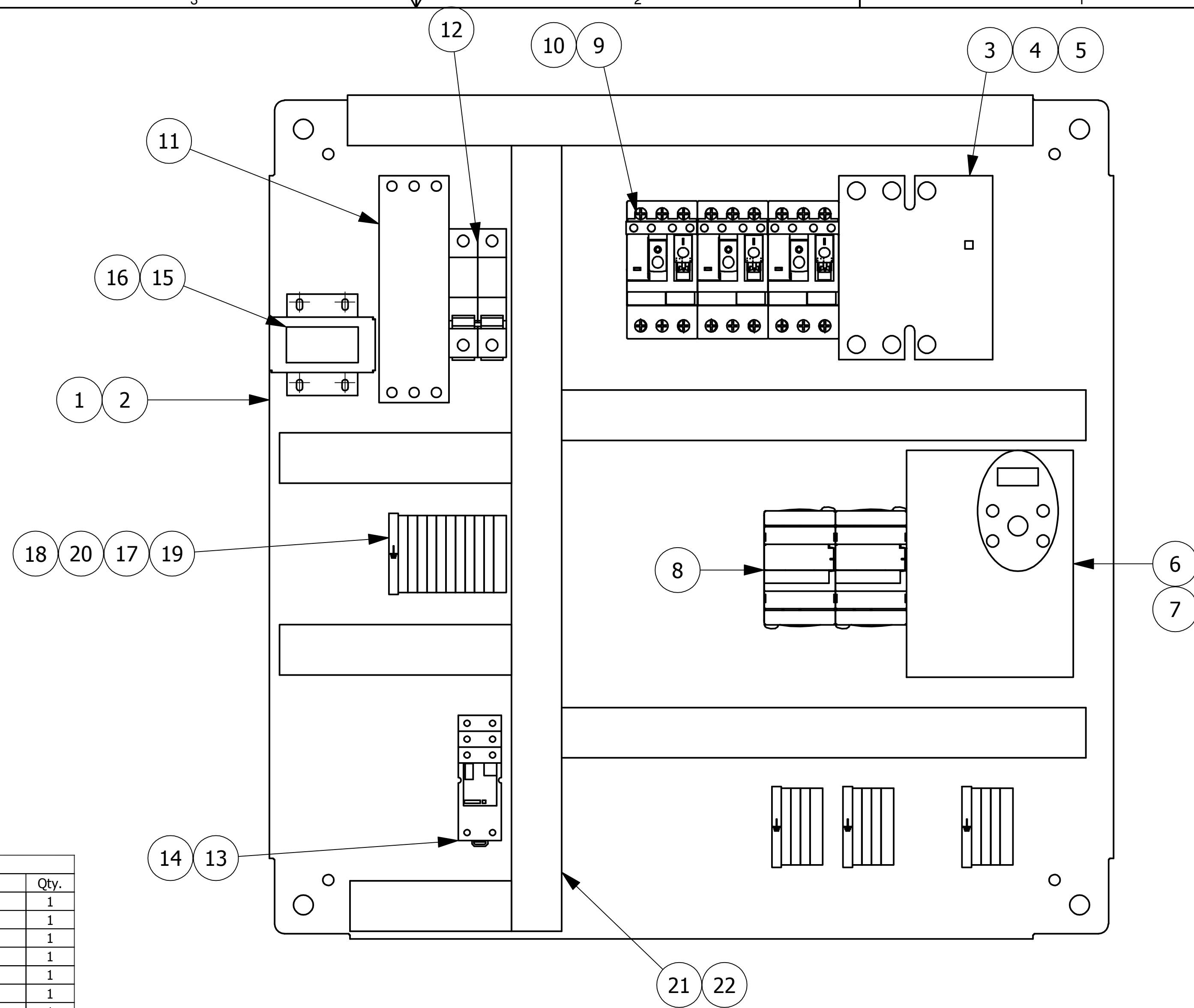
TOLERANCES:
XX = ± 0.01 Angular = ± 0.05°
XXX = ± 0.005 Fractional = ± 1/16

DRAWN BY: TChristian
CREATED: 2/10/2009
MACHINE NAME
MODEL #
MATERIAL
BLANK FILE NAME
QUANTITY PER MACHINE: 0
CLIENT/APPLICATION

PRIME
Equipment Group, Inc.
Columbus, Ohio USA

TITLE: ELECTRICAL,E-STOP,ASSY,

DWG NO: 14233
PART NO: 14233
SIZE: C SHEET 1 OF 2



Parts List

Item	Part No.	Description	Qty.
1	14211	ENCLOSURE,24X24X08,SS	1
2	14212	PANEL,BACK,24X24,ENCLOSURE	1
3	13732	SWITCH,D-CONNECT,FUSIBLE,30A,*	1
4	13733	ROD,D-CONNECT/13732,	1
5	11815	HANDLE,DISCONNECT SWITCH,30A	1
6	13734	VFD,1HP,3Ø,480VAC,TELE	1
7	13736	PLATE,MTG,VFD,DIN RAIL,F/13734	1
8	11822	CONTACTOR,5HP,24VDC	2
9	11823	CIRCUIT BREAKER,1.6-2.5A	3
10	11824	AUX. CONTACT,F/CIRC BREAKER*	3
11	11825	POWERSUPPLY,170-550VAC,24VDC*	1
12	11829	CIRCUIT BREAKER,2POLE,2AMP	1
13	12976	BASE,RELAY,DPDT,F/12975	1
14	12975	RELAY,DPDT,24VDC,	1
15	11816	BRKT,MTG,F/HOUR METER	1
16	11812	METER,HOUR,24VDC	1
17	11817	TERMINAL,4MM,GRAY	19
18	11818	END PLATE,TERMINAL,4MM	4
19	11819	TERMINAL,4MM,GROUNDING	4
20	11820	END CLAMP,TERMINAL,4MM	4
21	11826	WIRE DUCT,NARROW SLOT,1X2,WHT*	7
22	11827	PANDUIT COVER,F/11826	7

TOLERANCES:
XX = ± 0.01 Angular = ± 0.05'
XXX = ± 0.005 Fractional = ± 1/16

DRAWN BY	TChristian
CREATED	1/22/2009
MACHINE NAME	
MODEL #	
MATERIAL	
BLANK FILE NAME	
QUANTITY PER MACHINE	0
CLIENT/APPLICATION	

PRIME
Equipment Group, Inc.
Columbus, Ohio USA

TITLE

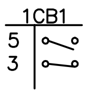
DWG NO

PART NO 14210

SIZE C SHEET 3 OF 3

OPTIONS — OPTIONS ARE POSSIBLE ADDITIONS THAT NEED TO BE ENABLED/DISABLED THROUGH THE TOUCHSCREEN. THEY ARE ACCESSED IN THE MAINTENANCE MENU(S) (8590).

5M1 — COMPONENT DESIGNATION
"5" REFERENCES PAGE NUMBER WHERE COIL IS LOCATED
"M" REFERENCES IT IS A MOTOR CONTACTOR COIL
"1" REFERENCES THE NUMBER OF THE CONTACTOR
"5M1" IS THE MOTOR 1 CONTACTOR, AND THE COIL IS ON PAGE 5



— AUXILIARY CONTACT REFERENCES
"1CB1" REFERENCES THE COMPONENT CONTAINING CONTACTS USED IN OTHER PLACES
"LEFT COLUMN (5,3)" REFERENCES THE PAGES THE CONTACTS CAN BE FOUND ON
"RIGHT COLUMN" REFERENCES THE TYPE OF CONTACT ON THE CORRESPONDING PAGE
"1CB1" CONTAINS ONE NORMALLY OPEN CONTACT ON PAGE 5, AND N.C. ON PAGE 3

X01 - OPERATOR CONTROL PANEL
X02 - JUNCTION BOX

REV.	DRAWN	DATE	DESCRIPTION

Material:

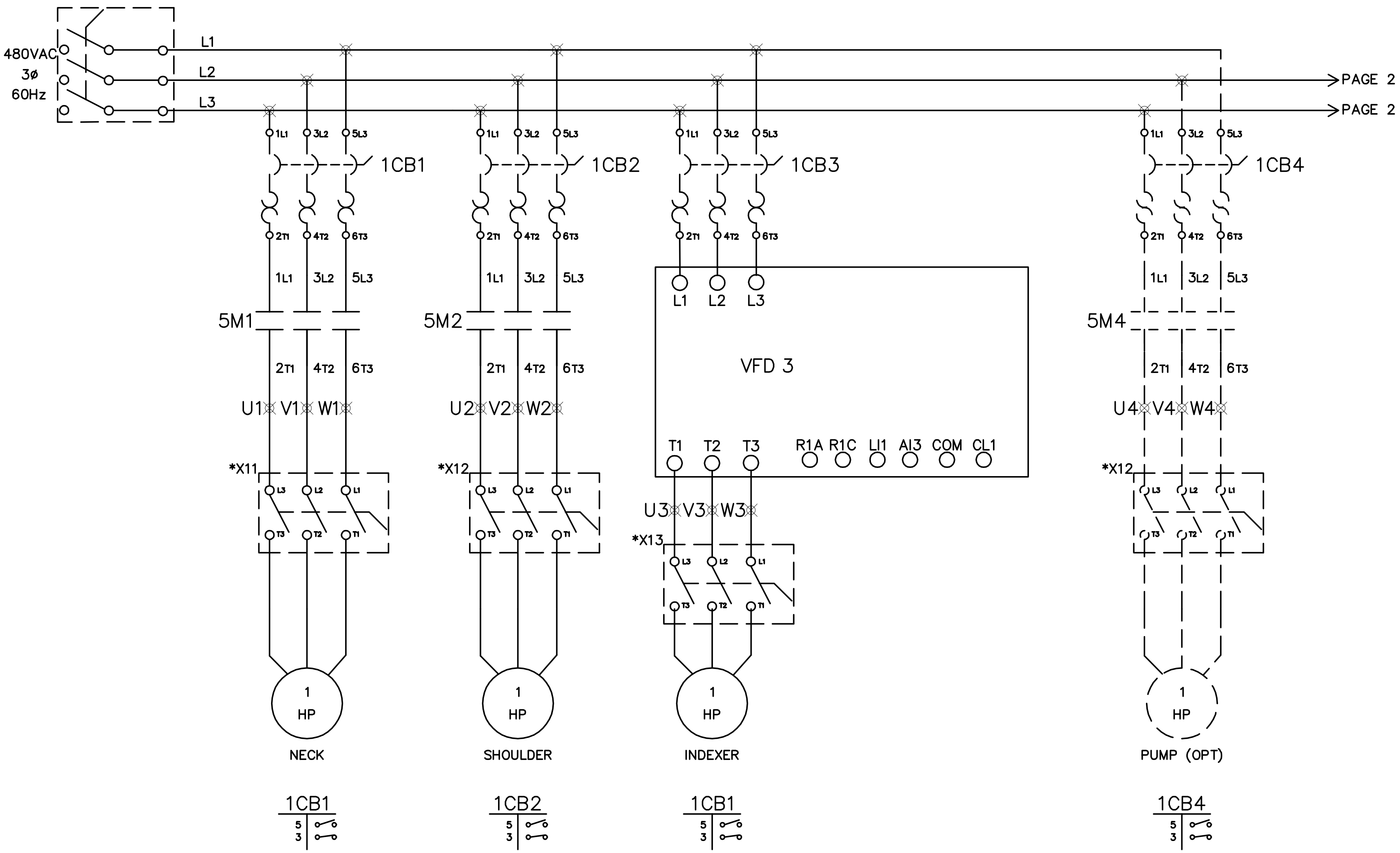
Quantity: LEGEND

Tolerances (Unless noted):
X = .5 XXX = .010 X/X = 1/16
XX = .03 XXXX = .0005 Angular = 1°

PRIME EQUIPMENT GROUP
Columbus, Ohio 43205 USA

WIRING SCHEMATIC
CHICKEN NECK SKINNER

Drawing By:	Scale:	Application:
TC	NONE	CSNS–1
Approved By:	**Date:**	**Drawing Number:**
	02/2009	14210



*NOTE: LOCAL MOTOR DISCONNECT SWITCHES ARE OPTIONAL AND CUSTOMER SUPPLIED.

REV.	DRAWN	DATE	DESCRIPTION

Material:

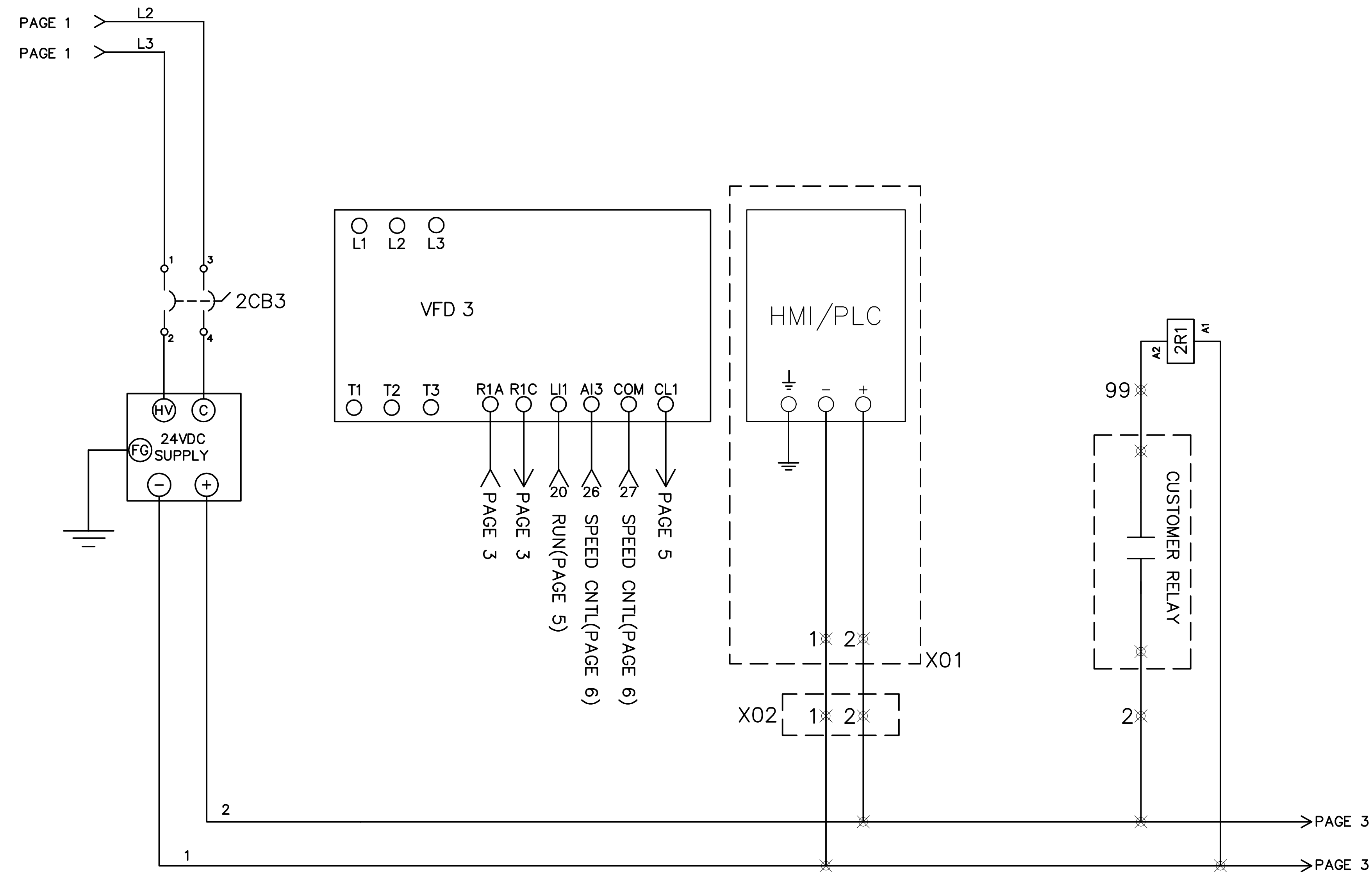
Quantity: PAGE 1 OF 7

Tolerances (Unless noted): X = .5 XX = .03 XXX = .010 XXXX = .0005 X/X = 1/16 Angular = 1°

PRIME EQUIPMENT GROUP
Columbus, Ohio 43205 USA

WIRING SCHEMATIC
CHICKEN NECK SKINNER

Drawing By:	Scale:	Application:
TC	NONE	CSNS-1
Approved By:	Date: 02/2009	Drawing Number: 14210



REV.	DRAWN	DATE	DESCRIPTION

Material:	PRIME EQUIPMENT GROUP Columbus, Ohio 43205 USA	
Quantity: PAGE 2 OF 7		
Tolerances (Unless noted): X = .5 XXX = .010 X/X = 1/16 XX = .03 XXXX = .0005 Angular = 1°	WIRING SCHEMATIC CHICKEN NECK SKINNER	
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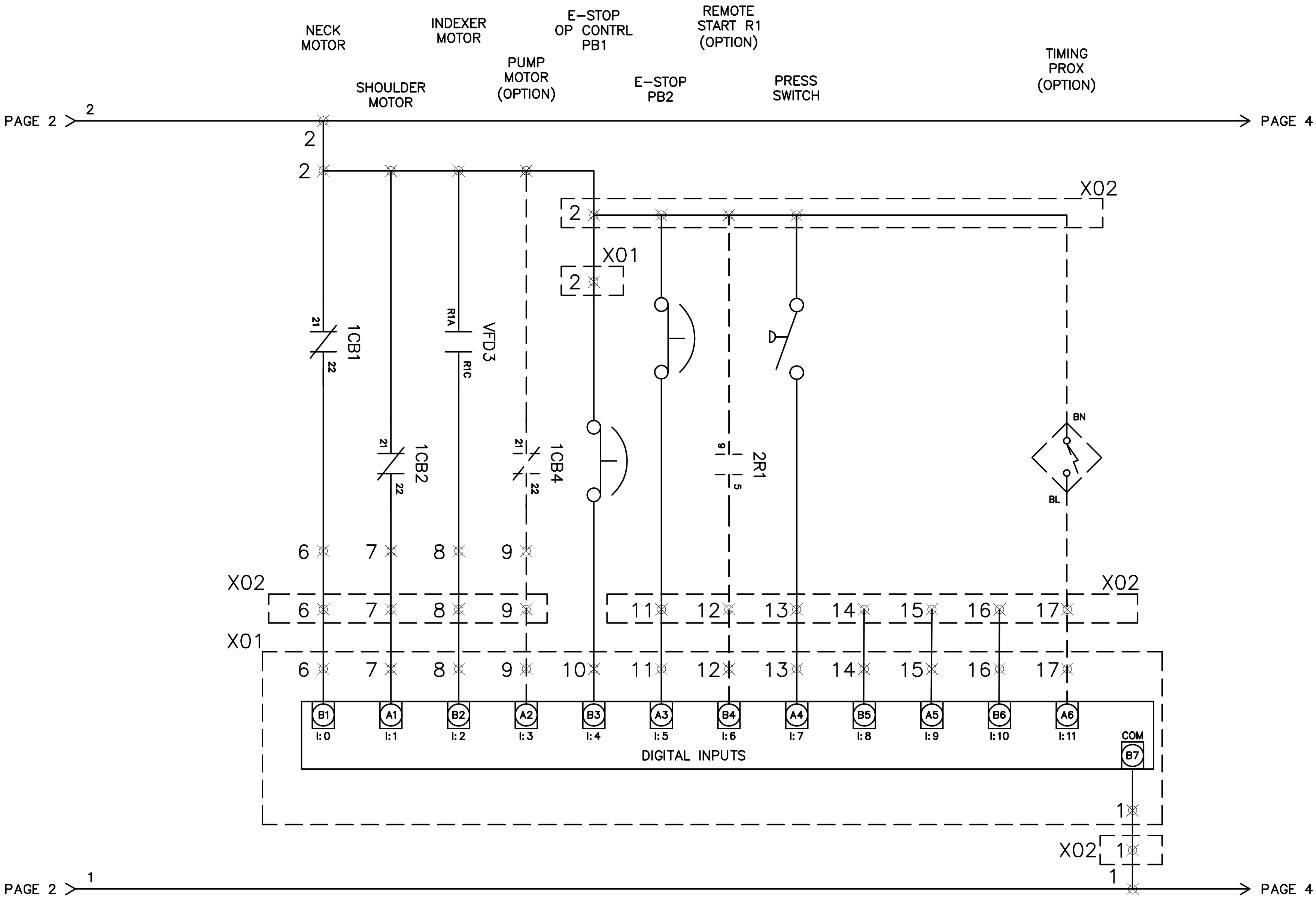
	Approved By:	Date: 02/2009	Drawing Number: 14210



REV.	DRAWN	DATE	DESCRIPTION

Material:

Quantity: PAGE 3 OF 7

Tolerances (Unless noted):
X = .5 XXX = .010 X/X = 1/16
XX = .03 XXXX = .0005 Angular = 1°

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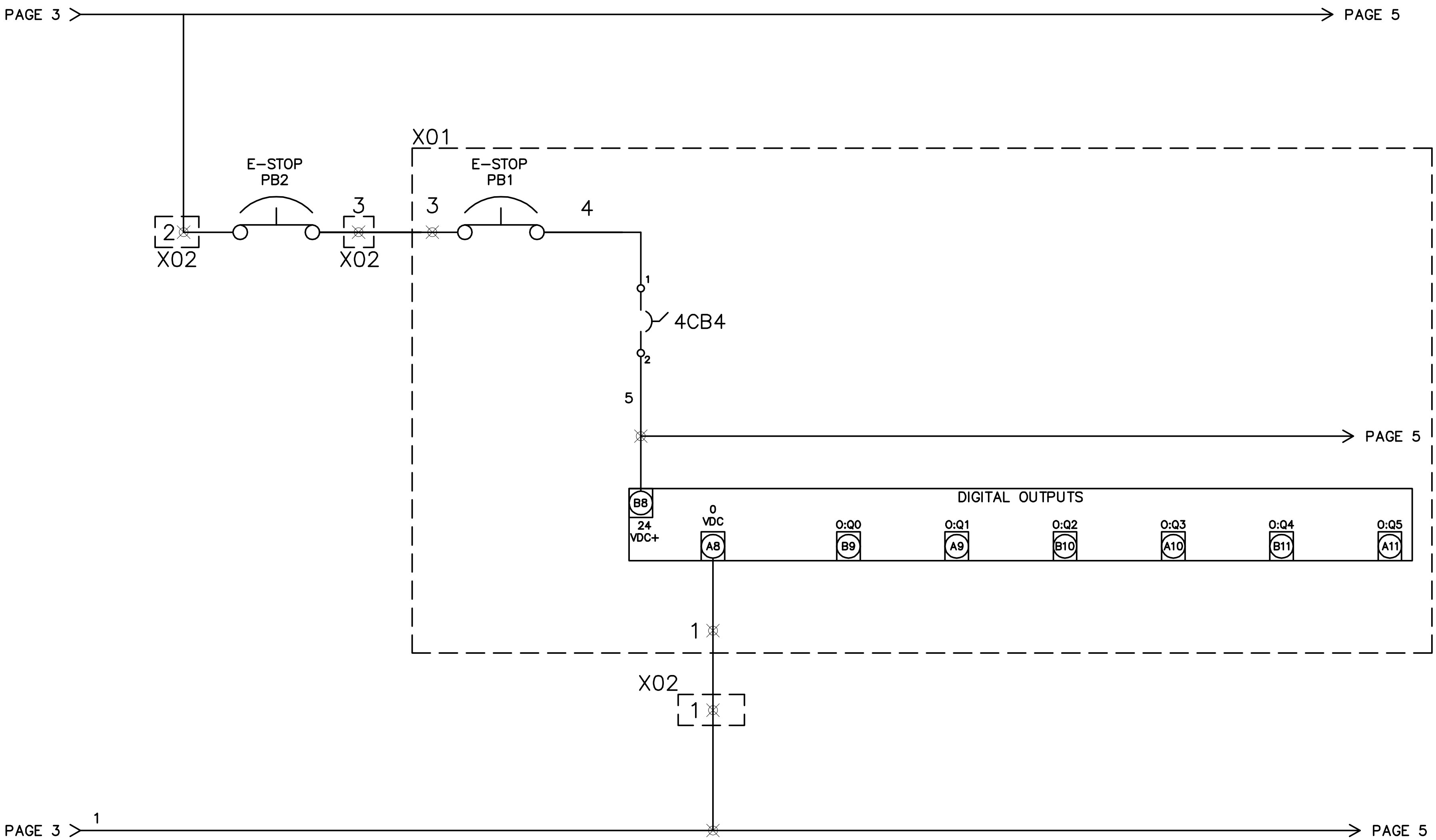
WIRING SCHEMATIC
CHICKEN NECK SKINNER

Drawing By:	Scale:	Application:
TC	NONE	CSNS-1
Approved By:	Date: 02/2009	Drawing Number: SKWSC_____



REV.	DRAWN	DATE	DESCRIPTION

Material:

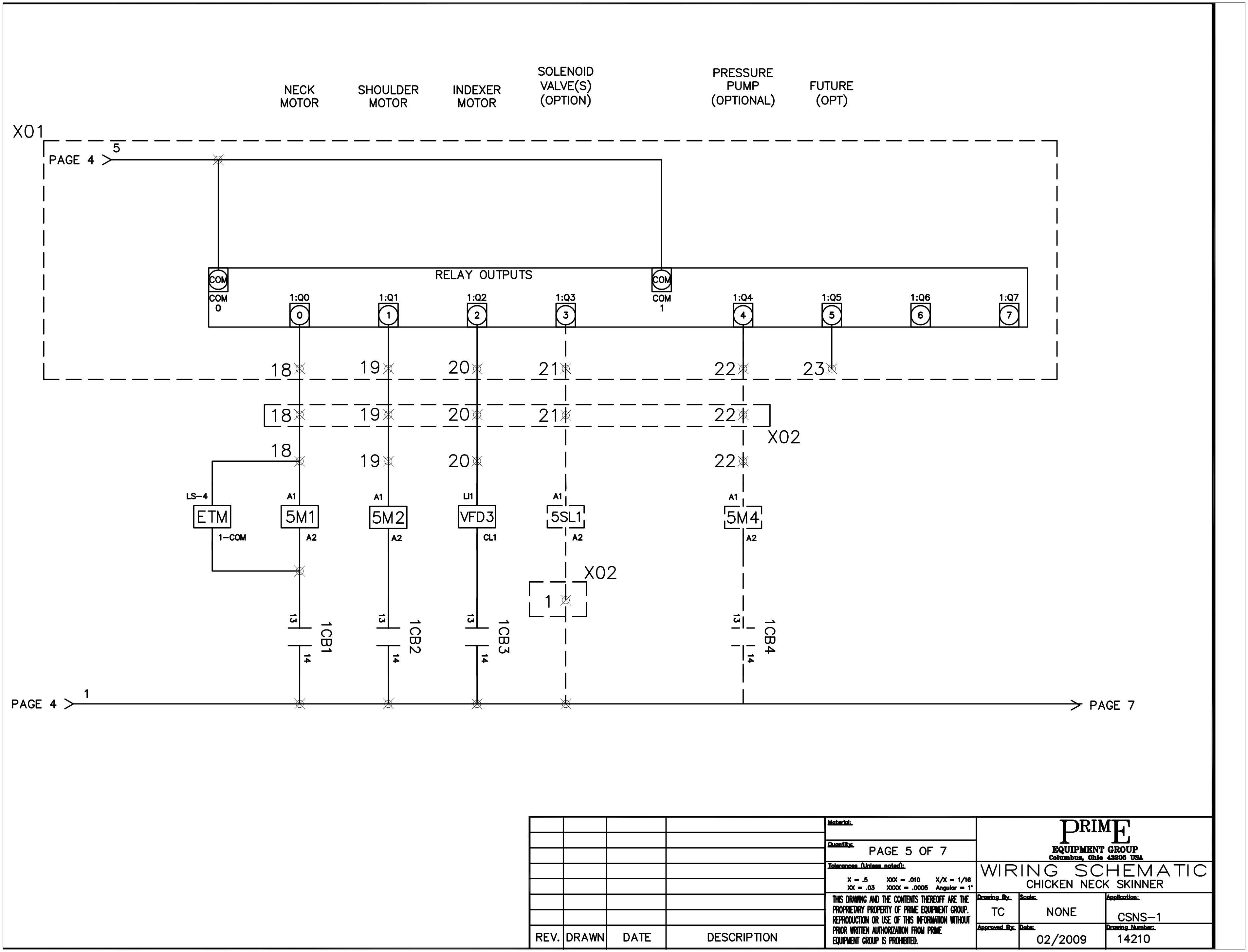
Quantity: PAGE 4 OF 7

Tolerances (Unless noted):
X = .5 XXX = .010 X/X = 1/16
XX = .03 XXXX = .0005 Angular = 1°

PRIME
EQUIPMENT GROUP
Columbus, Ohio 43205 USA

WIRING SCHEMATIC
CHICKEN NECK SKINNER

Drawing By:	Scale:	Application:
TC	NONE	CSNS-1
Approved By:	Date: 02/2009	Drawing Number: 14210


NECK MOTOR
SHOULDER MOTOR
INDEXER MOTOR
SOLENOID VALVE(S) (OPTION)
PRESSURE PUMP (OPTIONAL)
FUTURE (OPT)
X01
PAGE 4 5
RELAY OUTPUTS
COM
COM 0
1:Q0 1:Q1 1:Q2 1:Q3
COM 1
1:Q4 1:Q5 1:Q6 1:Q7
18 19 20 21 22 23
X02
LS-4
ETM
1-COM
A1 5M1 A2
A1 5M2 A2
LI1 VFD3 CL1
A1 5SL1 A2
A1 5M4 A2
1
13 1CB1 14
13 1CB2 14
13 1CB3 14
13 1CB4 14
PAGE 4 1
PAGE 7
REV. DRAWN DATE DESCRIPTION
Material:
Quantity: PAGE 5 OF 7
Tolerances (Unless noted): X = .5 XX = .03 XXX = .010 XXXX = .0005 X/X = 1/16 Angular = 1°
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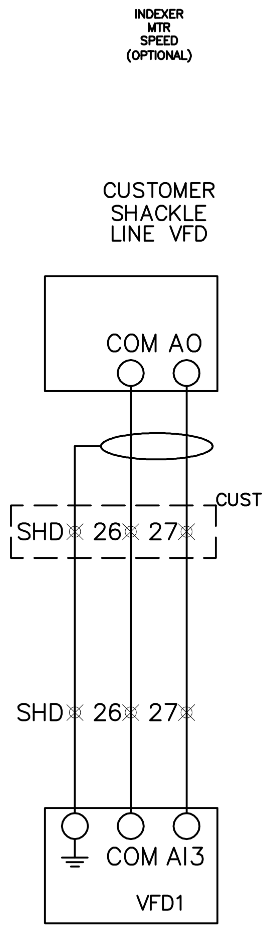
PRIME EQUIPMENT GROUP
Columbus, Ohio 43205 USA
WIRING SCHEMATIC
CHICKEN NECK SKINNER
Drawing By: TC
Scale: NONE
Application: CSNS-1
Approved By:
Date: 02/2009
Drawing Number: 14210



REV.	DRAWN	DATE	DESCRIPTION

Material: SHT 6 OF 7

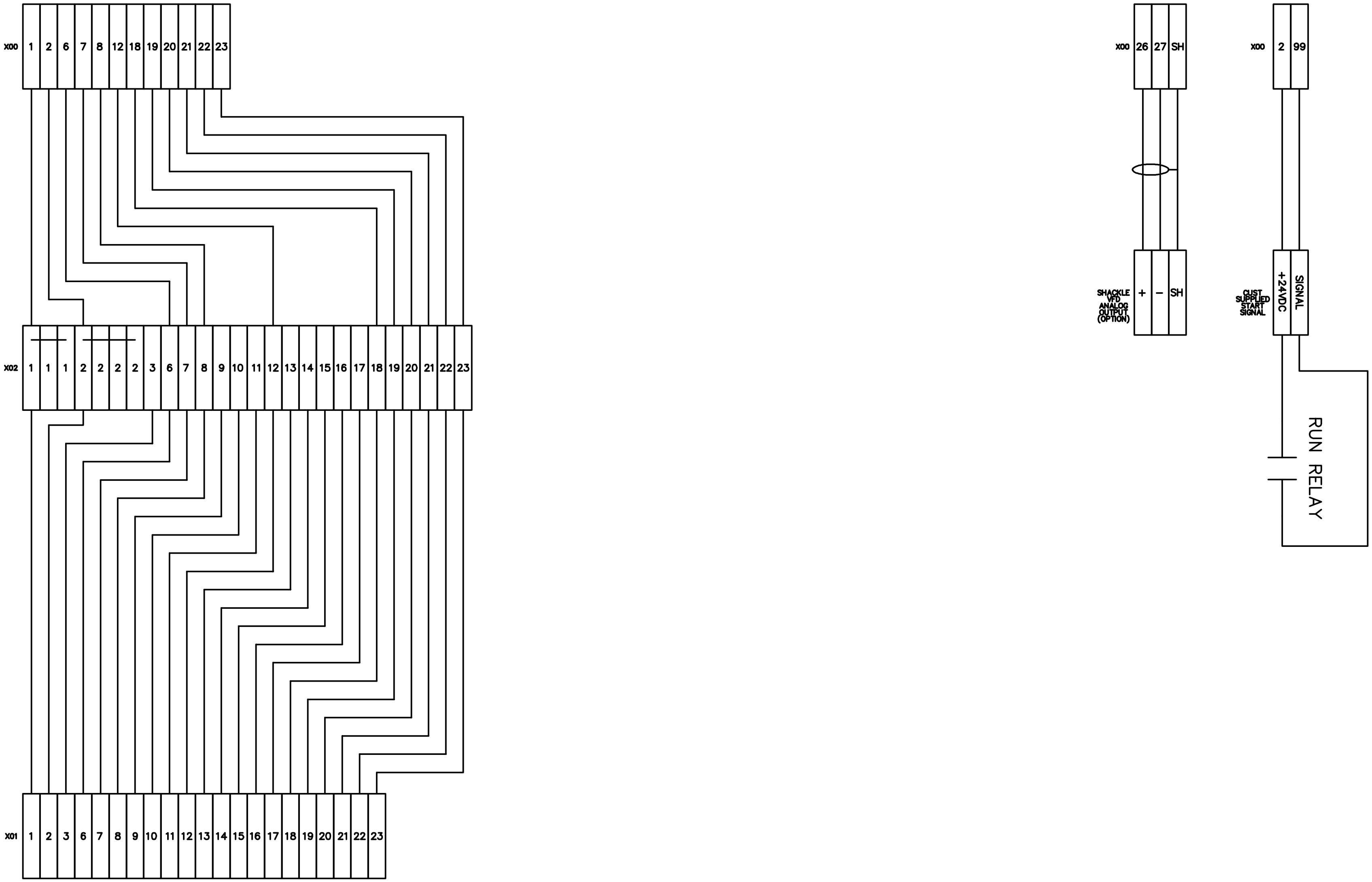
Quantity: –

Tolerances (Unless noted):
XX = ±0.01 X = .5 Angular = ±0.5° XXX = .010 X/X = 1/16
XXX = ±0.005 XX = ±1/16 XXXX = .0005 Angular = 1°

PRIME
EQUIPMENT GROUP
Columbus, Ohio 43205 USA

WIRING DIAGRAM
CHINCKEN NECK SKINNER

Drawing By:	Scale:	Application:
TC	NONE	CSNS–1
Approved By:	**Date:**	**Drawing Number:**
–	02/2009	14210



REV.	DRAWN	DATE	DESCRIPTION

Material: SHT 7 OF 7

Quantity: –

Tolerances (Unless noted):
XX = ±0.01 X = .5 Angular = ±0.5° XXX = .010 X/X = 1/16
XXX = XX ±0.005 XXXX = .0005 ±1/16 Angular = 1°

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PRIME EQUIPMENT GROUP
Columbus, Ohio 43205 USA

WIRING DIAGRAM
CHINCKEN NECK SKINNER

Drawing By:	Scale:	Application:
TC	NONE	CSNS–1
Approved By:	**Date:**	**Drawing Number:**
–	02/2009	14210

2.1 Offline Mode

Off-line mode does system setting and self diagnostic etc. Before operating, it prepares here.

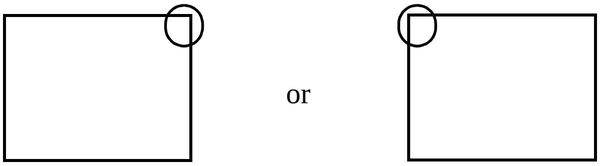
IMPORTANT
- OFFLINE mode is unavailable in a completely new GP until the necessary GP system data has been transferred from GP screen editor software.
  To do this, be sure the GP's power cord is plugged in and when transfer screen data from GP screen editor software to the GP, GP's system data will be automatically sent.
  For more information about data transfer, refer to GP-Pro EX Reference Manual.

2.1.1 Entering OFFLINE Mode

There is offline mode for various setup required at use in GP. There are two ways to enter OFFLINE mode. First, is immediately after plugging in the GP's power cord, and second, by using the Forced Reset feature.

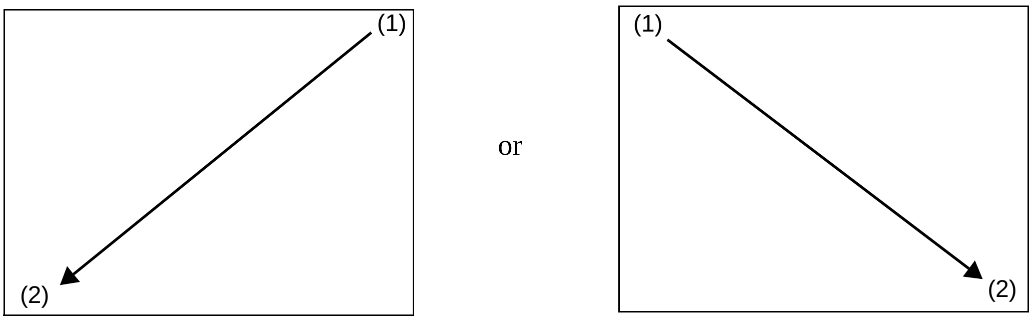
■ After Plugging in the Power Cord

Touch the upper right corner or upper left corner (within 40 pixels of the edges) of the panel for at least 3 seconds soon after the startup screen is displayed.

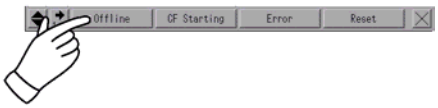


■ When Operating

1 Touch either the upper right and lower left corners, or the upper left and lower right corners (within 40 pixels of the edges) of the panel in this order within 0.5 seconds.



2 System menu is displayed and touch "Offline" button.
For details of the buttons of the system menu, see the following section.
☞ "2.15.12 System Menu" (page 2-171)



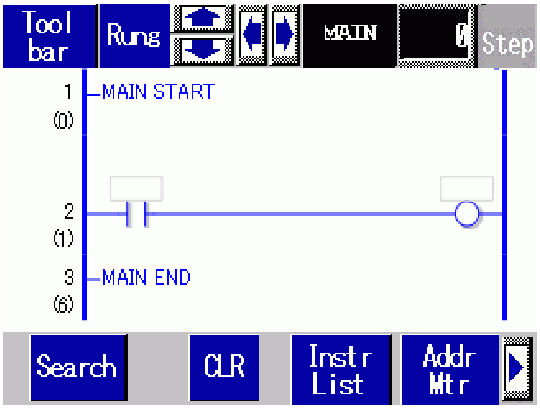
Continued

29.12.2 Logic Monitor Functions

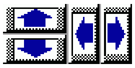
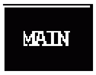
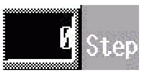
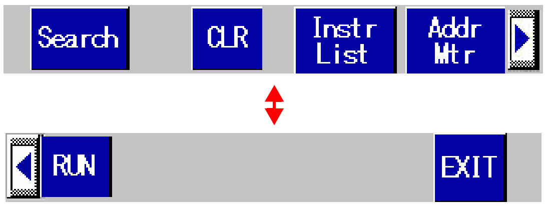
The following explains the logic monitor features.

■ Logic monitor

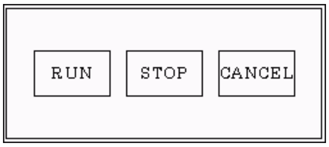
Monitors the entire logic. The logic monitor allows you to check the operational status and instruction layouts.



The logic monitor has the following features.

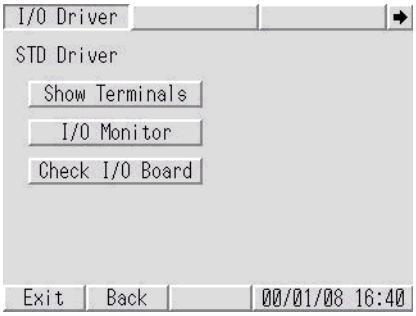
Feature		Details
Scroll 		Scroll the logic using [Rung] or [Column]. Rung: Scroll the logic using rungs. Column: Scroll the instructions one by one without the logic. For landscape, you can use only the [Column] scroll.
Zoom		Touch the displayed instruction to zoom the monitor display. ☞ " ■ Zoom Monitor" (page 29-94)
Logic Name Display 		Display the logic names being monitored. The names to be displayed are [INIT], [MAIN], [ERRH], and [SUB-01]-[SUB-32].
Step 		Display the top step number being monitored. When any change is made, the operation jumps to the rung with the specified step number.
Toolbar 		Switch the toolbar show/hide at the bottom of the screen. Page 1 Page 2  Click ▶ or ◀ to switch Page 1 with Page 2.
	Exit 	End the monitor.

Continued

Feature			Details
Toolbar	RUN/STOP	RUN	Switch the logic between RUN and STOP. Click to display the screen below. Use the buttons to run and stop the logic. 
	Address Monitor	Addr Mtr	Switch to the address monitor. ☞ " ■ Address Monitor" (page 29-95)
	Ladder Instructions	Instr List	Switch to the instruction list. ☞ " ■ Ladder Instructions" (page 29-97)
	Search	Search	Search the variables and instructions specified in the instruction list. ☞ " ■ Search" (page 29-98)

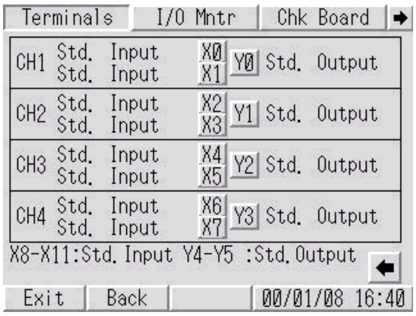
■ [Peripheral Settings] Setting Guide

◆ [I/O Driver] (STD Driver)



Setting item	Description
Show Terminals	The terminal configuration display screen is displayed.
I/O Monitor	The I/O monitor screen is displayed.
Check I/O Board	The I/O board check execution screen is displayed.

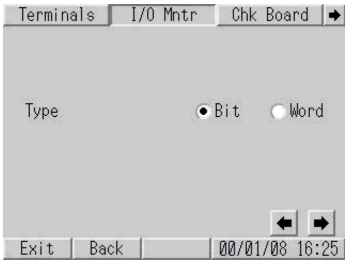
- [Show Terminals]



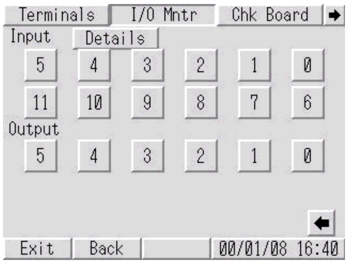
Setting item	Description
Terminals	Configurations of the specified input terminals (X0 to X11) and output terminals (Y0 to Y5) are displayed. **NOTE** • When the LT33** series is used, it displays the configuration of input terminals X0 to X15 and output terminals Y0 to Y15. The bottom of the screen shows [X8-X15: Std. Input] and [Y4-Y15: Std. Output].

- [I/O Monitor]

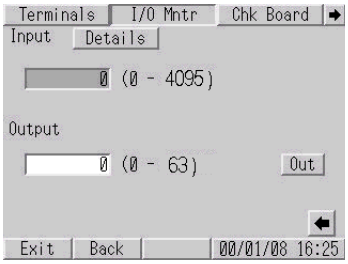
The monitoring results of the standard input and output can be displayed by selecting either bit or integer. The [Word] display cannot be selected when special I/O is used.

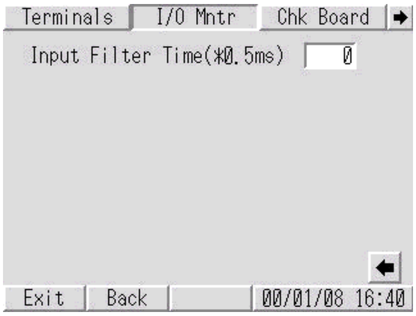


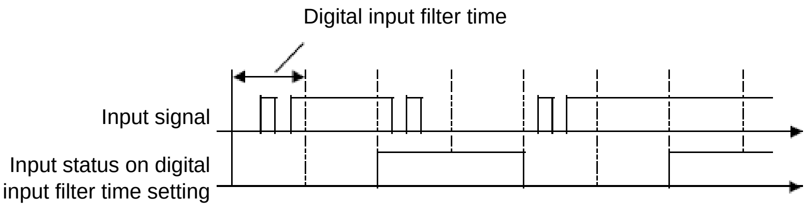
<I/O monitor input/output execution screen (Bit)>



<I/O monitor input/output execution screen (Word)>



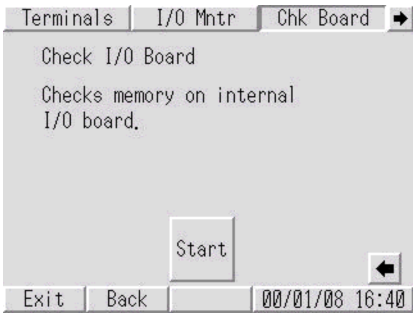
<table>
<tr><th>Setting item</th><th>Description</th></tr>
<tr><td>Details</td><td>The following details setting screen appears.


On the details setting screen, specify an input filter time in the range of 0 to 40. The input filter eliminates noise from an input signal. An input signal that does not reach the time duration specified here will not work. The system samples data at 0.5 ms intervals and stores the data internally, to verify the data by reading the input terminal status before the specified time at an I/O refresh interval of 2 ms. If the input terminal is always in the same state, this state is regarded as the value for the input terminal. If it is not in the same state, the previous value is regarded as the input value.

</td></tr>
</table>

Continued

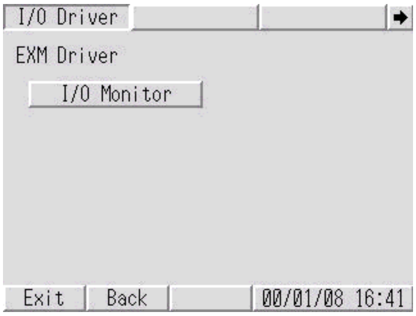
Setting item	Description
Input (0 to 11)	• When [Type] is set to [Bit] The terminal number specified for standard input is displayed. You can check the input value by bit. When the switch is not pressed, the OFF value is displayed. When the switch is pressed, the ON value is displayed. NOTE • With the LT33** series, you can check the status by using inputs 0 through 15. • When [Type] is set to [Word] You can check the input value with integers 0 to 4095 when the LT32** series is used, or with integers 0 to 65535 when the LT33** series is used.
Output (0 to 5)	• When [Type] is set to [Bit] The terminal number specified for standard output is displayed. You can check the output value by bit. NOTE • With the LT33** series, you can check the status by using outputs 0 through 15. • When [Type] is set to [Word] You can check the output value with integers 0 to 63 when the LT32** series is used, or with integers 0 to 65535 when the LT33** series is used.
Error display	If the I/O driver detects an error, the corresponding error code and error message are displayed. For details about error messages, refer to the following description. ☞ "1.7.3 Errors displayed with the LT3000 series ■ I/O driver errors" (page 1-199)

• [Check Board]



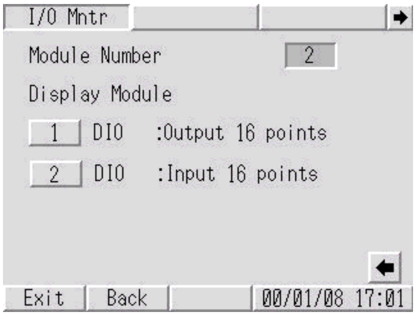
Setting item	Description
Start	The system starts checking if the I/O board is operating normally or not. If the I/O board operation is abnormal, the corresponding error message appears. For details about error messages, refer to the following description. ☞ "1.7.3 Errors displayed with the LT3000 series ■ I/O driver errors" (page 1-199)

◆ [I/O Driver] (EXM Driver)



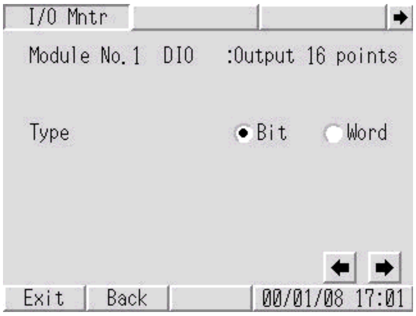
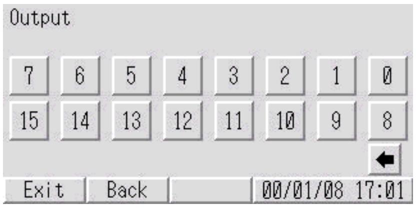
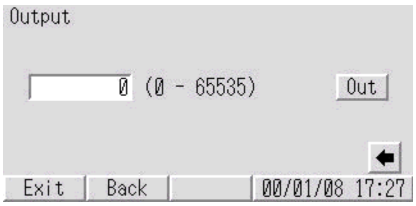
Setting item	Description
I/O Monitor	The I/O monitor connection module information screen is displayed.

- [I/O Monitor] (DIO settings)



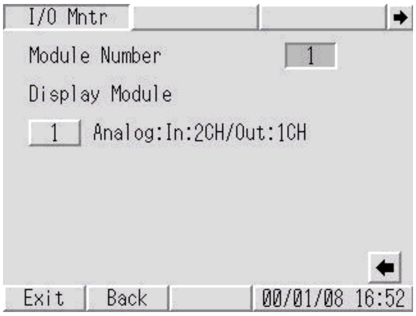
Setting item	Description
Module Number	The number of connected modules is displayed. (0 to 2) **NOTE** • When the LT33** series is used, it displays the number of connected modules with values 0 to 3.
Display Module	The number of connected modules is displayed. Module No. 1 indicates the module directly connected to the LT back panel.

Continued

Setting item	Description
Module No.	The I/O monitor setting screen is displayed. • I/O monitor setting screen Type  • I/O execution screen (Bit) Outputs signals to a destination module on a bit basis. Input status after I/O monitoring appears in a bit representation when selecting the input module.  • I/O execution screen (Word) Sets output values transmitted to a destination module and output values appear. Input values after I/O monitoring appear when selecting the input module. 

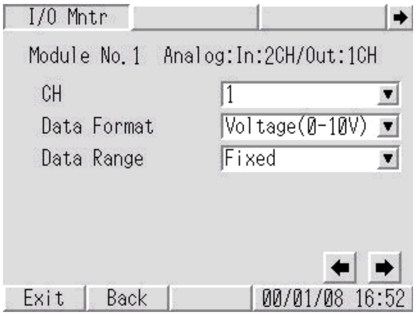
- [I/O Monitor] (Analog settings)

I/O monitor connection module information screen



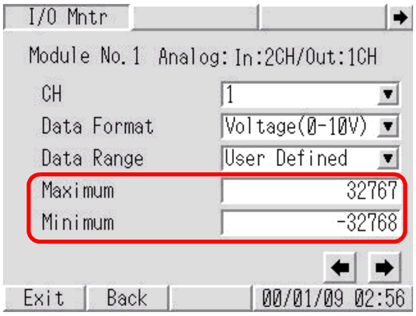
Setting item	Description
Module Number	The number of connected modules is displayed. (0 to 2) **NOTE** • When the LT33** series is used, it displays the number of connected modules with values 0 to 3.
Display Module	The number of connected modules is displayed. Module No. 1 indicates the module directly connected to the LT back panel.
Module No.	The I/O monitor setting screen is displayed.

I/O monitor connection module information screen

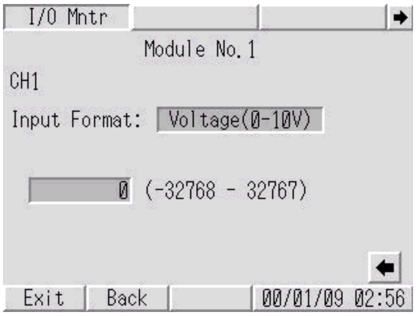


Setting item	Description
CH	Specify the number of channels for execution of the I/O monitor. **NOTE** • If “In: 2CH/Out: 1CH” is selected, the third channel is used for output.
Data Format	Select the data format for execution of the I/O monitor: “Voltage (0 - 10V)”, “Current (4 - 20 mA)”, “Pt100”, “K Thermocouple”, “J Thermocouple” or “T Thermocouple”.
Data Range	Select the data range for execution of the I/O monitor: “Fixed”, “Centigrade”, “Fahrenheit” or “User Defined”.

Continued

Setting item	Description
Maximum / Minimum	Specify the upper and lower limit values for execution of the I/O monitor. These parameters can be specified only when “User Defined” is selected for “Data Range”. 

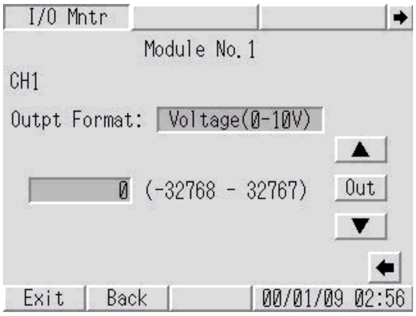
I/O monitor analog input execution screen



Setting item	Description
Input Format	Displays the “Data Format” specified on the I/O monitor setting screen.
Input value	Displays the input value.
Input data range	Displays the “Data Range” specified on the I/O monitor setting screen.

Continued

I/O monitor analog output execution screen



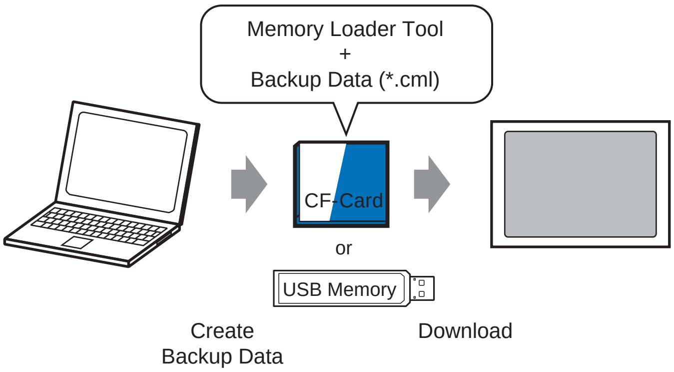
Setting item	Description
Output Format	Displays the “Data Format” specified on the I/O monitor setting screen.
Output value	Displays the output value. Touching the entry field displays numeric keys, allowing you to specify an output value.
Output Range	Displays the “Data Range” specified on the I/O monitor setting screen.
▲▼	Increases or decreases the output value.
Output	Outputs the value specified in “Output value”.

33.7 Transferring Project Files Using a CF Card or USB Storage

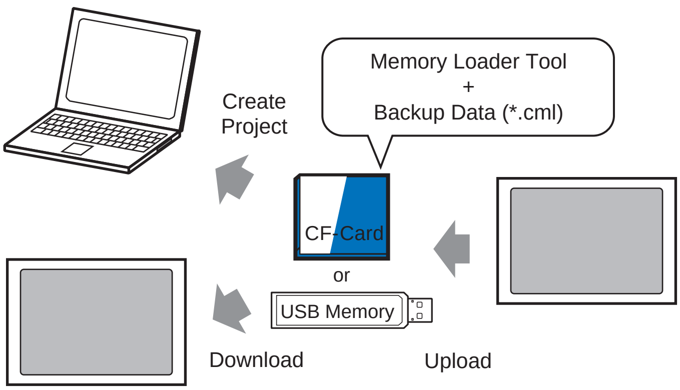
33.7.1 Introduction

Without connecting USB or Ethernet cables, you can transfer project files between the GP and your PC using an external memory such as a CF Card or USB storage device. You can also use the external memory to copy project files from one GP to another.

- Transfer from PC to GP
  On the GP, downloads and displays the project backup data stored on a CF Card or USB storage device.



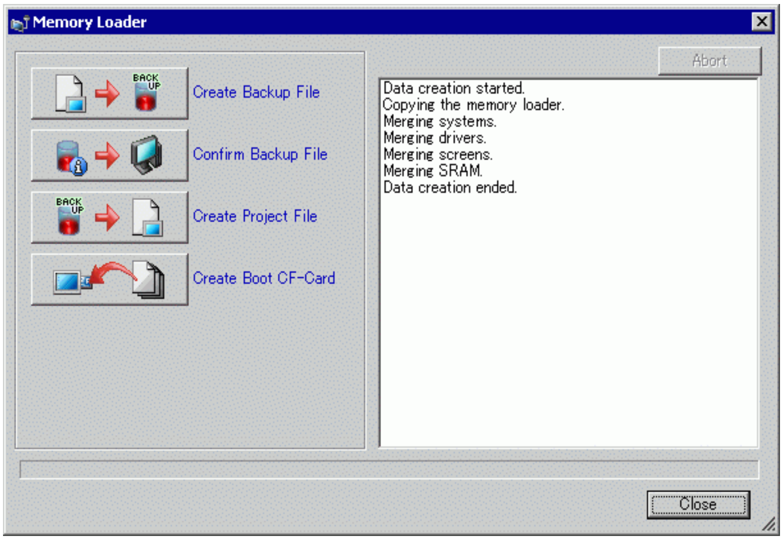
- Transfer from GP to PC
  Uploads the backup data from the GP to a CF Card or USB storage device and loads the data to the PC as a project file. You can also download and display the backup data on a different GP.



IMPORTANT

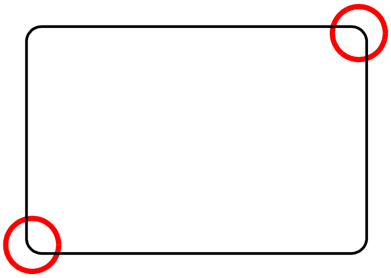
- To backup (upload) the project on the GP or start (download) the project from the backup data on a CF Card, the special Memory Loader Tool must also be copied to the CF Card
  To backup the project on a USB storage device, a different Memory Loader Tool specific for USB storage devices must be copied to the USB storage device.
- Once the data is saved in the CF Card folder, you cannot use the data in USB storage. Similarly, once the data is saved in the USB destination folder, you cannot use the data on a CF Card.

7 When the message "Data Creation Ended" appears, click [Close].



8 Copy or move all data created in the CF Card folder to the CF Card. The data, project backup data, and the CF Card Memory Loader Tool are all transferred to the CF Card.

9 Insert the created CF Card in the GP.
Touch the top right corner -> bottom left corner (or top left corner -> bottom right) on the GP screen in this order within 0.5 seconds and switch to offline mode.

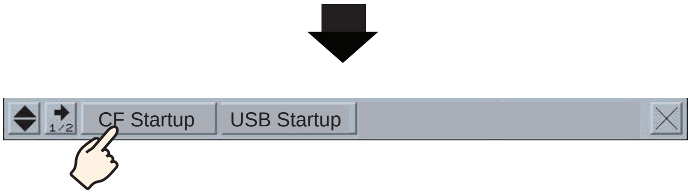


NOTE

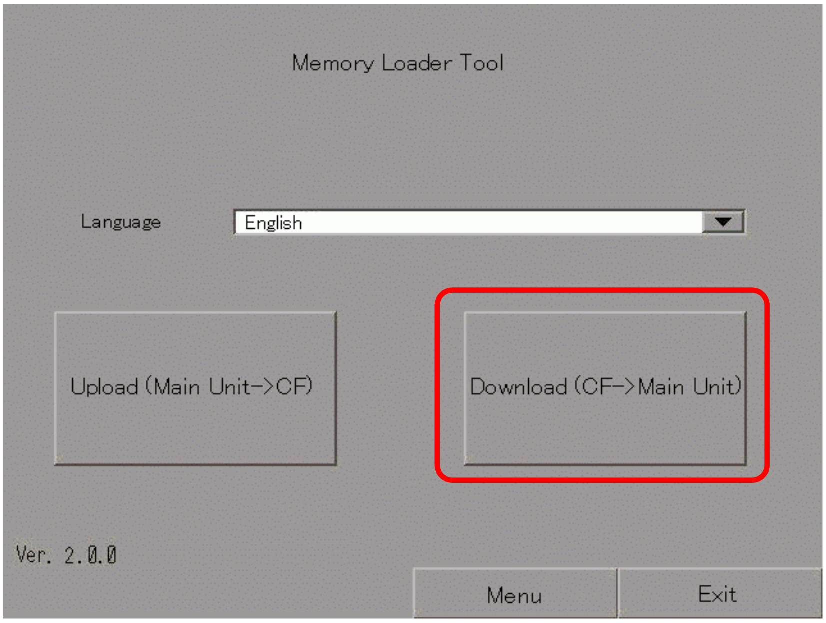
- To start the CF Card, you can either use the system menu or the DIP switches on the back of the GP. Turn the GP power OFF, turn ON DIP switch 1, and then turn the GP power back ON to launch the Memory Loader tool. You cannot use the DIP switches to start the CF Card when using USB Storage.
- To transfer project data to the GP for the first time (for a newly purchased GP), the [Initial Transfer Mode] screen appears when you turn the power ON.

10 Touch [CF/USB]. Touch [CF startup] to reset GP.

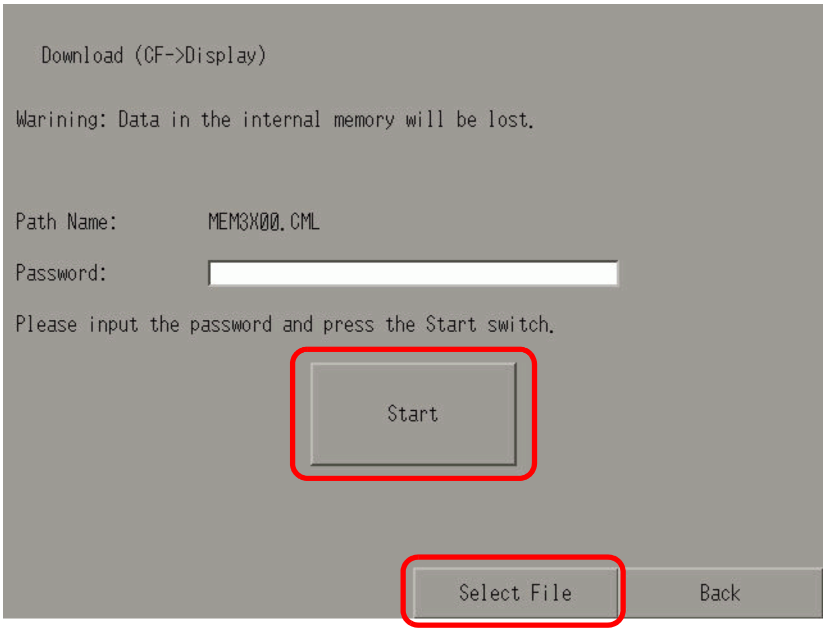
Touch [CF/USB].



11 The Memory Loader Tool launches. The following screen opens. Touch [Download].



12 In [Select File], select a file you want to transfer and touch [Start]. If there is a password for transferring, input the password first, then touch [Start].



IMPORTANT

- Once you start downloading, all project data on the GP is deleted, including the data in the backup SRAM.
- If you performed CF Start with the DIP switch, turn the Number 1 DIP switch back OFF.

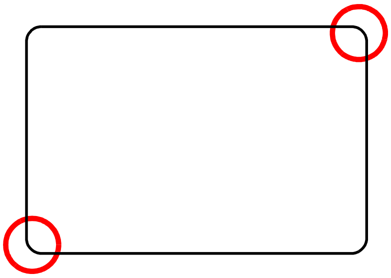
13 Once downloading is complete, touch [Back] and then [Exit] to exit the Memory Loader Tool.

4 Move or copy all the created data to the CF Card/USB storage. The CF Card/USB storage device is now ready.

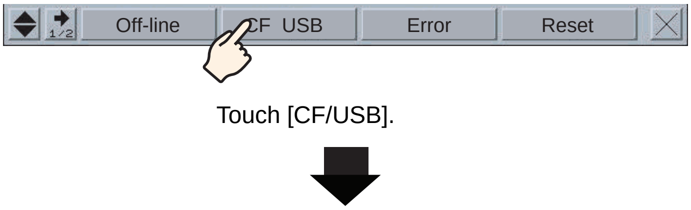
◆ Transfer (GP –> CF Card –> PC)

The following shows the process for using a CF Card. To use USB Storage, replace "CF Card" with "USB storage" in the process.

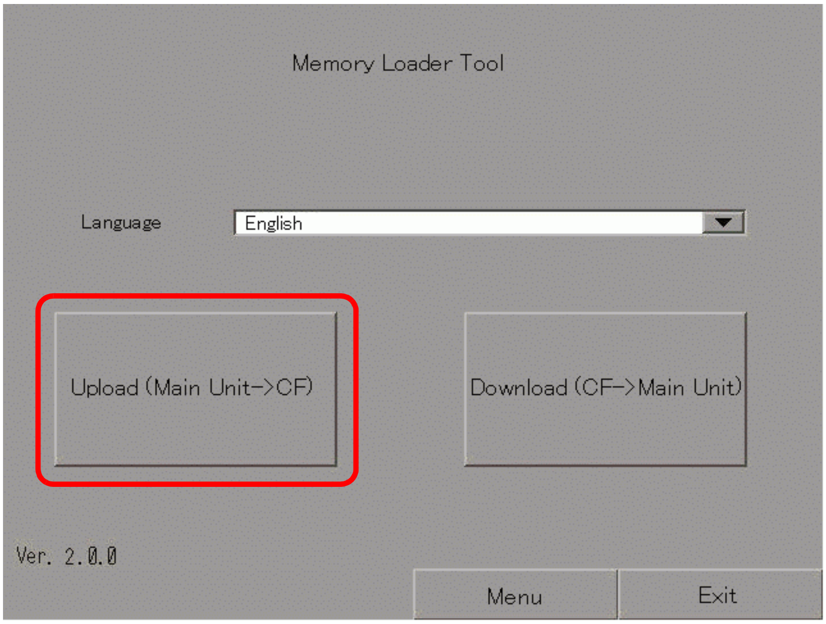
1 Insert the CF card with boot data.
To display the system menu, within 0.5 seconds of each touch, touch the top-right then the bottom-left corner, or touch the top-left then bottom-right corner.



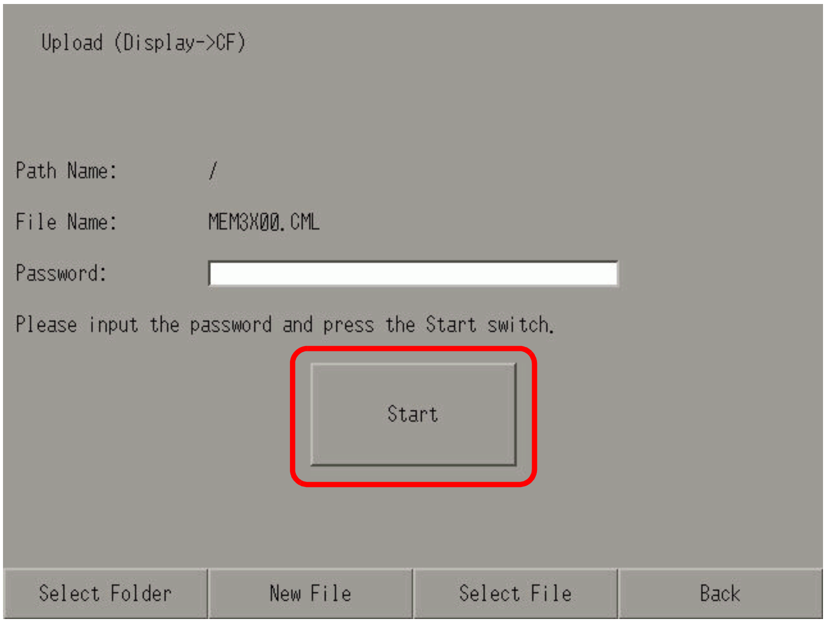
2 Touch [CF/USB]. Touch [CF startup] to reset GP.



3 The Memory Loader Tool launches. The following screen opens. Touch [Upload].



4 Touch [Start] to begin saving the GP project as backup data (*.cml). If there is a password for transferring, input the password first, then touch [Start].



NOTE

- Backup data is saved as displayed in the [File Name] field. If you are using AGP-3500T, by default this is "MEM3X00.CML." If you want to use another file name, touch [New File]. To overwrite an existing file name in the CF Card, touch [Select File].

5 Once upload is complete, touch [Back] then [Exit] to close the Memory Loader Tool.

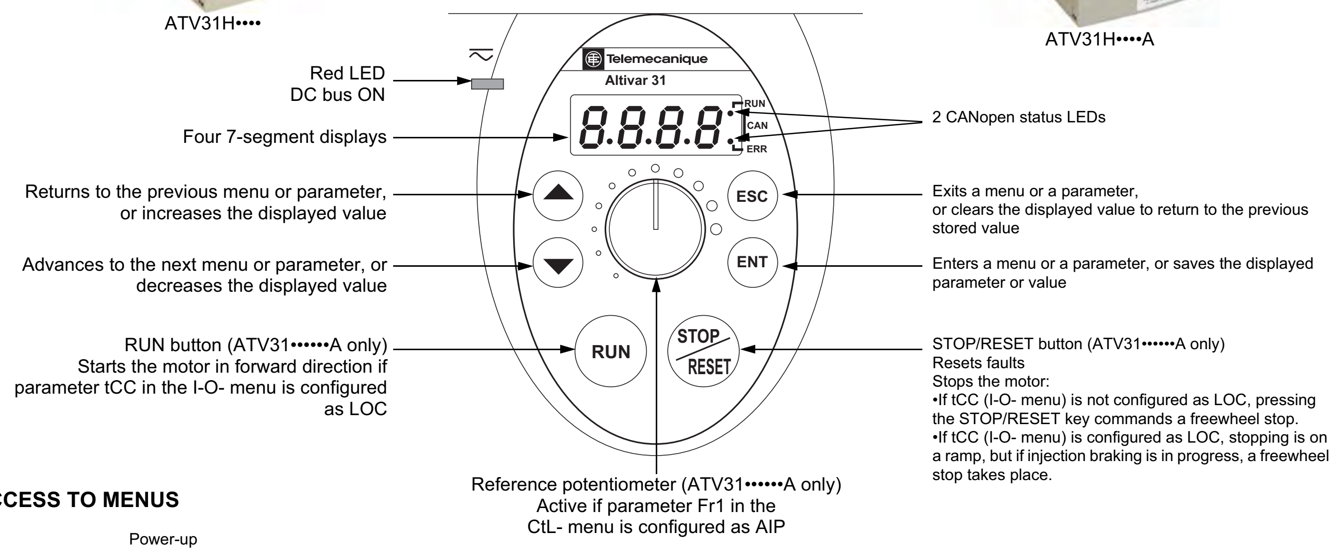
6 Remove the CF Card from the GP and insert it into the PC.

ATV31H••••

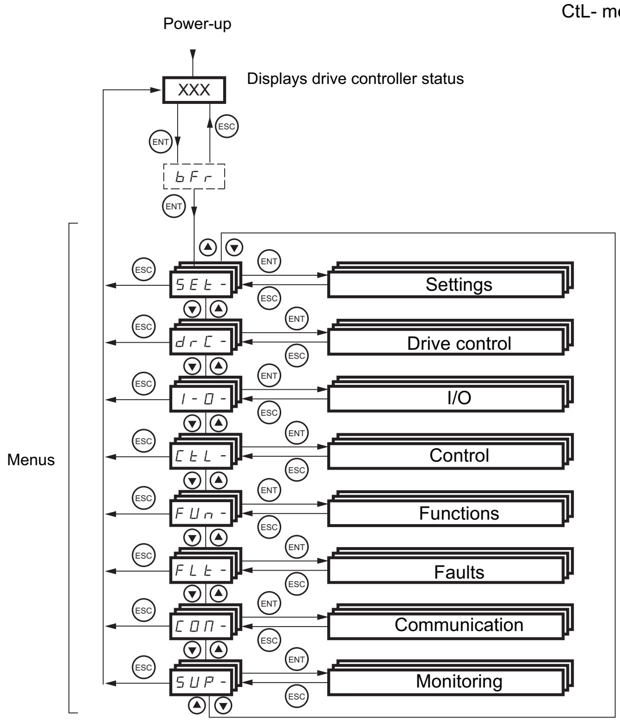
Note: Please refer to the ATV31 Installation Guide (VVDED303041US) and the ATV31 Programming Manual (VVDED303042US) for complete installation and programming instructions.

ATV31H••••A

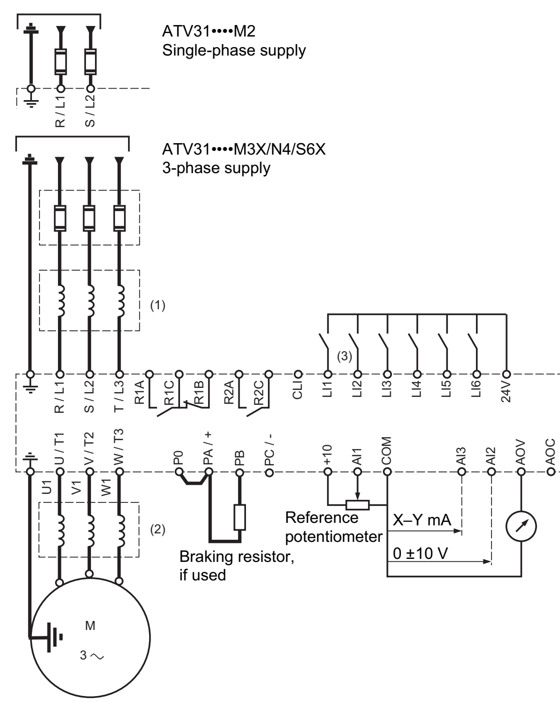
KEYPAD OPERATION



ACCESS TO MENUS

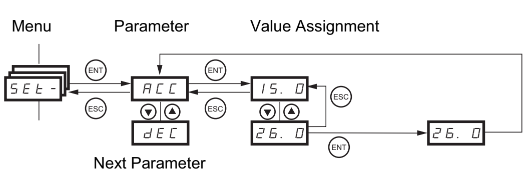


TYPICAL CONNECTIONS



(1) A line reactor is required for 575 V installations.
(2) A load reactor is recommended for 575 V installations.
(3) When configured for 2-wire control, LI1 is Run Forward (fixed). When configured for 3-wire control, LI1 is Stop (fixed) and LI2 is Run Forward (fixed).

ACCESS TO PARAMETERS



The display flashes when a value is stored. Some functions may require pressing the ENT button for two seconds to store the parameter change.

SEt - SETTINGS Menu

Parameter	Code	Factory Setting
Speed ref. from remote -Hz	LFr	
Internal PI regulator ref. -Hz	rPI	0 Hz
Acceleration ramp time -s	ACC	3 s
Acceleration ramp time 2 -s	AC2	5 s
Deceleration ramp time 2 -s	dE2	5 s
Deceleration ramp time -s	dEC	3 s
Start custom accel. ramp-%	tA1	10%
End custom accel. ramp -%	tA2	10%
Start custom decel. ramp-%	tA3	10%
End custom decel. ramp -%	tA4	10%
Low speed -Hz	LSP	0 Hz
High speed -Hz	HSP	bFr
Thermal current -A	ItH	Varies w/rating
IR compensation -%	UFr	20%
Gain -%	FLG	20%
Stability -%	StA	20%
Slip comp. -%	SLP	100%
DC injection curr -A	IdC	0.7 In
DC injection time -s	tdC	0.5 s
Auto. DC injection time -s	tdC1	0.5 s
Auto. DC injection curr -A	SdC1	0.7 In
Auto. DC injection time 2 -s	tdC2	0 s
Auto. DC injection curr 2 -A	SdC2	0.5 In
Skip freq. -Hz	JPF	0 Hz
Skip freq. 2 -Hz	JF2	0 Hz
Jog operating freq. -Hz	JGF	10 Hz
PI regulator prop. gain	rPG	1
PI regulator int. gain -/s	rIG	1/s
PID coeff	FbS	1
PID inversion	PiC	nO
2nd preset PI reference -%	rP2	30%
3rd preset PI reference -%	rP3	60%
4th preset PI reference -%	rP4	90%
Preset speed 2 -Hz	SP2	10 Hz
Preset speed 3 -Hz	SP3	15 Hz
Preset speed 4 -Hz	SP4	20 Hz
Preset speed 5 -Hz	SP5	25 Hz
Preset speed 6 -Hz	SP6	30 Hz
Preset speed 7 -Hz	SP7	35 Hz
Preset speed 8 -Hz	SP8	40 Hz
Preset speed 9 -Hz	SP9	45 Hz
Preset speed 10 -Hz	SP10	50 Hz
Preset speed 11 -Hz	SP11	55 Hz
Preset speed 12 -Hz	SP12	60 Hz
Preset speed 13 -Hz	SP13	70 Hz
Preset speed 14 -Hz	SP14	80 Hz
Preset speed 15 -Hz	SP15	90 Hz
Preset speed 16 -Hz	SP16	100 Hz
Current limit -A	CL1	1.5 In
Current limit 2 -A	CL2	1.5 In
Low speed oper. time -s	tLS	0 (no time limit)
Restart error threshold	rSL	0
Motor 2 IR compen. -%	UFr2	20%
Motor 2 freq. loop gain -%	FLG2	20%
Motor 2 freq. loop stabil.-%	StA2	20%
Motor 2 slip compen. -%	SLP2	100%
Frequency Lev.Att -Hz	Ftd	bFr
Thermal Level Att. -%	ttd	100%
Current Level Att. -A	Ctd	In
Display para. scale factor	SdS	30
Sw. Freq -kHz	SFr	4 kHz

drC - DRIVE CONTROL Menu

Parameter	Code	Factory Setting
Motor frequency -Hz	bFr	50 Hz
Nom. motor volt -V	UnS	Varies w/rating
Nom. motor frequency -Hz	FrS	50 Hz
Nom. motor current -A	nCr	Varies w/rating
Nom. motor speed -RPM	nSP	Varies w/rating
Motor CosPhi (power fact.)	CoS	Varies w/rating
Cool state stator resistance	rSC	nO
Auto tuning	tUn	nO
Auto tuning status	tUS	tAb
Voltage/frequency ratio	UFt	n
Noise reduction	nrd	YES
Switching frequency -kHz	SFr	4 kHz
Maximum frequency -Hz	tFr	60 Hz
Suppress speed loop filter	SrF	nO
Save the configuration	SCS	nO
Return to factory settings	FCS	nO

I-O - I/O Menu

Parameter	Code	Factory Setting
Terminal strip config	tCC	2C ATV31••••••A: LOC
Type 2 wire	tCt	trn
Reverse operation	rrS	If tCC=2C: L12 if tCC=3C: L13 if tCC=LOC:nO
AI3 low speed -mA	CrL3	4 mA
AI3 high speed -mA	CrH3	20 mA
Analog output config	AOIt	oA
Analog/logic output	dO	nO
Relay R1	r1	FLt
Relay R2	r2	nO

CtL - CONTROL Menu

Parameter	Code	Factory Setting
Function access level	LAC	L1
Ref 1 config	Fr1	AI1 ATV31••••••A: AIP
Ref 2 config	Fr2	nO
Ref switching	rFC	Fr1
Separate ctrl/ref channels	CHCF	SIM
Ctrl channel 1 config	Cd1	tEr ATV31••••••A: LOC
Ctrl channel 2 config	Cd2	Mdb
Ctrl channel switching	CCS	Cd1
Copy channel 1 to channel 2	COP	nO
Ctrl via remote keypad	LCC	nO
Stop priority	PSt	YES
Direction of operation	rOt	dFr

FUn - APPLICATION FUNCTIONS Menu

Parameter	Code	Factory Setting
rPC submenu		
Ramp type	rPt	LIn
Start CUS accel ramp-%	tA1	10%
End CUS accel ramp -%	tA2	10%
Start CUS decel ramp-%	tA3	10%
End CUS decel ramp -%	tA4	10%
Accel ramp time -s	ACC	3 s
Decel ramp time -s	dEC	3 s
Ramp switching	rPS	nO
Ramp switch. thresh-Hz	Frt	0
Accel ramp time 2 -s	AC2	5 s
Decel ramp time 2 -s	dE2	5 s
Decel ramp adaptation	brA	YES
StC submenu		
Normal stop	Stt	Stn
Fast stop	FSt	nO
Decel ramp coef.	dCF	4
DC injection stop	dCI	nO
DC injection current -A	IdC	0.7 In
DC injection time -s	tdC	0.5 s
Freewheel stop	nSt	nO
AdC submenu		
Auto DC injection	AdC	YES
Auto inject. time -s	tdC1	0.5 s
Auto inject. level -A	SdC1	0.7 In
Auto inject. time 2 -s	tdC2	0 s
Auto inject. level 2 -A	SdC2	0.5 In
SAI submenu		
Summing input 2	SA2	AI2
Summing input 3	SA3	nO
PSS submenu		
2 preset speeds	PS2	if tCC=2C/LOC: LI3 if tCC=3C: LI4
4 preset speeds	PS4	if tCC=2C/LOC: LI4 if tCC=3C: nO
8 preset speeds	PS8	nO
16 preset speeds	PS16	nO
JOG submenu		
Jog operation	JOG	if tCC=2C/LOC: nO if tCC=3C: LI4
Jog oper. reference -Hz	JGF	10 Hz
UPd submenu		
Plus speed	USP	nO
Minus speed	dSP	nO
Save references	Str	nO
PI submenu		
PI regulator feedback	PIF	nO
PI regul. proport. gain	rPG	1
PI regul. intergral gain	rIG	1
PI feedback coeff.	FbS	1
Reverse PI regul. direction	PIC	nO
2 preset PI references	Pr2	nO
4 preset PI references	Pr4	nO

FUn - APPL. FUNCTIONS Menu (cont.)

Parameter	Code	Factory Setting
PI submenu (cont.)		
Preset PI ref. 2 -%	rP2	30%
Preset PI ref. 3 -%	rP3	60%
Preset PI ref. 4 -%	rP4	90%
Restart after error thresh.	rSL	0
Internal PI regul. ref.	PII	nO
Internal PI regul. ref. -%	rPI	0
bLC submenu		
Brake control config.	bLC	nO
Brake release freq. -Hz	brL	Varies w/rating
Release current thresh.-A	Ibr	Varies w/rating
Brake release time -s	brt	0.5 s
Brake engage freq. thresh.	bEn	nO
Brake engage time -s	bEt	0.5 s
Brake release pulse	bIP	nO
LC2 submenu		
Current limit 2 switching	LC2	nO
Current limit 2 -A	CL2	1.5 In
CHP Motor Switching	CHP	nO
LSt Limit switch management		

FLt - FAULTS Menu

Parameter	Code	Factory Setting
Automatic restart	Atr	nO
Max restart duration	tAr	5
Reset fault	rSF	nO
Catch on fly	FLr	nO
External fault	EtF	nO
External fault stop mode	EPL	YES
Motor phase loss fault config.	OPL	YES
Line phase loss fault config.	IPL	YES
Drive overheat fault stop mode	OHL	YES
Mtr overload fault stop mode	OLL	YES
Modbus serial link fault stop	SLL	YES
CANopen serial link fault stop	COL	YES
Auto-tune fault config.	tnL	YES
Signal loss fault stop	LFL	nO
Fallback speed -Hz	LFF	10 Hz
Undervoltage derated oper.	drn	nO
Mains power loss stop	StP	nO
Fault inhibit	InH	nO
Reset oper. time to zero	rPr	nO

COn - COMMUNICATION Menu

Parameter	Code	Factory Setting
Modbus drive address	Add	1
Modbus transmission speed	tbr	19200
Modubus commun. format	tFO	8E1
Modbus timeout -s	ttO	10 s
CANopen drive address	AdCO	0
CANopen transmisson speed	bdCO	125
CANopen error registry	ErCO	
Forced local mode	FLO	nO
Ref & ctrl channel selection in forced local mode	FLOC	AI1 ATV31••••••A: AIP

SUP - DISPLAY Menu

Parameter	Code	Factory Setting
Speed ref. from remote -Hz	LFr	
Internal PI reference -%	rPI	
Freq. ref before ramp -Hz	FrH	
Output freq. at motor -Hz	rFr	
Output value in cust. units	SPd1 SPd2 SPd3	
Motor current -A	LCr	
Motor power -%	OPr	
Line voltage -V	ULn	
Motor thermal state -%	tHr	
Drive thermal state -%	tHd	
Last fault	LFt	
Motor torque -%	Otr	
Operating time -hr	rtH	

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Machine Tool Pressure Switches
Interruptores de presión para máquinas-herramienta
Manostats pour machines-outils

Class Clase Classe	Type / Tipo / Type		Series Serie Série
	Adjustable Differential Diferencial ajustable Différentiel réglable	**Non-adjustable Differential Diferencial fijo Différentiel non réglable**	
9012	GAW, GBW, GCW, GAWM, GBWM, GCWM	GDW, GEW, GFW, GDWM, GEWM, GFWM	C

Retain for future use. / Conservar para uso futuro. / À conserver pour usage ultérieur.

⚠ CAUTION / PRECAUCIÓN / ATTENTION

EXCESSIVE PRESSURE

Ensure that pressures applied to the switch, including surges, are within the stated range of the switch.

Failure to follow this instruction can result in injury or equipment damage.

PRESIÓN EXCESIVA

Asegúrese de que las presiones aplicadas al interruptor, incluyendo sobretensiones, se encuentren dentro de la gama especificada para el interruptor.

El incumplimiento de esta instrucción puede causar lesiones o daño al equipo.

PRESSION EXCESSIVE

Assurez-vous que les pressions appliquées sur le manostat, y compris les surpressions, se trouvent dans la gamme indiquée pour le manostat.

Si cette directive n'est pas respectée, cela peut entraîner des blessures ou des dommages matériels.

RATINGS

Pressure Ratings

Do not expose the pressure actuators to system or surge pressures greater than the maximum pressure rating printed on the device nameplate, to avoid leakage from the actuator or a change in operating setpoints.

Do not use the switch directly on steam that exceeds 15 psig (1 bar). For indirect use, refer to "Indirect Use" on page 5.

Temperature Ratings

Refer to Table 1 on page 1. The switch may not operate properly if the media fluid freezes or if frost or ice forms inside the switch.

VALORES NOMINALES

Valores nominales de presión

No exponga los accionadores de presión al sistema ni a presiones excesivas mayores que el valor nominal máximo de la presión especificado en la placa de datos del dispositivo, para evitar fugas del accionador o un cambio en los puntos de referencia de funcionamiento.

No utilice el interruptor directamente en vapor que exceda 15 psig (1 bar) de presión. Para un uso indirecto, consulte la sección "Uso indirecto" en la página 5.

Valores nominales de temperatura

Consulte la tabla 1 en la página 1. Es posible que el interruptor no funcione adecuadamente si se congela el líquido o si se forma hielo o escarcha dentro del interruptor.

VALEURS NOMINALES

Valeurs nominales de pression

N'exposez pas les actionneurs de pression au système ou à des surpressions supérieures à la pression nominale maximale imprimée sur la plaque signalétique du dispositif, afin d'éviter une fuite de l'actionneur ou une modification des points de consigne de fonctionnement.

Ne pas utiliser le manostat directement sur les lignes de vapeur qui dépasse 15 lb/po^2 (1 bar) de pression. Pour une utilisation indirecte, consulter la section « Utilisation indirecte » à la page 5.

Valeurs nominales de température

Consulter le tableau 1 à la page 1. Le manostat peut ne pas fonctionner correctement si le fluide sous pression gèle ou si du givre ou de la glace se forme à l'intérieur du manostat.

Table / Tabla / Tableau 1 : Continuous-Use Temperature Ratings / Valores nominales en temperaturas de uso continuo Valeurs nominales de température en régime d'utilisation continu

Ambient Temperature –10 to +185 °F (–25 to +85 °C)	**Temperatura ambiente** –25 a +85 °C (–10 a +185 °F)	**Température ambiante** –25 à +85 °C (–10 à +185 °F)
Pressure Media Temperature –10 to +250 °F (–25 to +120 °C)	**Temperatura de los medios de presión** –25 a +120 °C (–10 a +250 °F)	**Température des milieux sous pression** –25 à +120 °C (–10 à +250 °F)

⚠ DANGER / PELIGRO / DANGER

HAZARDOUS VOLTAGE

Turn off all power supplying this pressure switch before working on or inside the switch.

Failure to follow this instruction will result in death or serious injury.

TENSION PELIGROSA

Desenergice el interruptor de presión antes de efectuar cualquier tabajo dentro o fuera de él.

El incumplimiento de esta instrucción podrá causar la muerte o lesiones serias.

TENSION DANGEREUSE

Coupez l'alimentation du manostat avant d'y travailler.

Si cette directive n'est pas respectée, cela entraînera la mort ou des blessures graves.

MOUNTING

Mount the pressure switch by the surface mounting holes (**F** or **G**, Figures 1 and 2) and by the pressure connection. Do not mount the switch by its pressure connection only. Turn the switch onto the pressure system pipe using a wrench on the actuator's hexagonal pipe connector. Do not apply leverage through the switch housing. Periodically torque the actuator mounting screws of Types GAW, GAWM, GDW, and GDWM switches to 8–10 lb-in (0.9–1.13 N•m).

NOTE: Do not obstruct the 1/4 in. (6 mm) vent hole on Types GBW, GEW, GCW, GFW, GEWM, GBWM, GCWM, and GFWM switches.

MONTAJE

Monte el interruptor de presión en los agujeros de montaje (**F** o **G**, figuras 1 y 2) y a la conexión de presión. No instale el interruptor por la conexión de presión solamente. Gire el interruptor en el tubo del sistema de presión con una llave para tuercas en el conector de tubo hexagonal del accionador. No haga palanca en la caja del interruptor. Regularmente, apriete los tornillos de montaje del accionador de los interruptores tipos GAW, GAWM, GDW y GDWM en un par de apriete de 0,9–1,13 N•m (8–10 lbs-pulg).

NOTA: No bloquee el agujero de ventilación de 6 mm (1/4 pulg) en los interruptores tipos GBW, GEW, GCW, GFW, GEWM, GBWM, GCWM y GFWM.

MONTAGE

Monter le manostat par les trous de montage (**F** ou **G**, figures 1 et 2), et aussi au raccordement de pression. Ne pas monter le manostat uniquement par le raccordement de pression. Faire tourner le manostat sur le tuyau du système de pression en utilisant une clé, sur le connecteur de tuyau hexagonal de l'actionneur. Ne pas appliquer d'effet de levier sur l'enveloppe du manostat. Serrer périodiquement les vis de montage de l'actionneur des manostats types GAW, GAWM, GDW et GDWM à un couple entre 0,9 à 1,13 N•m (8 à 10 lb-po).

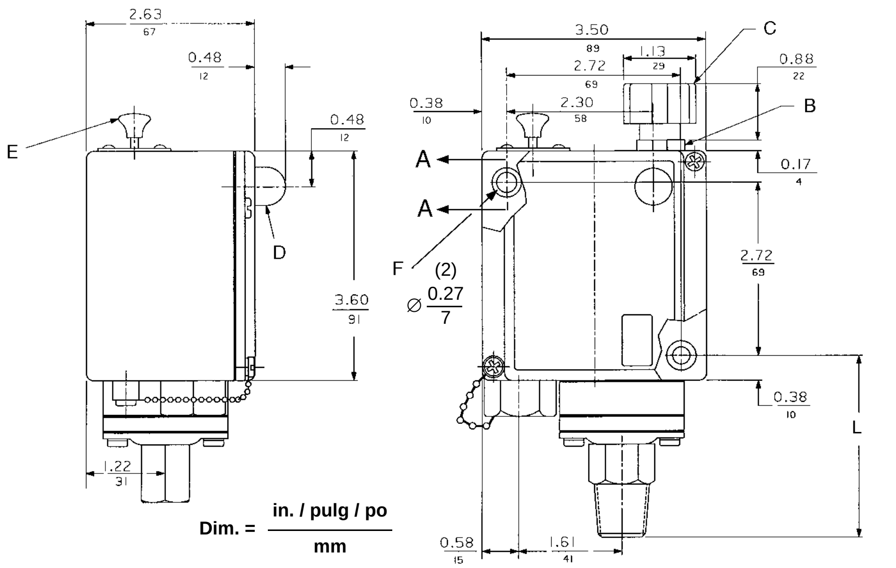
REMARQUE : Ne pas boucher le trou d'aération de 6 mm (1/4 po) sur les manostats types GBW, GEW, GCW, GFW, GEWM, GBWM, GCWM et GFWM.

Table / Tabla / Tableau 2 : Pressure Connections / Conexiones de presión / Raccordements de pression

Form Forma Forme	Type Tipo Type	Connection	Conexión	Raccordement
Standard Estándar Standard	G•W	1/4–18 NPTF. The dry seal thread should seal against a new external 1/4–NPT thread without sealing tape or compounds.	1/4–18 NPTF. La rosca de cierre hermético en seco deberá cerrar herméticamente con una nueva rosca externa de 1/4–NPT sin necesidad de cinta o de los compuestos de sellado.	1/4–18 NPTF. Le filetage de type sec devrait fermer hermétiquement avec un nouveau filetage externe de 1/4–NPT sans besoin de ruban ou de composés de scellement.
Z	GAW, GDW	1/4–18 NPT external thread	Rosca externa de 1/4–18 NPT	Filetage externe de 1/4–18 NPT
Z16	GAW, GDW	1/2–14 NPT external and 1/4–18 NPTF internal thread	Rosca externa de 1/2–14 NPT e interna de 1/4–18 NPTF	Filetage externe de 1/2–14 NPT et interne de 1/4–18 NPTF
Z18	G•W	7/16–20 UNF-2B	7/16–20 UNF-2B	7/16–20 UNF-2B
Metric / Métrica / Métrique	G•WM	G 1/4 BS2779	G 1/4 BS2779	G 1/4 BS2779

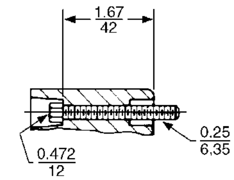
DIMENSIONS | DIMENSIONES | DIMENSIONS

Figure / Figura / Figure 1 : Standard Dimensions / Dimensiones estándar / Dimensions normales



Mounting Detail / Detalle de montaje / Détail de montage

Section A–A / Sección A–A
Section A–A

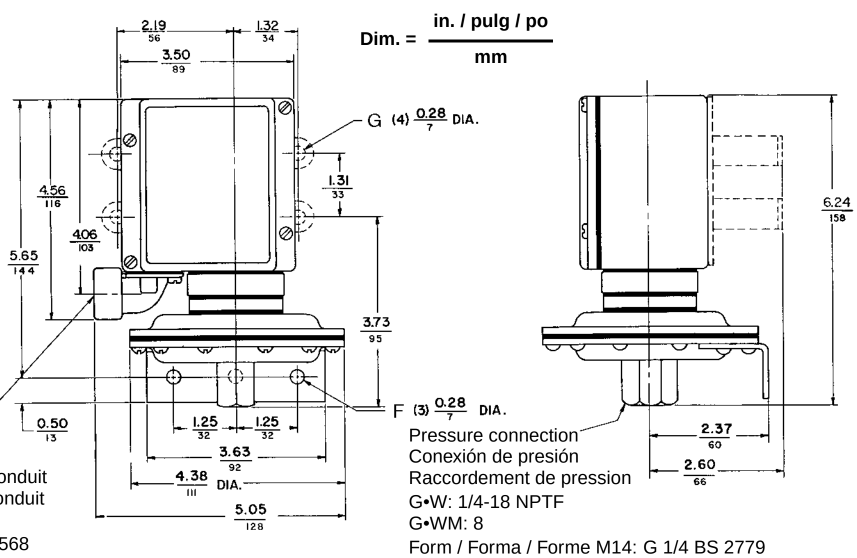


Max. head dia.
Diá. máx de la cabeza
Diam. max. de la tête

Item/ Art.	Forms / Formas / Formes
B	K1
C	K
D	G17–G22
E	E2–E4

Type / Tipo / Type	L in. / pulg / po (mm)	Turn Radius / Radio de rotación / Rayon de courbure in. / pulg / po (mm)
GAW, GDW	2.33 (59)	2.86 (73)
GBW, GEW	2.23 (57)	2.63 (67)
GCW, GFW	3.15 (80)	2.86 (73)

Figure / Figura / Figure 2 : Types / Tipos / Types : GAW-, GAWM-, GDW-, GDWM-1, -21



F	Mounting holes (standard)	Agujeros de montaje (estándar)	Trous de montage (standard)
G	Mounting holes (Form F)	Agujeros de montaje (forma F)	Trous de montage (forme F)

Turn Radius = 2.86 in. (73 mm) | **Radio de rotación = 73 mm (2,86 pulg)** | **Rayon de courbure = 73 mm (2,86 po)**

WIRING

- **Wire:** #16–12 AWG ($1.5–2.5\ mm^2$) solid or stranded **copper** (not aluminum)
- **Tightening torque:** 6–9 lb-in (0.7–1.0 N•m)
- **Grounding** (⏚): see Figure 4
- **SPDT snap switch** (single pole, double throw): contains 1 N.O. and 1 N.C. double break element that must be used on circuits of the same polarity
- **DPDT snap switch** (double pole, double throw): contains two electrically separated sets of contact elements for use on circuits of opposite polarity. Each set contains 1 N.O. and 1 N.C. double break element that must be used on circuits of the same polarity.
- **Pilot lights (neon):**
  - — 120 V (Forms G17 and G18)
  - — 240 V (Forms G19 and G20)
- **Pilot lights (LED):**
  - — 24 Vdc (Forms G21 and G22)

ALAMBRADO

- **Conductor:** calibre $1,5–2,5\ mm^2$ (16–12 AWG) de **cobre** (no de aluminio) sencillo o trenzado
- **Par de apriete:** de 0,7 a 1,0 N•m (6 a 9 lbs-pulg)
- **Tierra** (⏚): vea la figura 4
- **Interruptor de resorte 1P2T** (un polo, doble tiro): contiene unidades de apertura doble, 1 N.A. y 1 N.C., que deben usarse en circuitos con la misma polaridad
- **Interruptor de resorte 2P2T** (dos polos, doble tiro): incluye dos juegos de unidades de contactos eléctricamente separados para usarse en circuitos con polaridad opuesta. Cada juego contiene unidades de apertura doble, 1 N.A. y 1 N.C., que deben usarse en circuitos con la misma polaridad.
- **Lámparas piloto (neón):**
  - — 120 V∿/V⎓ (formas G17 y G18)
  - — 240 V∿/V⎓ (formas G19 y G20)
- **Lámparas piloto (LED)**
  - — 24 V⎓ (c.d.) (formas G21 y G22)

CÂBLAGE

- **Fil :** calibre 16 à 12 AWG ($1,5–2,5\ mm^2$) en **cuivre** (pas en aluminium) rigide ou toronné
- **Couple de serrage :** entre 0,7 et 1,0 N•m (6 à 9 lb-po)
- **Mise à la terre** (⏚) : voir la figure 4
- **Interrupteur à rupture brusque UPBD** (unipolaire, bidirectionnel) : contient des éléments d’ouverture double, 1 N.O. et 1 N.F., qui doivent être utilisés avec des circuits de la même polarité
- **Interrupteur à rupture brusque BPBD** (bipolaire, bidirectionnel) : contient deux jeux électriquement séparés d’éléments de contact pour l’utilisation avec des circuits de polarité opposée. Chaque jeu contient des éléments d’ouverture double, 1 N.O. et 1 N.F., qui doivent être utilisés avec des circuits de la même polarité.
- **Lampes témoins (néon) :**
  - — 120 V (formes G17 et G18)
  - — 240 V (formes G19 et G20)
- **Lampes témoins (DÉL):**
  - — 24 Vcc, (formes G21 et G22)

INDIRECT USE

For indirect use on steam, attach a minimum of 10 ft (3.05 m) of capillary tubing with an outer diameter (OD) of 1/8 in. (3.2 mm) between the steam source and the actuator. Ensure that the tubing is rated for use on steam up to 250 psig (17 bars). (Tubing is not available from Schneider Electric). Do not exceed the maximum pressure and temperature ratings of the switch. Coil the tubing in the pressure line in several loops of 4–8 in. (100–200 mm) diameter, to serve as a heat exchanger and to form a static water head as a buffer to steam temperature. See Figure 3.

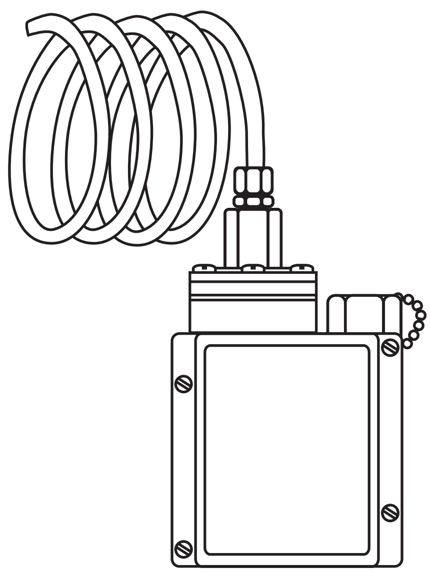
USO INDIRECTO

Para un uso indirecto en vapor, conecte un tubo capilar de 3 m (10 pies) de largo como mínimo y un diámetro exterior de 3,2 mm (1/8 pulgada) entre la fuente de vapor y el accionador. Asegúrese que el tubo sea adecuado para usarse en vapor de hasta 250 psig (17 bars) de presión, (el tubo capilar no se encuentra disponible de Schneider Electric). No exceda los valores nominales máximos de la presión ni de la temperatura especificados para el interruptor. Enrolle el tubo en la línea de presión en varios bucles de 100 a 200 mm (4 a 8 pulgadas) de diámetro; para que sirva como termointercambiador y que forme un tope de altura de elevación para la temperatura del vapor. Consulte la figura 3.

UTILISATION INDIRECTE

Pour une utilisation indirecte sur de la vapeur, attacher un tube capillaire d'une longueur minimale de 3 m (10 pi) et d'un diamètre extérieur de 3,2 mm (1/8 po) entre la source de vapeur et l'actionneur. S'assurer que le tube est d'une valeur nominale pour être utilisé sur de la vapeur allant jusqu'à 250 lb/po^2 (17 bars) de pression. (Le tube n'est pas disponible chez Schneider Electric). Ne pas dépasser la pression maximale et les températures nominales du manostat. Bobiner le tube de la ligne de pression dans plusieurs boucles de 100 à 200 mm (4 à 8 po) de diamètre, pour servir d'échangeur thermique et former une tête de pression statique d'eau servant de tampon à la température de la vapeur. Voir la figure 3.

Figure / Figura / Figure 3 : Capillary Tubing / Tubo capilar / Tube capillaire



SETPOINT ADJUSTMENTS

Refer to Figure 4.

The pressure switch is factory set to the operating setpoints marked on the outside of the switch housing. Before readjusting the switch, cycle it to determine the actual operating setpoints.

Non-Adjustable Differential (GDW, GDWM, GEW, GEWM, GFW, GFWM)

The range adjustment nut **(H)** or knob **(C)** adjusts both setpoints by the same amount. To adjust the setpoints:

1. Place a flat-blade screwdriver in a slot of the range adjustment nut.
2. To raise the setpoints, rotate the nut to the left. To lower the setpoints, rotate the nut to the right.

Adjustable Differential (GAW, GAWM, GBW, GBWM, GCW, GCWM)

The range adjustment nut or knob determines the decreasing setpoint. The adjusting screw **(J)** determines the increasing setpoint. To adjust the setpoints:

1. Adjust the decreasing setpoint (see Steps 1–2 above).
2. To raise the increasing setpoint, turn the adjusting screw to the right. To lower the increasing setpoint, turn the adjusting screw to the left.

   This adjustment does not affect the decreasing setpoint.

AJUSTE DE LOS PUNTOS DE REFERENCIA

Consulte la figura 4.

El interruptor de presión se calibra en la fábrica de acuerdo con los puntos de referencia de funcionamiento marcados en el exterior de la caja. Antes de volver a ajustar el interruptor, páselo por un ciclo para determinar los puntos de referencia de funcionamiento reales.

Diferencial fijo (GDW, GDWM, GEW, GEWM, GFW, GFWM)

La tuerca **(H)** o perilla **(C)** de ajuste de la gama ajusta ambos puntos de referencia en el mismo valor. Para ajustar los puntos de referencia:

1. Coloque un desatornillador de punta plana en una de las ranuras de la tuerca de ajuste de la gama.
2. Para aumentar los puntos de referencia, gire la tuerca hacia la izquierda. Para disminuir los puntos de referencia, gire la tuerca hacia la derecha.

Diferencial ajustable (GAW, GAWM, GBW, GBWM, GCW, GCWM)

La tuerca o perilla de ajuste de la gama determina el punto de referencia de disminución. El tornillo de ajuste **(J)** determina el punto de referencia de aumento. Para ajustar los puntos de referencia:

1. Ajuste el punto de referencia de disminución (vea los pasos 1–2 anteriores).
2. Para aumentar el punto de referencia de aumento, gire el tornillo de ajuste hacia la derecha. Para disminuir el punto de referencia de aumento, gire el tornillo de ajuste hacia la izquierda.

   Este ajuste no altera el punto de referencia de disminución.

RÉGLAGES DES POINTS DE CONSIGNE

Voir la figure 4.

Le manostat est réglé à l’usine aux points de consigne de fonctionnement indiqués à l’extérieur de l’enveloppe. Avant de régler de nouveau le manostat, effectuer un cycle de manœuvre pour déterminer les points de consigne de fonctionnement actuels.

Différentiel non réglable (GDW, GDWM, GEW, GEWM, GFW, GFWM)

L’écrou **(H)** ou le bouton **(C)** de réglage de la gamme permet de régler les deux points de consigne à la même valeur. Pour régler les points de consigne :

1. Introduire un tournevis à lame plate dans une fente de l’écrou de réglage de la gamme.
2. Pour augmenter les points de consigne, tourner l’écrou vers la gauche. Pour diminuer les points de consigne, tourner l’écrou vers la droite.

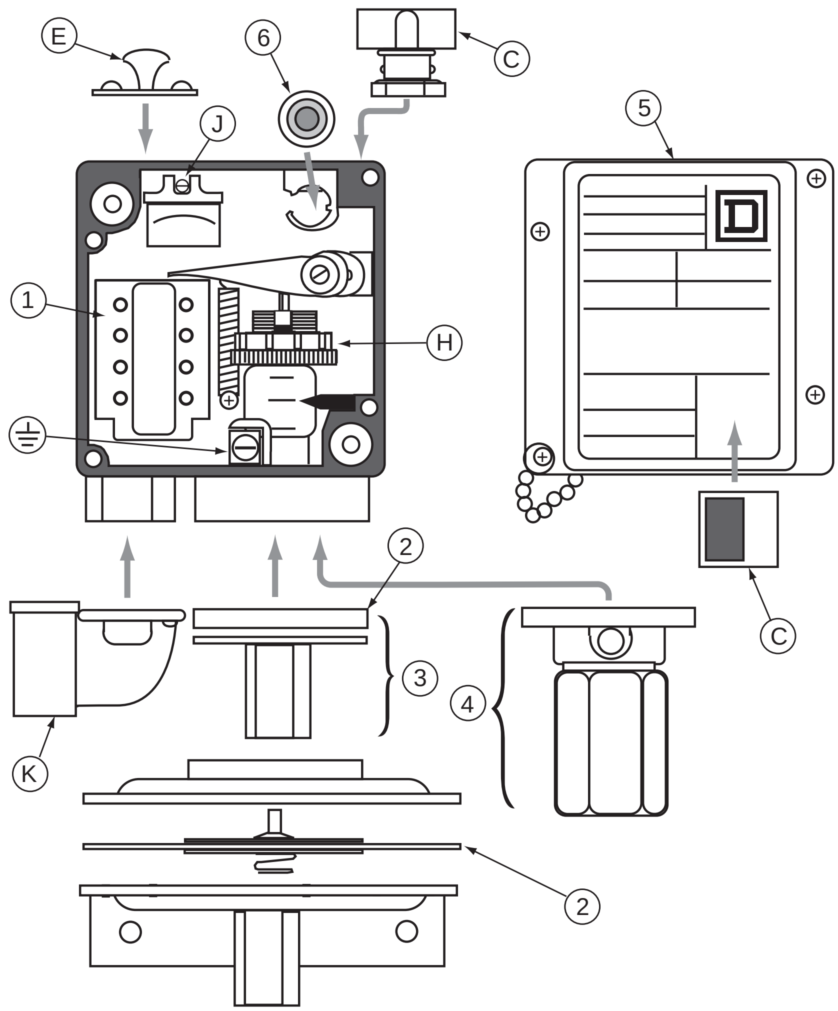
Différentiel réglable (GAW, GAWM, GBW, GBWM, GCW, GCWM)

L’écrou ou le bouton de réglage de la gamme détermine le point de consigne descendant. La vis de réglage **(J)** détermine le point de consigne ascendant. Pour régler les points de consigne:

1. Régler le point de consigne descendant (voir les étapes 1–2 ci-dessus).
2. Pour augmenter le point de consigne ascendant, tourner la vis de réglage vers la droite. Pour diminuer le point de consigne ascendant, tourner la vis de réglage vers la gauche.

   Ce réglage n’affecte pas le point de consigne descendant.

Figure / Figura / Figure 4 : Pressure Switch Assembly / Montaje del interruptor de presión / Assemblage du manostat



Item Art.	Description	Descripción	Description
1	Snap switch	Interruptor de resorte	Interrupteur à rupture brusque
2	Diaphragm assembly	Ensamble del diafragma	Assemblage de membrane
3	Diaphragm actuator (only available as an assembly)	Accionador del diafragma (solamente está disponible como ensamble)	Actionneur de la membrane (seulement disponible en tant qu'assemblage)
4	Piston actuator (only available as an assembly)	Accionador del pistón (solamente está disponible como ensamble)	Actionneur du piston (seulement disponible en tant qu'assemblage)
5	Cover	Cubierta	Couvercle
6	Pilot light (Forms G17–G22)	Lámpara piloto (formas G17–G22)	Lampe témoin (formes G17–G22)
C	Range adjustment knob and window (Form K)	Perilla de ajuste de la gama y ventana (forma K)	Bouton de réglage de la gamme et fenêtre (forme K)
E	Pull or push knob (Form E2–E4)	Perilla de empujar y jalar (forma E2–E4)	Bouton pousser-tirer (formes E2–E4)
H	Range adjustment nut (not replaceable—**do not remove**)	Tuerca de ajuste de la gama (no se puede reemplazar—**no la desmonte**)	Écrou de réglage de la gamme (non remplaçable—**ne pas démonter**)
J	Adjustment screw	Tornillo de ajuste	Vis de réglage
K	Side conduit (Form B2)	Tubo conduit lateral (forma B2)	Conduit latéral (forme B2)

REPLACEMENT PARTS

When ordering a replacement part, always specify the Class, Type, and Form of the switch. For item numbers, refer to Figure 4 on page 7.

PIEZAS DE REPUESTO

Cuando solicite piezas de repuesto, siempre especifique la clase, el tipo y la forma del interruptor. Consulte la figure 4 en la página 7 para conocer los componenetes.

PIÈCES DE RECHANGE

Pour commander une pièce de rechange, il faut toujours spécifier la classe, le type et la forme du manostat. Pour les numéros d'articles, voir la figure 4 à la page 7.

Table / Tabla / Tableau 1 : Machine Tool Pressure Switch Replacement Parts / Piezas de repuesto del interruptor de presión para máquinas-herramienta / Pièces de rechange du manostat pour machines-outils

Item Art.	Description Descripción Description	Use on Types: Para usar en tipos: Utiliser avec types :	Class 9998 Type Clase 9998 Tipo Classe 9998 Type	Form Forma Forme
1	Snap switch assembly Ensamble del interruptor de resorte Assemblage de l'interrupteur à rupture brusque	G•W 1–6 G•W 21–26 G•WM 1–6 G•WM 21–26	PC313 PC314 PC339 PC340	–
2	Diaphragm assembly (only the diaphragm can be replaced) Ensamble del diafragma (solamente se puede reemplazar el diafragma) Assemblage de membrane (seule la membrane peut être remplacée)	GAW, GAWM, GDW, GDWM 1, 21 [1] GAW, GAWM, GDW, GDWM 2, 22 [1] GAW, GAWM, GDW, GDWM 4, 24 [1]	PC265 PC266 PC267	–
3	Diaphragm actuator assembly Ensamble del accionador del diafragma Assemblage de l'actionneur de la membrane	GAW, GAWM, GDW, GDWM 5, 25 GAW, GAWM, GDW, GDWM 6, 26 GBW, GBWM, GEW, GEWM 1, 21 GBW, GBWM, GEW, GEWM 2, 22	PC268 PC269 PC177 PC178	–
4	Piston actuator assembly Ensamble del accionador del pistón Assemblage de l'actionneur du piston	GCW, GCWM, GFW, GFWM 1, 21 GCW, GCWM, GFW, GFWM 2, 22 GCW, GCWM, GFW, GFWM 3, 23 GCW, GCWM, GFW, GFWM 4, 24	PC270 PC271 PC272 PC273	–
5	Cover assembly Ensamble de cubierta Assemblage de couvercle	Include the Class, Type, and Form to be printed on the nameplate. Incluir la clase, el tipo y la forma para que se impriman en la placa de datos. Inclure la classe, le type et la forme à être imprimé sur la plaque signalétique.	PC302	–
6	Pilot light kits Accesorios de lámpara piloto Kits de lampe témoin	Modification Kit for… / Accesorio de modificación para… Kit de modification pour… G•WM (120 V∿/V⎓) G•WM (240 V∿/V⎓) G•WM (24 V⎓) Replacement for… / Repuesto para… / Remplacement pour… 120 V∿/V⎓ 240 V∿/V⎓ 24 V⎓ Field conversion for… / Conversión en campo para… Conversion sur place pour… G•W (120 V∿/V⎓) G•W (240 V∿/V⎓) G•W (24 V⎓)	 PC278 PC279 PC276 PC303 PC304 PC305 PC306 PC307 PC308	 G17, G18 G19, G20 G21, G22 G17, G18 G19, G20 G21, G22 G17, G18 G19, G20 G21, G22
–	Gasket kit (not shown) Accesorio de empaques (no se muestra) Kit de joint (non représenté)	All / Todos / Tous	PC184	–

[1] Except Form O1 / Excepto la forma O1 / Sauf la forme O1

Schneider Electric USA
8001 Highway 64 East
Knightdale, NC 27545
1-888-SquareD (1-888-778-2733)
www.us.SquareD.com

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881AS
SERVICE BULLETIN

Class 9012 Type A and Type G
MAINTENANCE AND TROUBLE-SHOOTING TIPS

All maintenance recommended by the manufacturer of the equipment shall be performed. If the equipment fails to operate properly, carefully re-read the manufacturers' instructions and recommendations to see that they have been correctly followed. If the operating problems persist and it is suspected that the pressure switch is the cause, the following chart may be of assistance in identifying and correcting the most common pressure switch problems.

PROBLEM	POSSIBLE CAUSE	CORRECTION
Switch will not trip at desired pressure.	1. Defective pressure gauge. 2. Incorrect pressure settings. 3. Settings outside of switch pressure limits. 4. Damaged actuator. 5. Damaged snap switch. 6. Contacts overloaded (welded). 7. Surge pressure in system exceeds maximum allowable pressure rating of switch.	1. Replace pressure gauge. 2. Readjust to correct pressure setting (see Service Bulletin). 3. Replace with switch with correct range and differential (see catalog). 4. Replace actuator. 5. Replace snap switch. 6. Correct overload condition and replace snap switch. If normal load is higher than snap switch rating, install relay. 7. Replace switch with one with correct maximum allowable pressure (see catalog). 8. Replace with switch with correct differential (see catalog). 9a. Place restriction in pressure line to smooth out pressure fluctuations. 9b. If switch is adjustable differential type, increase differential until chatter or bounce stops. 9c. If switch is non-adjustable differential type, change to adjustable differential type and increase differential until chatter or bounce stops. 10. Move switch to new location or shock mount.
Switch will not reset at desired pressure.	1. Defective pressure gauge. 2. Incorrect pressure settings. 8. Differential either too wide or too narrow. 4. Damaged actuator. 5. Damaged snap switch. 6. Contacts overloaded (welded). 7. Surge pressure in system exceeds maximum allowable pressure rating of switch.	
Contact chatter or bounce (trips and resets rapidly).	9. Peaks and valleys of ripples in excess of switch setting. 10. Excessive vibration.	
Nuisance tripping on falling pressure.	11. Momentary dip in pressure before cycle ends.	
Switch operates mechanically but not electrically.	12. No power to switch. 13. Wired to wrong contact. 14. Corroded or loose connections. 5. Damaged snap switch. 6. Contacts overloaded (welded).	

881AS
SERVICE BULLETIN

<table>
<tr><th>PROBLEM</th><th>POSSIBLE CAUSE</th><th>CORRECTION</th></tr>
<tr><td>Erratic operation.</td><td>15. Diaphragm or O-ring swell (pressure medium non-compatible with diaphragm or O-ring material.
10. Excessive shock.
16. Foreign matter in pressure medium.
6. Contacts overloaded (welded).</td><td rowspan="6">11. Widen differential or replace with wider differential.
12. Get power to switch.
13. Wire to correct contact. (See wiring diagram).
14. Make new connections or tighten existing connections.
15. Replace with switch with correct diaphragm or O-ring material.
16. Install filter in system and replace actuator.
17. Remove plug from drain hole and replace diaphragm assembly.
18. Check drain line to be sure it is always open to atmosphere and not connected into another line. Replace diaphragm assembly.
19. If leakage is objectionable, install drain line.
20. Replace bulb or bulb assembly.
21. Change system pressure by amount of differential of switch.</td></tr>
<tr><td>Oil leaking past diaphragm into switch housing on 9012 Type A piston type switch.</td><td>17. Plug has not been removed from drain hole.
18. Drain line not open to atmosphere.</td></tr>
<tr><td>Oil leaking from drain hole on 9012 Type A piston type switch.</td><td>19. Some bypass leakage is normal on unsealed piston type switch.</td></tr>
<tr><td>Oil leaking from vent hole on 9012 Type G.</td><td>4. Small amount of seepage is normal to ensure seal lubrication. Excessive leakage indicates seal failure.</td></tr>
<tr><td>Pilot light does not light.</td><td>13. Wired to wrong contact.
14. Loose connections.
5. Damaged snap switch.
20. Pilot light burned out.</td></tr>
<tr><td>Manual reset button will not reset switch.</td><td>5. Damaged snap switch.
21. System pressure has not changed by amount of differential of switch.</td></tr>
</table>

ROUTINE MAINTENANCE

During the normal maintenance period for the machine, the following steps should be followed for all pressure switches.

1. Remove cover
2. Visually inspect for leakage
3. Check all electrical connections
4. Check all pneumatic or hydraulic connections
5. If switch is used as a safety shutdown device, check setting
6. Replace cover

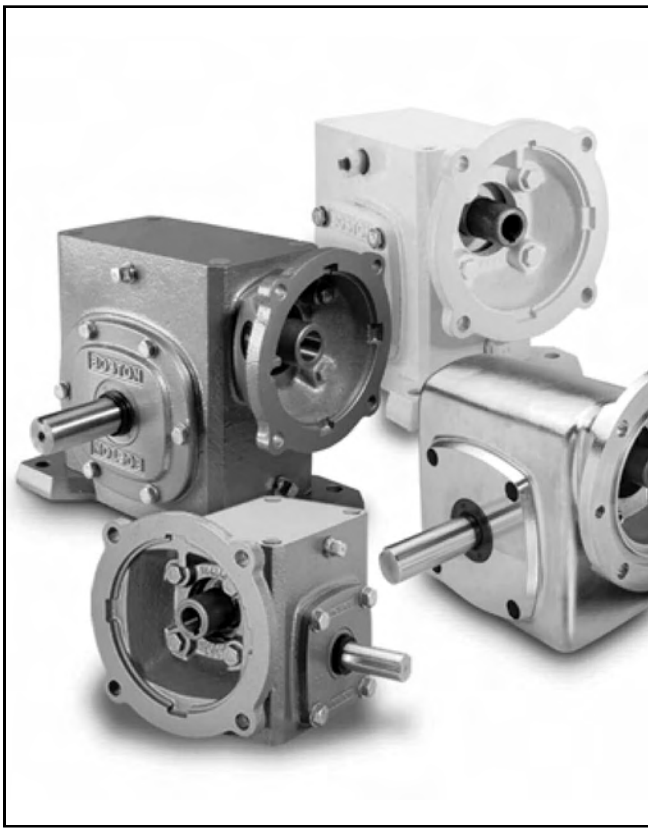
Boston Gear®

Worm Gear Speed Reducers

P-3009-BG *Installation and Operation*

Doc. No. 57746

700 Series
Single and Double
Reductions



Boston Gear

An Altra Industrial Motion Company

Inadequate lubrication is by far the largest contributor to premature failures of worm gear drives, applied within proper selection practice.

While lubricant selection is important to all gear reducers, it is critical for the worm gear type. Sliding action at the tooth mesh dictates use of a relatively high viscosity oil with special characteristics. A recent survey indicated improper lubricants were used in two-thirds of the applications. Although lightly loaded drives may survive, optimum performance is not obtained.

Boston Gear now offers Klubersynth UH1 6-460 Synthetic Lubricating Oil as a premium lubricant with many outstanding benefits for worm gear applications.

- Reduction in maintenance costs due to extended time between oil changes.
- Increased ratings providing for smaller drive selection or longer gear life.
- Lower energy consumption from improved efficiency.
- Broad ambient temperature range due to high viscosity temperature.
- Longer seal life, based on lower operating temperature.
- Multi-purpose application, including most other types of gear drives.

Lubrication Instructions

⚠ WARNING Boston Gear speed reducers are normally shipped without lubricant. They must be filled to the proper level with the recommended lubricant for your application before operation.

The recommended lubricant table indicates the type and viscosity of lubricant suitable for reducers operating at various temperatures.

Lubrication and maintenance instructions are provided with each speed reducer. These instructions should be followed for best results. It is important that the proper type of oil be used since many oils are not suitable for the lubrication of gears. Various types of gearing require different types of lubricants.

The lubricant must remain free from oxidation and contamination by water or debris, since only a very thin film of oil stands between efficient operation and failure. To assure long service life, the reducer should be periodically drained (preferably while warm) and refilled to the proper level with a recommended gear oil. Under normal environmental conditions oil changes are suggested after the initial 250 hours or every 6 months.

Synthetic lubricants will allow extended lubrication intervals due to its increased resistance to thermal and oxidation degradation. It is suggested that the initial oil change be made at 1500 hours and, thereafter, at 5000 hour intervals.

During the initial period of operation, higher than normal operating temperatures may be seen. This is due to the initial break-in of the worm gear set. The temperature of Double Reduction Worm Gear Reducers may reach 160°F and Single Reduction Worm Gear Reducers approximately 225°F.

These instructions must be read thoroughly before installing or operating speed reducers. File instructions for future reference and for ordering of replacement parts.

General Instructions

1. Align all shafts accurately. Improper alignment can result in failure. Use of flexible couplings is recommended to compensate for slight misalignment.

2. When mounting, use maximum possible bolt size and secure reducer to a rigid foundation. Periodic inspection of all bolts is recommended.

3. Auxiliary drive components (such as sprockets, gears and pulleys) should be mounted on the shafts as close as possible to the housing to minimize effects of overhung loads. Avoid force fits that might damage bearings or gears.

4. For hollow-shaft speed reducers, place speed reducer as close as possible to supporting bearing on drive shaft. Spot-drill driven shaft for setscrews in severe applications. See kit instructions for reaction rod assembly.

5. Check and record gear backlash at installation and again at regular intervals. This should be done by measuring the rotary movement of the output shaft (rotating alternately clockwise and counterclockwise) at a suitable radius while holding the input shaft stationary. Gears should be replaced when the backlash exceeds four times the measurement taken at installation.

6. Gear drives are rated for 1750 input RPM and Class I Service (Service Factor 1.0), using Klubersynth UH1 6-460 synthetic lubricant. For lower input speeds or for different service classes or lubricants, see catalog selection pages for rating information.

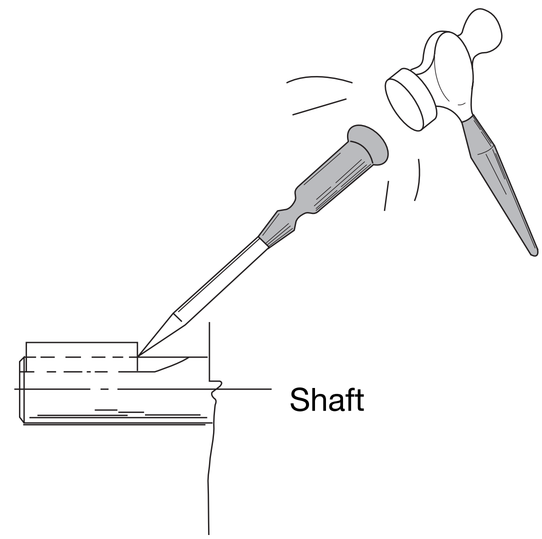
7. Initial operating temperatures may be higher than normal during the break-in period of the gear set. FOR MAXIMUM LIFE, DO NOT ALLOW THE SPEED REDUCER TO OPERATE CONTINUOUSLY ABOVE 225°F AT THE GEAR CASE. In the event of overheating, check for overloads or high ambient temperatures. Keep shafts and vent plugs clean to prevent foreign particles from entering seals or gear housing.

8. All reducers should be checked to see if they have been lubricated. Prelubed 700 Series reducers will have a solid plug in the vent hole which must be replaced by the vent plug at time of installation.

⚠CAUTION If the motor does not readily seat itself, check to determine if key has moved axially along motor shaft, causing interference. Staking of the keyway adjacent to the motor key will facilitate this procedure.

Key Staking Instructions

Lightly tap area of keyway adjacent to key. This will upset material and not allow key to move axially when assembling to speed reducer.



⚠CAUTION

- **For safe operation of any gear drive, all rotating shafts and auxiliary components must be shielded to conform with applicable safety standards. You must consider overall operational system safety at all times.**

- **When using a speed reducer to raise or lower a load, such as in hoisting applications, provision must be made for external braking. Under no conditions should a speed reducer be considered self-locking.**

- **Mounting of speed reducers in overhead positions may be hazardous. Use of external guides or supports is strongly recommended for overhead mounting.**

Instructions for Flanged Models

F700 (Quill Type Input)

1. Assemble the key to the motor shaft and coat the shaft with anti-seize compound. Insert the motor shaft into the reducer input shaft.

2. Rotate the motor to proper position and firmly secure to flange with four hex-head cap screws.

RF700 (Coupling Input – 3-Jaw Type)

1. Coat reducer input and motor shaft with anti-seize compound.
2. Position coupling half on input shaft with shaft flush to end of coupling bore.
3. Locate remaining half on motor shaft, with 1/32" clearance between jaw surfaces.
4. Tighten setscrews securely. For reversing applications, a thread-locking compound is recommended.
5. Install coupling insert and position motor. Rotate motor to proper position and firmly secure to flange.

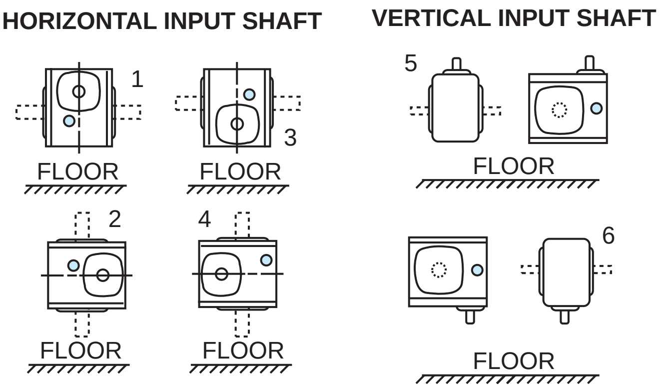
QC700 (Coupling Input-3-Jaw Quick Connect Type)

1. Coat motor shaft with anti-seize compound.
2. Install motor coupling half onto motor shaft. Use a straight edge to align coupling jaw top end flush with motor shaft except 738-B9 which will be flush with bottom of jaw. Secure with set screw.
3. Install urethane spider insert on motor coupling half.
4. Insert D-Bore coupling half into urethane spider element.
5. Rotate reducer input shaft so “milled flats” are either vertical or parallel. Rotate motor coupling D-Bore to match the reducer milled flats. Coat “D” flats with anti-seize compound furnished with speed reducer.
6. Insert motor assembly into reducer flange assembly. Minor rotating of the motor may be necessary to facilitate D-Bore alignment.
7. Once aligned, push motor towards reducer until properly seated against the face of the reducer flange.
8. Insert (4) hex head cap screws into the designated locations and securely tightened.

Oil Capacities

Single Reduction Models Only

Oil Levels for typical mounting positions



⚠CAUTION Avoiding those positions where the high speed oil seal is immersed in oil will provide greater security against high speed input seal wear.

Oil Capacity in Fluid Ounces

Unit Size	Positions				
	1	2	3	4	5 & 6
710	2.2	3.3	3.3	3.3	3.3
713	5.5	7.0	7.0	7.0	5.5
715	10.0	15.0	15.0	13.5	13.5
718	12.0	16.0	18.5	16.0	16.0
721	15.0	20.5	20.5	19.0	19.0
724	18.0	24.5	28.5	24.5	24.5
726	28.0	36.0	43.0	36.0	36.0
730	44.0	60.0	67.0	60.0	60.0
732	58.0	84.0	90.0	80.0	80.0
738	85.0	120.0	130.0	120.0	107.0
752	204.0	240.0	245.0	240.0	215.0
760	330.0	400.0	415.0	400.0	370.0

Note: SS718, SS721, SS726, SS732 models are filled to position one (1) only and are listed on Page 6.

Double Reduction Models

The variety of mounting possibilities for double reduction drives makes it impractical to illustrate positions for these models. In general, the vent filler is at the uppermost plug position, and the drain plug at the lowest possible position. The oil level must be at the approximate centerline of the uppermost gear.

Recommended Lubricants

Enclosed Worm Gear Reducers

Ambient (Room) Temperature 1.00 SF	Recommended Oil (or equivalent)	Viscosity Range SUS @ 100°F	Lubricant AGMA No.	ISO Viscosity Grade No. +
-30° to 225°F** (-34° to 107°C)	Klubersynth* UH1 6-460 Synthetic	1950/2500	-----	460
30° to 225°F (-34° to 107°C)	Mobil SHC634 Synthetic	1950/2500	-----	320/460

Worm Gear Lubricants Available from Boston Gear

Order By Item Code

Type	Klubersynth	Mobil SHC634
Size	QT.	QT.
Item Code	65159	51493

⚠CAUTION **Relubricate more frequently if drive operated in high ambient temperatures or unusually contaminated atmosphere. High loads and operating temperatures will also require more frequent lubrication.**

***Food Grade Synthetic recommendation is exclusively for Klubersynth UH1 6-460.**

+Other lubricants corresponding to AGMA/ISO numbers are available from all major oil companies.

****The synthetic lubricant will perform at temperatures considerably higher than 225°F. However, the factory should always be consulted prior to operating at higher temperatures as damage may occur to oil seals and other components.**

Lubricant Interchange

Lubricants are compounded for use in worm gears. Some contain non-corrosive, extreme pressure additives. DO NOT USE lubes that contain sulphur and/or chlorine which are corrosive to bronze gears. Extreme pressure lubes, in some cases contain materials that are toxic. Avoid use of these lubes where they can result in harmful effects. If in doubt, consult your lube supplier.

Manufacturer	Lubricant Name	AGMA Rating
Getty Refining Co.	Veedol Asreslube 98	8 EP
Getty Refining Co.	Veedol Asreslube 95	7 EP
Getty Refining Co.	Veedol Asreslube 90	6 EP
Lubrication Engr. Inc.	Almasol 609	8
Lubrication Engr. Inc.	Almasol 608	7
Mobil Oil Corp.	Mobilgear 634	8 EP
Mobil Oil Corp.	Mobil Extra Hecla Super	8
Mobil Oil Corp.	Mobil Cylinder 600W	7
Shell Oil Co.	Omala 460	7 EP
Shell Oil Co.	Valvala J460	7
Shell Oil Co.	Omala 680	8 EP
Shell Oil Co.	Valvala J680	8
Texaco Inc.	Meropa 680	8 EP
Texaco Inc.	Meropa 460	7 EP

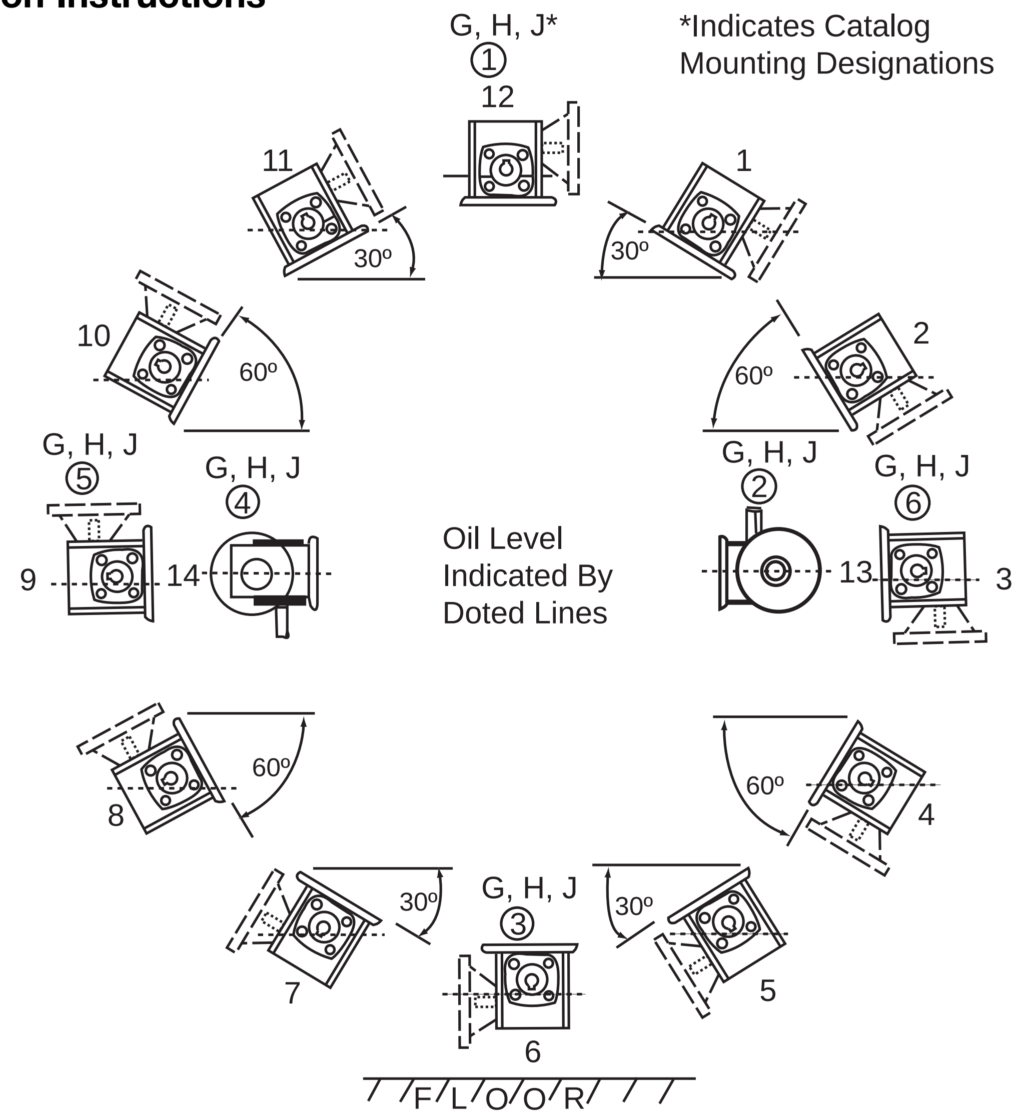
PosiVent® Option

Units supplied with this option are Lubed-for-Life and do not require oil change.

These units are factory filled with Klubersynth UH1 6-460 for universal mounting. To ensure that the system operates properly, DO NOT REMOVE THE VENT PLUG.

SS718-SS732 models are factory-filled with Klubersynth UH1 6-460 for universal mounting and are non-vented and sealed for life.

Lubrication Instructions

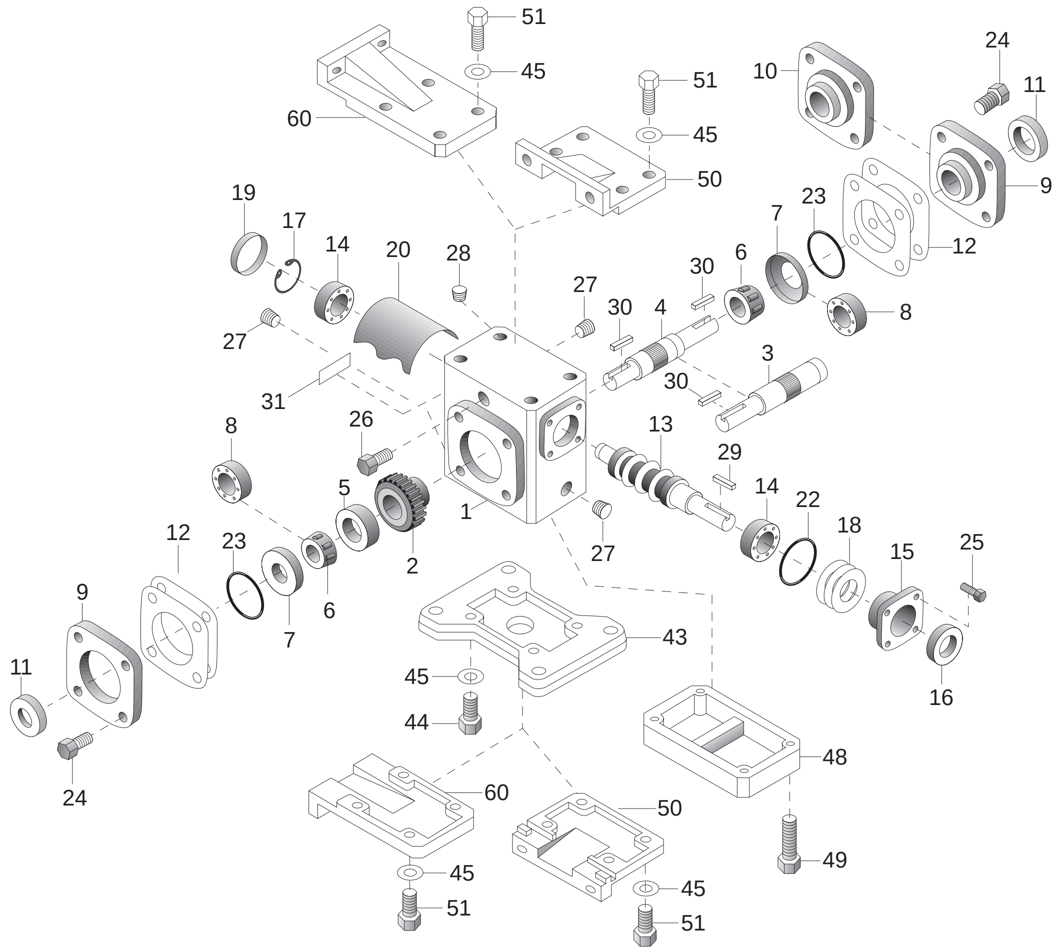


For Single Reduction Only: Refer To Lubrication & Installation Instructions

Oil Capacity In Ounces	Unit Size											
	Position	**710**	**713**	**715**	**718**	**721**	**724**	**726**	**730**	**732**	**738**	**752**	**760**
	1	3.00	6.37	9.30	12.85	14.04	16.97	28.50	42.60	56.80	71.60	162.0	255.7
	2	3.30	7.16	9.25	15.04	16.81	19.93	32.00	50.00	67.30	79.70	195.2	265.9
	3	3.30	5.50	13.50	16.00	19.00	24.50	36.00	60.00	80.00	107.0	215.0	370.0
	4	3.55	7.98	12.06	18.00	20.28	24.95	37.70	57.00	77.10	102.0	209.5	321.7
	5	3.98	8.18	12.30	18.69	21.45	26.95	40.00	60.40	80.90	106.7	192.0	357.0
	6	3.30	7.00	15.00	18.50	20.50	28.50	43.00	67.00	90.00	130.0	245.0	415.0
	7	3.98	8.18	12.30	18.69	21.45	26.95	40.00	60.40	80.90	106.7	192.0	357.0
	8	3.55	7.96	12.06	18.00	20.28	24.95	37.70	57.00	77.10	102.0	209.5	321.7
	9	3.30	5.50	13.50	16.00	19.00	24.50	36.00	60.00	80.00	107.0	215.0	370.0
	10	3.31	7.16	9.25	15.04	16.81	19.93	32.00	50.00	67.30	79.70	195.2	265.9
	11	3.00	6.37	9.30	12.85	14.04	16.97	28.50	42.60	56.80	71.60	162.0	255.7
	12	2.20	5.50	10.00	12.00	15.00	18.00	28.00	44.00	58.00	85.00	204.0	330.0
	13	3.30	7.00	15.00	16.00	20.00	24.50	36.00	60.00	84.00	120.0	240.0	400.0
	14	3.30	7.00	13.50	16.00	19.00	24.50	36.00	60.00	80.00	120.0	240.0	400.0

Notes: PosiVent® Models are factory-filled with Klubersynth UH1 6-460 for multi-positional mounting.
For Models SS718 through SS732, see page 22.

Parts List – Single Reduction Models



Part No.	Description
1	HOUSING
2*	WORM GEAR
3*	SINGLE PROJECTING OUTPUT SHAFT
4*	DOUBLE PROJECTING OUTPUT SHAFT
5*	GEAR SPACER
6*	OUTPUT BEARING (CONE) – MODELS 713-760
7	OUTPUT BEARING (CUP) – MODELS 713-760
8	OUTPUT BEARING – MODEL 710 ONLY
9	BEARING CARRIER (OPEN)
10	BEARING CARRIER (CLOSED)
11*	OUTPUT OIL SEAL
12*	ADJUSTMENT SHIMS
13	INPUT WORM SHAFT
14	INPUT BEARING – MODELS 710-730
15	INPUT BEARING RETAINER
16	INPUT OIL SEAL – MODELS 710-760
17	RETAINING RING
18	ADJUSTMENT SHIMS
19	BORE PLUG – MODELS 710-730
20	INTERNAL BAFFLE – MODELS 713-732
22	INPUT “O” RING
23*	OUTPUT “O” RING
24	HEX HEAD CAP SCREW
25	HEX HEAD CAP SCREW
26**	VENT PLUG - 2 PIECE

Part No.	Description
27	PIPE PLUG
28	PROTECTIVE CAP PLUG
29	INPUT KEY
30	OUTPUT KEY
31	NAMEPLATE
32	INPUT BEARING (CUP) – MODELS 732-760
33	INPUT BEARING (CONE) – MODELS 732-760
34	GREASE CUPS – MODELS 732-760
35	HEX HEAD CAP SCREW
37	OUTPUT SHAFT KEY – MODELS 730-760
38	RETAINING RING – MODELS 710-738
39	MOTOR SHAFT – MODELS 710-738
40	MOTOR FLANGE – MODELS 710-738
41	OIL SEAL – MODELS 710-738
42	HEX HEAD CAP SCREW
43	HORIZONTAL BASE
44	HEX HEAD CAP SCREW
45	LOCKWASHER
46	2 PIECE FC COUPLING – WITH INSERT
47	RETAINING MOTOR FLANGE
48	RISER BLOCK
49	HEX HEAD CAP SCREW
50	VERTICAL BASE (HIGH OR LOW)
51	HEX HEAD CAP SCREW

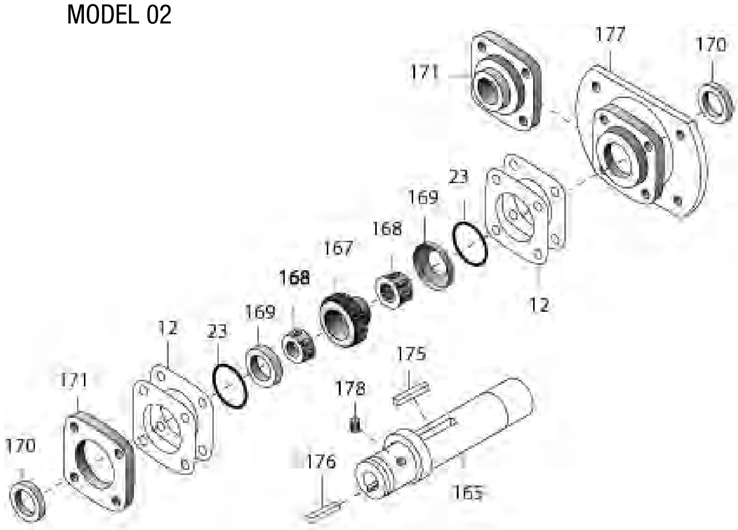
Part No.	Description
60	VERTICAL BASE (X & Y ASSEMBLY)
101	FAN
102	SPACER
103	HEX HEAD CAP SCREW
104	FAN GUARD
105	HEX HEAD CAP SCREW
106	WASHER
165	HOLLOW OUTPUT SHAFT (S VERSION ONLY)
166	HOLLOW OUTPUT SHAFT (H VERSION ONLY)
167	WORM GEAR
168	OUTPUT BEARING (CONE)
169	OUTPUT BEARING (CUP)
170	OIL SEAL
171	BEARING CARRIER
172	HOLLOW SHAFT MTG. BRACKET
173	HEX HEAD CAP SCREW
174	LOCKWASHER
175	KEY (INTERNAL)
176	KEY (EXTERNAL)
177	“V” TYPE BASE-MODEL 718, 721, 726, 732)
178	SOCKET SETSCREW

*For Models 710 to 726, these parts are available as complete assemblies. See Part Ordering Information, page 8.
**Extension not required on single reduction Models 713 through 732.

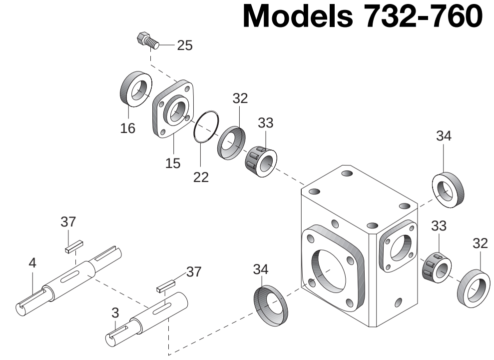
Options & Accessories – Single Reduction Models

Hollow Output Shaft Models S and SF718-732*

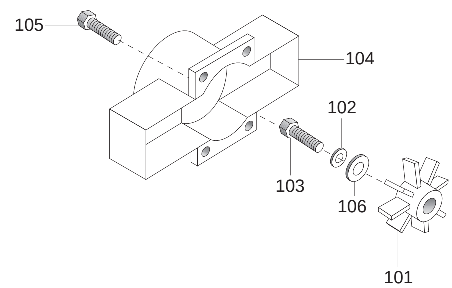
MODEL 02



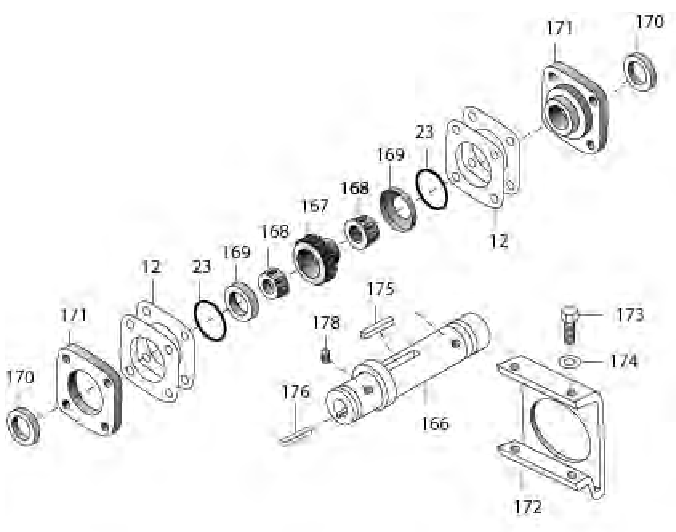
Models 732-760



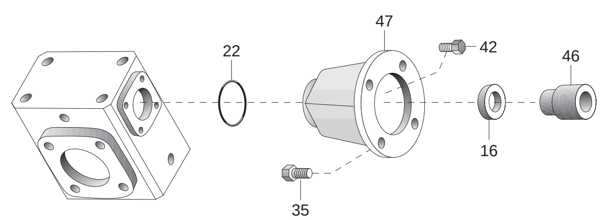
Fan kit for models 732-760



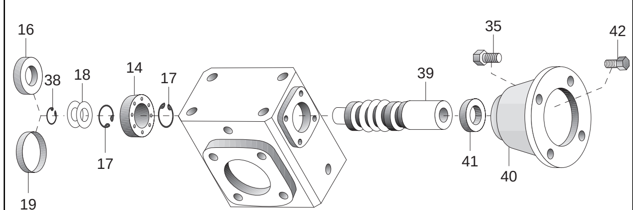
HOLLOW OUTPUT Shaft Models H, HF, and HQC713-738



Models QC710-QC738, RF752-RF760



Models F710-F738

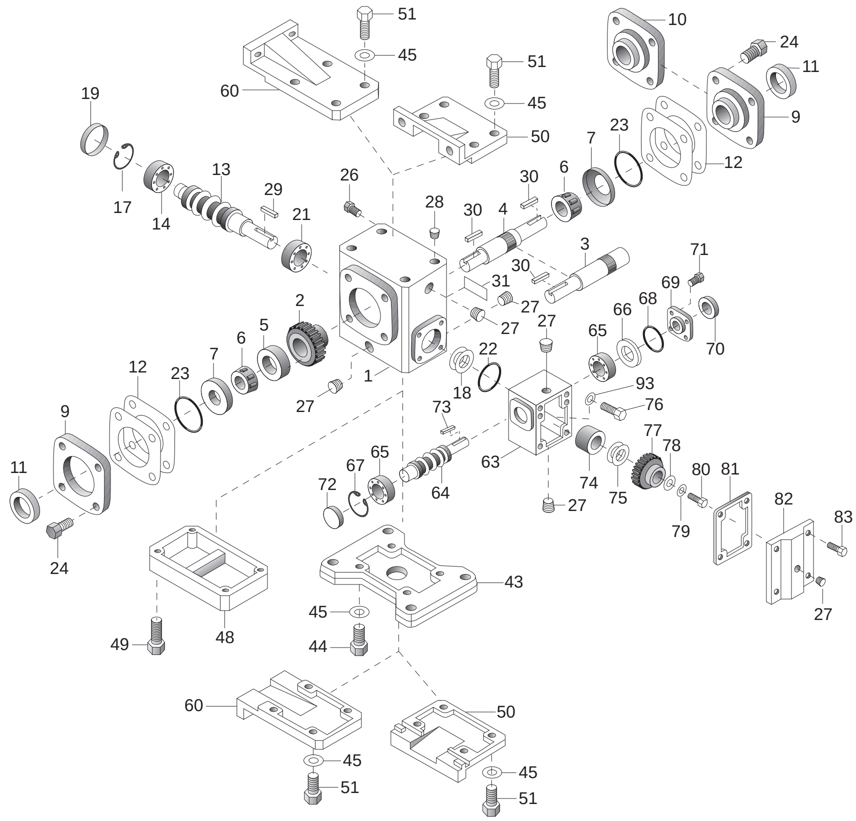


Part ordering information

1. Be sure to provide complete Boston Gear catalog number from speed reducer nameplate, along with part description and number. For example, "One output oil seal, Part No. 11, for RF718-30-B5-G".
2. Output shaft components for Boston Gear models 710 through 726 are available only as complete assemblies that include Parts 2, 3, 5, 6, 11, 12 and 23 for single projecting shafts; and Parts 2, 4, 5, 6, 11, 12 and 23 for double projecting shafts. When ordering, specify "output shaft assembly" and full Boston Gear catalog number from nameplate.

* Not available in the 730 center distance, see H series.

Parts List – Double Reduction Models



Part No.	Description
1	HOUSING
2*	WORM GEAR
3*	SINGLE PROJECTING OUTPUT SHAFT
4*	DOUBLE PROJECTING OUTPUT SHAFT
5*	GEAR SPACER
6*	OUTPUT BEARING (CONE)
7	OUTPUT BEARING (CUP)
9	BEARING CARRIER (OPEN)
10	BEARING CARRIER (CLOSED)
11*	OUTPUT OIL SEAL
12*	ADJUSTMENT SHIMS
13	INTERMEDIATE WORM SHAFT
14	INTERMEDIATE BEARING–MODELS W713-W730
15	INTER. BEARING RETAINER–MODELS W732-W760
16	INTER. OIL SEAL – MODELS W732-W760
17	RETAINING RING – MODELS W713-W730
18	ADJUSTMENT SHIMS
19	BORE PLUG – MODELS W713-W730
21	INTERMEDIATE BEARING
22	INTERMEDIATE “O” RING
23*	OUTPUT “O” RING
24	HEX HEAD CAP SCREW
25	HEX HEAD CAP SCREW
26	VENT PLUG – 2 PIECE
27	PIPE PLUG
28	PROTECTIVE CAP PLUG
29	INTERMEDIATE KEY
30	OUTPUT KEY

Part No.	Description
31	NAMEPLATE
32	INTER. BEARING (CUP) – MODELS W732-W760
33	INTER. BEARING (CONE) – MODELS W732-W760
34	GREASE CUPS – MODELS W732-W760
35	HEX HEAD CAP SCREW
37	OUTPUT SHAFT KEY – MODELS W730-W760
43	HORIZONTAL BASE
44	HEX HEAD CAP SCREW
45	LOCKWASHER
48	RISER BLOCK
49	HEX HEAD CAP SCREW
50	VERTICAL BASE (HIGH OR LOW)
51	HEX HEAD CAP SCREW
60	VERTICAL BASE (ASSEMBLY X & Y)
63	ATTACHMENT HOUSING
64	INPUT WORM SHAFT
65	INPUT BEARING
66	ADJUSTMENT SHIMS
67	RETAINING RING
68	“O” RING
69	BEARING RETAINER
70	OIL SEAL
71	HEX HEAD CAP SCREW
72	BORE PLUG – MODELS W713-W738
73	INPUT WORM SHAFT KEY
74	GEAR SPACER
75	ADJUSTMENT SHIMS
76	HEX HEAD CAP SCREW
77	INTERMEDIATE WORM GEAR

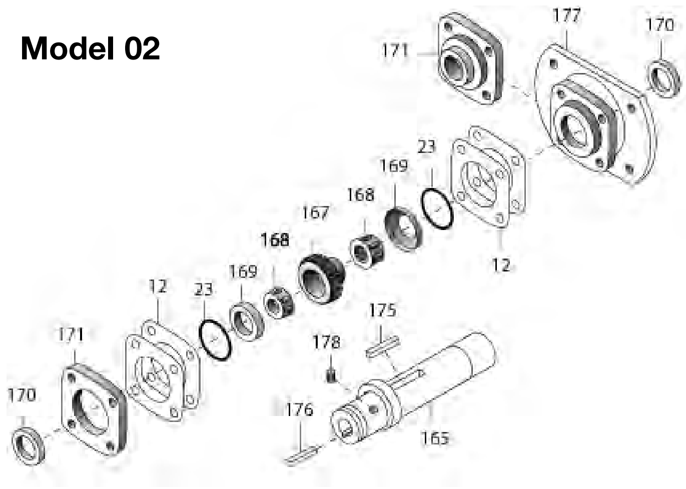
Part No.	Description
78	WASHER
79	LOCKWASHER
80	HEX HEAD CAP SCREW
81	ATTACHMENT COVER GASKET
82	ATTACHMENT COVER
83	HEX HEAD CAP SCREW
84	INPUT BEARING (CONE) – MODEL W760 ONLY
85	INPUT BEARING (CUP) – MODEL W760 ONLY
86	TWO PIECE FC COUPLING WITH INSERT
87	MOTOR FLANGE
88	HEX HEAD CAP SCREW
89	MOTOR FLANGE
90	INPUT WORM SHAFT
91	EXTERNAL RETAINING RING
92	OIL SEAL – MODELS FW713-FW738
93	WASHER
165	HOLLOW OUTPUT SHAFT (S VERSION ONLY)
166	HOLLOW OUTPUT SHAFT (H VERSION ONLY)
167	WORM GEAR
168	OUTPUT BEARING (CONE)
169	OUTPUT BEARING (CUP)
170	OIL SEAL
171	BEARING CARRIER
172	HOLLOW SHAFT MTG. BRACKET
173	HEX HEAD CAP SCREW
174	LOCKWASHER
175	KEY (INTERNAL)
176	KEY (EXTERNAL)
177	“V” TYPE BASE MODEL (718, 721, 726, 732)
178	SOCKET SETSCREWS

*For Models 710 to 730, these parts are available as complete assemblies. See Part Ordering Information, Page 10.

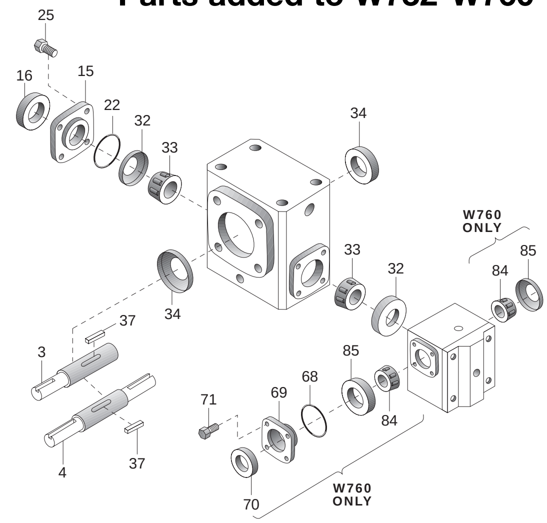
Options & Accessories – Double Reduction Models

Hollow Output Shaft Models SW, SFW, and SRFW718-732*

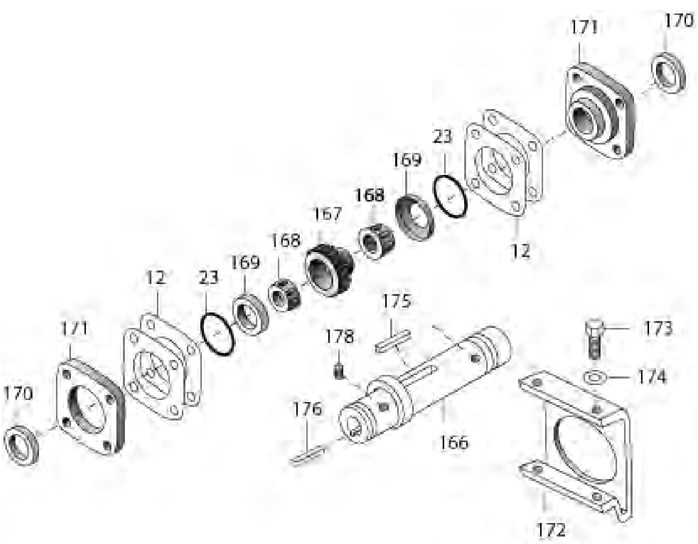
Model 02



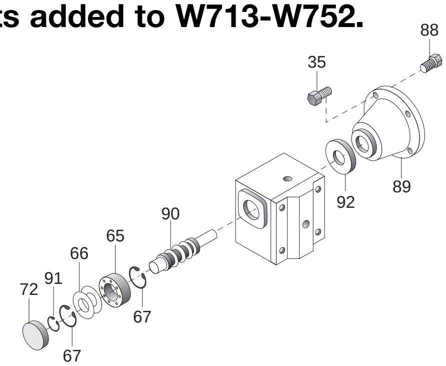
Models W732-W760 Parts added to W732-W760



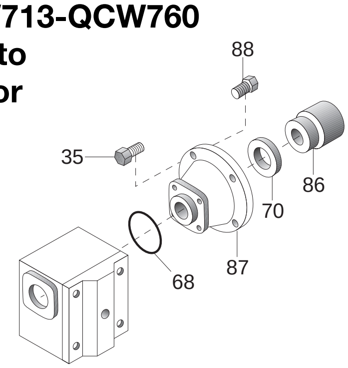
Hollow Output Shaft Models HW, HFW, and HQCW713-738



Models FW713-FW752 Parts added to W713-W752.



Models QCW713-QCW760 Parts added to W713-W726 or W732-W760. These parts available in kit form.



Part ordering information

1. Be sure to provide complete Boston Gear catalog number from speed reducer nameplate, along with part description and number. For example, "One output oil seal, Part No. 11, for W713-150-G".
2. Output shaft components for Boston Gear models 710 through 726 are available only as complete assemblies that include Parts 2, 3, 5, 6, 11, 12 and 23 for single projecting shafts; and Parts 2, 4, 5, 6, 11, 12 and 23 for double projecting shafts. When ordering, specify "output shaft assembly" and full Boston Gear catalog number from nameplate.

* Not available in 730 center distance, see H series.

F700 Series

Disassembly and Reassembly Procedures

(For Item Identification, Refer To Exploded View)

Output Shaft Disassembly

1. Remove vented filler (Item 26), and the most convenient pipe plug (Item 27) and completely drain oil.

2. Remove bearing carrier screws (Item 24) from projecting shaft bearing carrier (Item 9). Remove carrier by CAREFULLY sliding it over the projecting shaft diameter.

3. Output shaft assembly (Items 2, 3, 5 & 6) can now be removed from the unit. Exercise care not to nick or scratch worm gear or shaft diameters.

4. Visually examine the output shaft assembly. Check tapered roller bearings (Item 6) for signs of metallic contamination or discoloration. Rollers should have continuously smooth action and should not bind or exhibit “flat-spots”.

5. Note that replacement parts for output gear (Item 2) will include an output shaft assembly (Items 3 OR 4) for sizes F710 through F726. Also included are items 5, 6, 11, 12 and 23.

Input Shaft Disassembly

Models F710 through F730

1. With a screwdriver or suitable tool, pierce the bore plug (Item 19) and remove from the housing bore.

Models F732 and F738 only

2. Remove fan guard (Item 104). Remove fan retaining screw (Item 103), fan (Item 101) and spacer. With a screw-driver or suitable tool, pierce and fan end oil seal (Item 16) and remove from the housing bore.

3. Remove the outboard retaining ring (Item 17) from the housing bore. Remove the metal shims (Item 18), located between the retaining ring and the ball bearing (Item 14).

4. Remove four (4) screws (Item 42) from the motor flange (Item 40) and remove the flange from the housing.

5. Remove the input worm assembly (Items 14, 38 and 39) through the bore opposite the flange side. Remove the oil seal (Item 41) from the housing.

6. Check the condition of the ball bearing (Item 14). The bearing should roll smoothly and not bind. If the bearing needs replacement, remove the snap ring (Item 38) and press the shaft through the bearing. Install new bearing onto the shaft and re-assemble the snap ring. If the bearing is not pre-packed with grease, pack at least 50% full with Mobilux EP #2 All Purpose Grease or equivalent.

Input Shaft Reassembly

Models F710 through F738

1. Insert input worm assembly in the housing. Seat the ball bearing against the inner retaining ring (Item 17).

2. Install the metal shims (Item 18) and assemble outboard snap ring (Item 17).

3. Clean the housing bore(s) in the area where oil seals will be installed.

4. Coat the oil seals as follows:

   A. Rubber Clad Oil Seals - Apply ALL-PURPOSE grease (NLGI #2 consistency) to the I.D.

   B. Steel Clad Oil Seals - Apply ALL-PURPOSE grease to the ID and coat the bore evenly with Permatex Form-A-Gasket #3 sealant or equivalent.

5. Insert the new oil seal (Item 41) over the shaft until it contacts the housing. CARE MUST BE TAKEN NOT TO DAMAGE THE OIL SEAL LIP.

6. With a small hammer, tap around the face of the oil seal casing - near the outside diameter. Locate the seal as shown in Figure 2 Page 19. Use a suitable driving tool.

Model F710 through F730

7. Install a new bore plug (Item 19). Coat the bore with PERMATEX FORM-A-GASKET #3 OR EQUIVALENT SEALANT. Using a small hammer, lightly tap around the plug face near the outside diameter. CAUTION should be exercised not to distort or cock the bore plug.

Model F732 and F738

8. Install new oil seal on the fan end (Item 16). With a small hammer, tap around the face of the oil seal casing - near the outside diameter. Use a suitable tool to assure squareness of the seal to the bore. Drive the seal flush to 1/16th inch projection. Reinstall fan spacer, fan, washer, lockwasher, and bolt. Tighten the bolt per the chart in Figure 5, Page 20. Reinstall the fan guard and tighten the screws per chart in Figure 5.

Worm Gear Replacement

Model F710 through F726

1. Replace the entire output assembly.

Model F730 through F760

2. Place the output gear assembly into a pressing fixture and remove the worm gear from the shaft.

3. With the gear key assembled in the shaft key seal, press the new gear onto the shaft to the dimension shown in Figure 1, Page 19.

4. Install the shaft spacer (Item 5) and grease cups (Item 34).

5. Press the bearing cones (Item 6) onto the shaft making sure that the assembly is tight (no space between items).

6. Remeasure from the end of the shaft to the gear face as shown in Figure 1, Page 19. Readjust, if necessary, by pressing on the bearing cone until desired dimension is achieved.

7. Where grease cups are used, pack with Mobilux EP #2 All Purpose Grease or equivalent.

Output Shaft reassembly into Housing

1. Remove the existing oil seal (Item 11) from the bearing carrier (Item 9).

2. Coat the gear teeth with blue or red Dykme mixture or similar coating and install the output assembly into the housing.

3. Slide the bearing carrier over the projecting shaft and bolt the carrier to the housing, making sure the metal shims (Item 12) are between the carrier and the housing. Rotate the input shaft to properly seat the tapered bearings.

4. Shim adjustment must be made at this time. If the output shaft is excessively loose, measure the endplay of the output shaft and remove shims evenly from BOTH carriers until the endplay is within the limits shown in Figure 3, Page 20.

   If the carrier does NOT meet the housing face, measure the gap and add shims evenly to BOTH carrier locations until the endplay is within the limits specified.

5. ASSEMBLE THE MOTOR TO THE UNIT. Check the worm gear centrality. Apply a slight load on the output shaft then rotate the input shaft for one to two minutes. Remove the bearing carrier and remove the gear shaft assembly. Check the gear teeth for correct bearing pattern as shown in Figure 6, Page 21.

6. If adjustment is required, all adjustments MUST be made to the carrier which is located on the side OPPOSITE the GEAR HUB. All adjustment for centrality must be made from the TOTAL shim pack which has already been determined.

Example: If a shim is REMOVED from the centralized side, it must be ADDED to the opposite side.

7. When a good bearing pattern has been established, assemble carrier(s) to the unit. Replace all O-rings (Item 23) where required. (Install all shims on the carriers BEFORE installing O-rings.) If replacement O-rings are not available, apply a large bead of Permatex FORM-A-GAS-KET #3 or equivalent on the pilot diameter and flange.

8. Install a new oil seal (Item 11) into the carrier bore. Use procedures as described above. Press the seal flush to 1/16th inch projecting. Install the carrier over the projecting shaft (recommend placing masking tape over the sharp edges of the shaft keyseat to prevent cutting the oil seal lip).

9. Tighten all screws as shown in Figure 5, Page 20.

10. Fill the unit to the proper oil level with the recommended lubricant.

11. Install the vent plug (Item 26) and the drain pipe plug (Item 27).

700 and RF700 Series Flanged

Disassembly and Reassembly Procedures

(For Item Identification, Refer To Exploded View)

Output Shaft Disassembly

1. Remove vented filler (Item 26) and the most convenient pipe plug (Item 27) and completely drain oil.

2. Remove bearing carrier screws (Item 24) from projecting shaft bearing carrier (Item 9). Remove carrier by CAREFULLY sliding it over the projecting shaft diameter.

3. Output shaft assembly (Items 2, 3, 5 & 6) can now be removed from the unit. Exercise care not to nick or scratch worm gear or shaft diameters.

4. Visually examine the output shaft assembly. Check tapered roller bearings (Item 6) for signs of metallic contamination or discoloration. Rollers should have continuously smooth action and should not bind or exhibit “flat-spots”.

5. Note that replacement parts for output gear (Item 2) will include an output shaft (Items 3 OR 4) for sizes RF/710 through RF/726.

Input Shaft Disassembly

Model RF/710 through RF/730

1. With a screwdriver or suitable tool, pierce the bore plug (Item 19) and remove from the housing.

2. Remove the snap ring (Item 17) from the housing. Remove the retainer at the projecting shaft (Item 15) and the shims (Item 18).

3. With a soft mallet, tap lightly on the projecting shaft, removing the shaft assembly from the bore plug end.

4. Check the condition of the ball bearings (Item 14). The bearing should roll smoothly and not bind. If the bearing needs replacement, press the shaft through the bearing. Install new bearings on the shaft. Install bearings so that the shields face inward. If the bearing is not prepacked with grease, pack at least 50% full with Mobilux EP #2 All Purpose Grease or equivalent.

Model RF/732 through RF/760

5. Remove the fan guard (Item 104). Remove the fan retaining screw (Item 103), fan (Item 101), and spacer. Remove the retainer(s) (Item 15) at both ends, if applicable or remove the motor flange (Item 47) and shims (Item 18) and O-ring (Item 22).

6. With a soft mallet, tap lightly on the projecting shaft, removing the bearing cup and input shaft through the fan end.

7. Remove the bearing cup from the projecting shaft end.

8. Check the conditions of the bearing cones and cups (Items 32 and 33). The rollers should not exhibit pitting. The cage should show no wear or distortion. The bearings, when supported in the cups, should run smoothly with no binding.

9. If the bearings need replacement, press the shaft through the bearing. Hand-pack the new cones with grease and install onto the shafts.

Input Shaft Reassembly

Model RF/710 through RF/730

1. Install the retaining ring (Item 17) in the outboard housing groove. Install the input shaft assembly through the projecting shaft end of the housing. Tap lightly to seat the bearing against the snap ring retainer.

2. Remove the old oil seal (Item 16) from the bearing retainer (Item 15). Using a small hammer, install a new oil seal by tapping around the face of the seal casing - near the outside diameter. The seal should be flush to 1/16th inch projection above the retainer surface.

3. Install the bearing cup (Item 32) and shims (Item 18), the bearing retainer (Item 15) and the O-ring (Item 22). If no replacement O-ring is available, apply a heavy bead of PERMATEX FORM-A-GASKET #3 or equivalent in the housing groove and also form a fillet at the retainer pilot diameter and flange. It is recommended to protect the oil seal lip by using masking tape on the keyseat edges.

Model RF/732 through RF/760

4. Install a new oil seal in the far retainer (Item 16). With a small hammer, tap around the face of the oil seal casing - near the outside diameter. Use a suitable tool to assure squareness of the seal to the bore. Drive the seal flush to 1/16th inch projection.

5. Repack the roller bearings (Items 32 and 33) or replace if required.

6. Reinsert the input shaft assembly, assuring that the far bearing cup (Item 32) is in place. Lightly tap the end of the projecting shaft to seat the bearing.

7. Insert the front bearing cup (Item 32) and the shim pack. Replace the front oil seal in the retainer as per step #4 above.

8. Reinstall the front retainer using the O-ring or Permatex. Tighten all bolts per table Figure 5, Page 20.

9. Rotate the shaft several times to seat the bearings and check for shaft endplay per Figure 4, Page 20. Adjust if required.

10. Replace fan and fan guard. Tighten bolts per Figure 5, Page 20.

Worm Gear Replacement

Model RF/710 through RF/726

1. Replace the entire output assembly.

Model RF/730 through RF/760

2. Place the output gear assembly into a pressing fixture and remove the worm gear from the shaft.

3. With the gear key assembled in the shaft key seat, press the new gear onto the shaft to the dimension shown in Figure 1, Page 19.

4. Install the shaft spacer (Item 5) and grease cups (Item 34).

5. Press the bearing cones (Item 6) onto the shaft making sure that the assembly is tight (no space between items).

6. Remeasure from the end of the shaft to the gear face as shown in Figure 1, Page 19. Readjust, if necessary, by pressing on the bearing cone until desired dimension is achieved.

7. Where grease cups are used, pack with Mobilux EP #2 All Purpose Grease or equivalent.

Output Shaft Reassembly into Housing

1. Remove the existing oil seal (Item 11) from the bearing carrier (Item 9).

2. Coat the gear teeth with blue or red Dykme mixture or similar coating and install the output assembly into the housing.

3. Slide the bearing carrier over the projecting shaft and bolt the carrier to the housing, making sure the metal shims (Item 12) are between the carrier and the housing. Rotate the input shaft to properly seat the tapered bearings.

4. Shim adjustment must be made at this time. If the output shaft is excessively loose, measure the endplay of the output shaft and remove shims evenly from BOTH carriers until the endplay is within the limits shown in Figure 4. If the carrier does NOT meet the housing face, measure the gap and add shims evenly to BOTH carrier locations until the endplay is within the limits specified.

5. Check the worm gear centrality. Apply a slight load on the output shaft and rotate the input shaft for one to two minutes. Remove the bearing carrier and remove the gear shaft assembly. Check the gear teeth for correct bearing pattern as shown in Figure 6, Page 21.

6. If adjustment is required, all adjustments MUST be made to the carrier which is located on the side OPPOSITE the GEAR HUB. All adjustment for centrality must be made from the TOTAL shim pack which has already been determined.

Example: If a shim is REMOVED from the centralized side, it must be ADDED to the opposite side.

7. When a good bearing pattern has been established, assemble carrier(s) to the unit. Replace all O-rings (Item 23) where required. (Install all shims on the carriers BEFORE installing O-rings). If replacement O-rings are not available, apply a large bead of Permatex FORM-A-GAS-KET #3 or equivalent on the housing face and also form a fillet at the carrier pilot diameter and flange.

8. Install a new oil seal (Item 11) into the carrier bore. Use procedures as described above. Press the seal flush to 1/16th inch projecting. Install the carrier over the projecting shaft (recommend placing masking tape over the sharp edges of the shaft keyseat to prevent cutting the oil seal lip).

9. Tighten all screws as shown in Figure 5, Page 20.

10. Fill the unit to the proper oil level with the recommended lubricant.

11. Install the vent plug (Item 26) and the drain pipe plug (Item 27).

FW713 - FW752 Series

Disassembly and Reassembly Procedures

(For item identification, refer to exploded view)

FW713 - FW752 Series

Output Shaft Disassembly

1. Remove vented filler (Item 26), and the most convenient pipe plug (Item 27) and completely drain oil.

2. Remove bearing carrier screws (Item 24) from both bearing carriers (Items 9 & 10). Remove both carriers.

Note: Carefully slide open carrier (Item 9) over projecting shaft diameter.

3. Output shaft assembly (Items 2, 3, 5 & 6) can now be removed from unit. Exercise care not to nick or scratch worm gear or shaft diameters.

4. Output shaft assembly can now be visually examined. Check tapered roller bearings (Item 6) for signs of any metallic contamination or discoloration. Rollers should have continuously smooth action and should not bind or exhibit “flat-spots”.

Intermediate Worm Shaft Disassembly

Models FW713 - FW730

1. With a screwdriver or other similar tool, pierce input bore plug (Item 19) and remove from housing bore.

Models FW732 - FW752

Remove bearing retainer screws (Item 25) and remove bearing retainer (Item 15).

Models FW713 - FW752

2. Remove outboard retaining ring (Item 17) from housing bore.

Models FW713 - FW738

3. Remove four (4) screws (Item 83) from attachment cover (Item 82) and remove from housing.

4. Remove screw (Item 80), lock washer (Item 79) and washer (Item 78).

5. Remove intermediate worm assembly (Item 13) through bore opposite attachment housing.

Models FW713 - FW730

6. Check condition of ball bearings (Items 14 & 21). Bearings should roll smoothly and not bind. If bearings need replacement, press shaft through bearings. Install new bearings onto shaft. If not already packed with grease, bearings should be packed at least 50% full with Mobilux EP #2 All Purpose Grease or equivalent.

Models FW732 - FW752

Check tapered roller bearings (Items 32 & 33) for signs of any metallic contamination or discoloration. Rollers should have continuously smooth action and should not bind or exhibit “flat-spots”. Repack bearings with Mobilux EP #2 All Purpose or equivalent.

Intermediate Worm Gear

Now that intermediate shaft has been removed, intermediate gear (Item 77) may be removed from attachment housing. Shims (Item 75) and gear spacer (Item 74) should be set aside for reassembly.

Note: Do not attempt to remove intermediate worm gear (Item 77) prior to removal of intermediate shaft as gear teeth will be damaged from being pried out from under the engaged worm threads.

Input Shaft Disassembly

1. With a screwdriver or other similar tool, pierce input bore plug (Item 72) and remove from housing bore.

2. Remove outboard retaining ring (Item 67) from housing bore. Remove metal shims (Item 66) located between snap ring and ball bearing (Item 65).

3. Remove four (4) screws (Item 88) from motor flange (Item 89) and remove flange from housing.

4. Remove input worm assembly (Item 90) through bore opposite flange side. Remove oil seal (Item 92) from housing bore. Inboard retaining ring (Item 67) will remain in housing.

5. Check condition of ball bearing (Item 65). Bearing should roll smoothly and not bind. If bearings need replacement, remove snap ring (Item 91) and press shaft through bearing.

Install new bearing onto shaft and reassemble snap ring (Item 91). If not already packed with grease, bearing should be packed at least 50% full with Mobilux EP #2 All Purpose Grease or equivalent.

Input Shaft Reassembly

1. Insert input worm shaft assembly (Item 90) into housing with retaining ring (Item 67) used to seat ball bearing.
2. Install metal shims (Item 66) and assemble outboard snap ring (Item 67).
3. Clean housing bore(s) in area where oil seal is to be inserted.
4. Oil Seal Assembly:

   A. Rubber Clad Oil Seal

      Apply All Purpose Grease (NLGI #2 consistency) to I.D. and O.D.

   B. Steel Clad Oil seal

      Lightly apply All Purpose Grease to I.D. only and coat bore evenly with Permatex "Aviation FORM-A-GASKET" #3 or equivalent.
5. Insert new oil seal (Item 92) over the shaft (care must be taken not to damage oil seal lip) until it contacts the housing.
6. With small hammer, tap around the face of seal casing near the outside diameter. Oil seal location as follows:

Flange end oil seal - Refer to Figure (2) Page 19 (use suitable driving tool to recess seal).

7. Install new bore plug (Item 72). Coat bore with PERMATEX "Aviation FORM-A-GASKET" #3 or equivalent sealant. If rubber clan O.D. no sealant is required. Using small hammer, lightly tap around plug face near the outside diameter. Caution should be exercised not to distort or cock plug during installation.
8. Assemble motor flange (Item 89).

Intermediate Worm Reassembly

1. Insert intermediate worm assembly (Item 13) through bore opposite attachment housing.

Note: Spacer (Item 74) and shims (Item 75) should be put on shaft and worm gear (Item 77) held in mesh with input worm (Item 90) while sliding intermediate worm assembly into position.

2. Assemble attachment cover (Item 82).

Models FW713 - FW730

3. Install new bore plug (Item 19). Coat bore with PERMATEX "Aviation FORM-A-GASKET" #3 or equivalent sealant. If rubber clad O.D. no sealant is required. Using small hammer, lightly tap around plug face near the outside diameter. Caution should be exercised not to distort or cock plug during installation.

Models FW732 - FW752

Assemble intermediate bearing retainer (Item 15).

Worm Gear Replacement

Models FW713 - FW726

1. Replace entire output assembly.

Models FW730 - FW752

1. Place output gear assembly (Item 2) into a pressing fixture and remove worm gear from shaft.
2. With gear key assembled in shaft keyseat, press new gear onto shaft to dimension shown in Figure 1, Page 19.
3. Install shaft spacer (Item 5) and grease cups (Item 34) when applicable.
4. Press bearing cones (Item 6) onto shaft making sure the assembly is tight.
5. Remeasure from end of shaft to worm gear face as shown in Figure 1, Page 19. If adjustment is necessary, press bearing cone (Item 6) until required dimension is achieved.
6. Where grease cups are used, pack with Mobilus EP #2 All Purpose Grease or equivalent.

Output Shaft Reassembly into Housing

1. Remove existing oil seal (Item 11) from bearing carrier (Item 9).

2. Coat gear teeth (Item 3) with red-lead mixture or similar coating and install output gear assembly into housing.

3. Slide bearing carrier (Item 9) over projecting shaft (Item 4) diameter and bolt carrier to housing. Make sure metal shims (Item 12) are between carrier and housing face. Rotate input shaft to properly seat tapered roller bearings.

4. Adjustments of shims (Item 12) must be made at this time. If output shaft is excessively loose, measure endplay of output shaft and remove shims (Item 12) evenly from both carriers (Items 9 and 10) until endplay is within limits specified on Figures 3 or 4, Page 20.

   If bearing carrier (Item 9) does not meet housing face, measure gap and add shims (Item 12) evenly to each side (Items 9 and 10).

5. Assemble motor to unit. Check worm gear centrality. Apply slight load to output shaft and rotate input shaft for 1 or 2 minutes. Remove output bearing carrier (Item 9) and remove output gear assembly. Check gear teeth for bearing pattern. Optimum bearing pattern is shown in Figure 3, Page 20.

   If gear requires adjustment for centrality, all adjustments must be made from side opposite gear hub.

Note: All adjustment for gear centrality to be made from the already established total shim pack (Ref. - Step 4). For example - If a shim is removed from centralized side, it must be added to the opposite side.

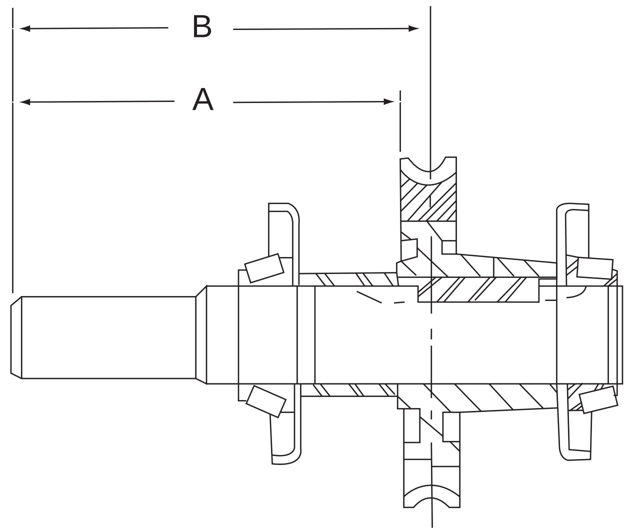
6. When good bearing pattern or gear teeth is established, assemble carriers(s) to unit. Apply PERMATEX “Aviation FORM-A-GASKET” #3 or equivalent sealant to form fillet at pilot diameter of carrier. Install new oil seal (Item 12) into bearing carrier bore. Seal assembly same as shown in “Input Shaft Reassembly” - Steps 4, 5 and 6. (Recommend masking tape over sharp keyseat edges so seal lip is not cut or damaged.) Oil seal to be flush with carrier face.

7. Fill unit to proper level with recommended gear lubrication. (Ref. Page 5.)

8. Install vent (Item 26) and pipe plug (Item 27).

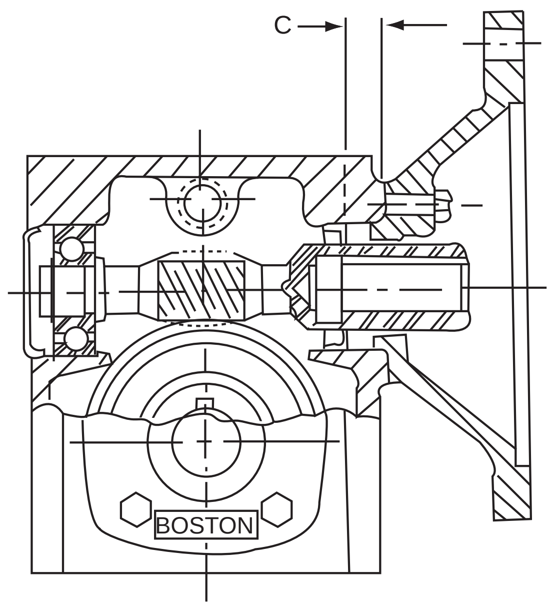
Important: All screw tightening torques listed in Figure 5, Page 20.

Note: For RFW and W700 Series Ref. to Single Reduction Instructions for 700/RF Series Input Disassembly and reassembly Page 13.



Unit	"A" ± 1/64	"B" Ref. Only
730	6.310	6-3/4
732	7.469	7-1/16
738	7.25	7-3/4
752	8.44	9-1/16
760	9.28	10

Figure No. 1



Unit Size	"C" Dimension
F710/FW713	5/16
FW718	
F713/FW721	3/8
FW726	
F715	3/8
F718/FW732	3/8
F721/FW738	3/8
F724	3/8
F726	3/8
F730	3/8
F732	5/8
F738	5/8

Figure No. 2

Note: "C" Dimensions shown are original oil seal locations. When seal is replaced add or subtract 1/16th of an inch to dimension shown. This will allow seal to wear on a new surface for extended life.

ENDPLAY - FLANGED REDUCTOR

UNIT SIZE	INPUT SHAFT ENDPLAY	OUTPUT SHAFT ENDPLAY
F710, F713	.0005 to .0075 Max.	.0005 to .003 Max.
F715 Through F730	.0005 to .009 Max.	.0005 to .003 Max.
F732 Through F738	.0005 to .003 Max.	.0005 to .003 Max.

Figure No. 3

ENDPLAY - REDUCTOR & RF TYPE

UNIT SIZE	INPUT SHAFT ENDPLAY	OUTPUT SHAFT ENDPLAY
710, 713	.0005 to .002 Max.	.0005 to .003 Max.
715 Through 730	.0005 to .003 Max.	.0005 to .003 Max.
732 Through 738	.0005 to .005 Max.	.0005 to .003 Max.

Figure No. 4

Note: Endplays adjusted by:
1. Input Shaft - Adding or subtracting metal shims (Item 18)
2. Output Shaft - Adding or subtracting metal shims (Item 12)

SCREW TIGHTENING TORQUES* (IN-LBS.)

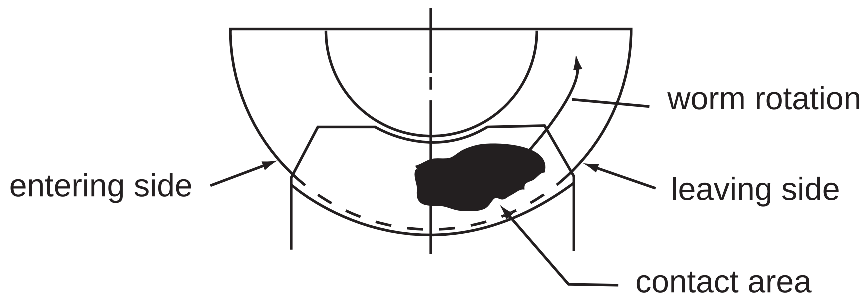
UNIT SIZE	CARRIER SCREW	FLANGE SCREW	FAN GUARD SCREW	FAN ATTACH. SCREW	PIPE PLUG
F710	40-65	45-65			90
F713	132-156	132-156			132
F715	264-312	264-312			132
F718	264-312	264-312			132
F721	264-312	264-312			132
F724	264-312	264-312			132
F726	264-312	264-312			132
F730	264-312	264-312			132
F732	264-312	264-312	85-105	140-160	240
F738	480-552	264-312	85-105	140-160	240
F752	840-948	840-948	132-156	140-160	240
F760	1200-1368	840-948	132-156	140-160	240

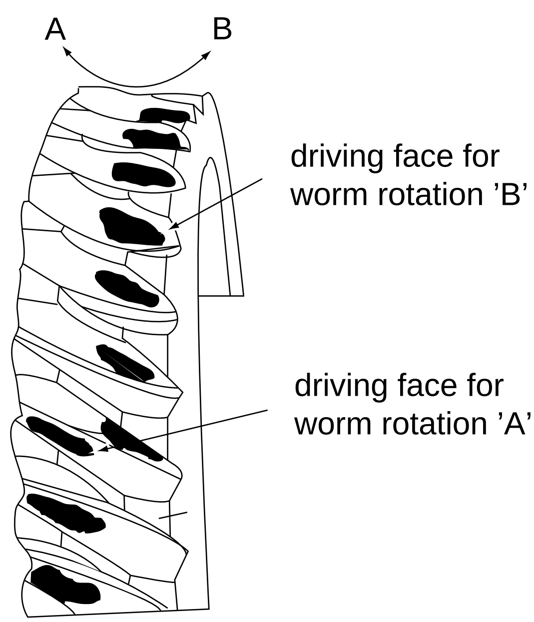
Figure No. 5

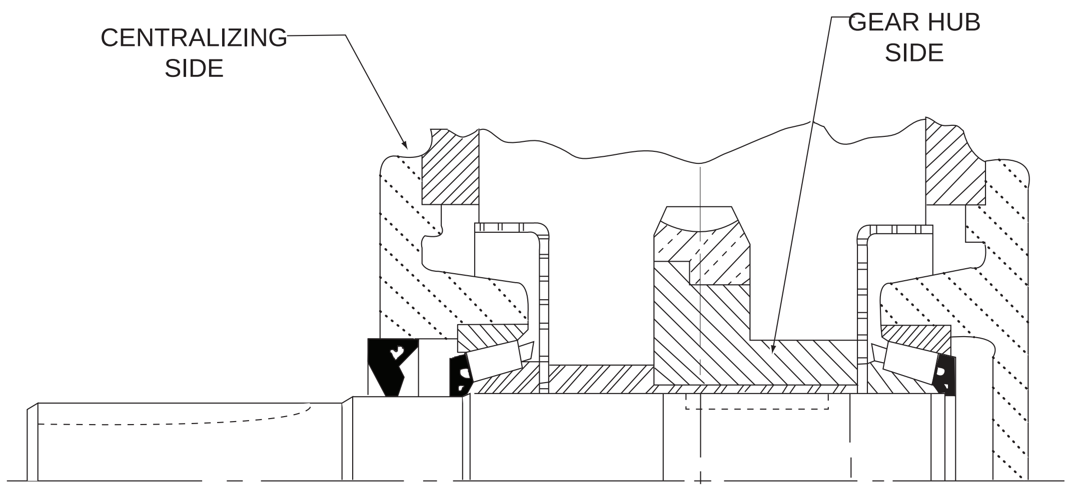
Note: Does not apply to SSF or SSHF reducers

IDEAL POSITIONING OF WORM GEAR CONTACT

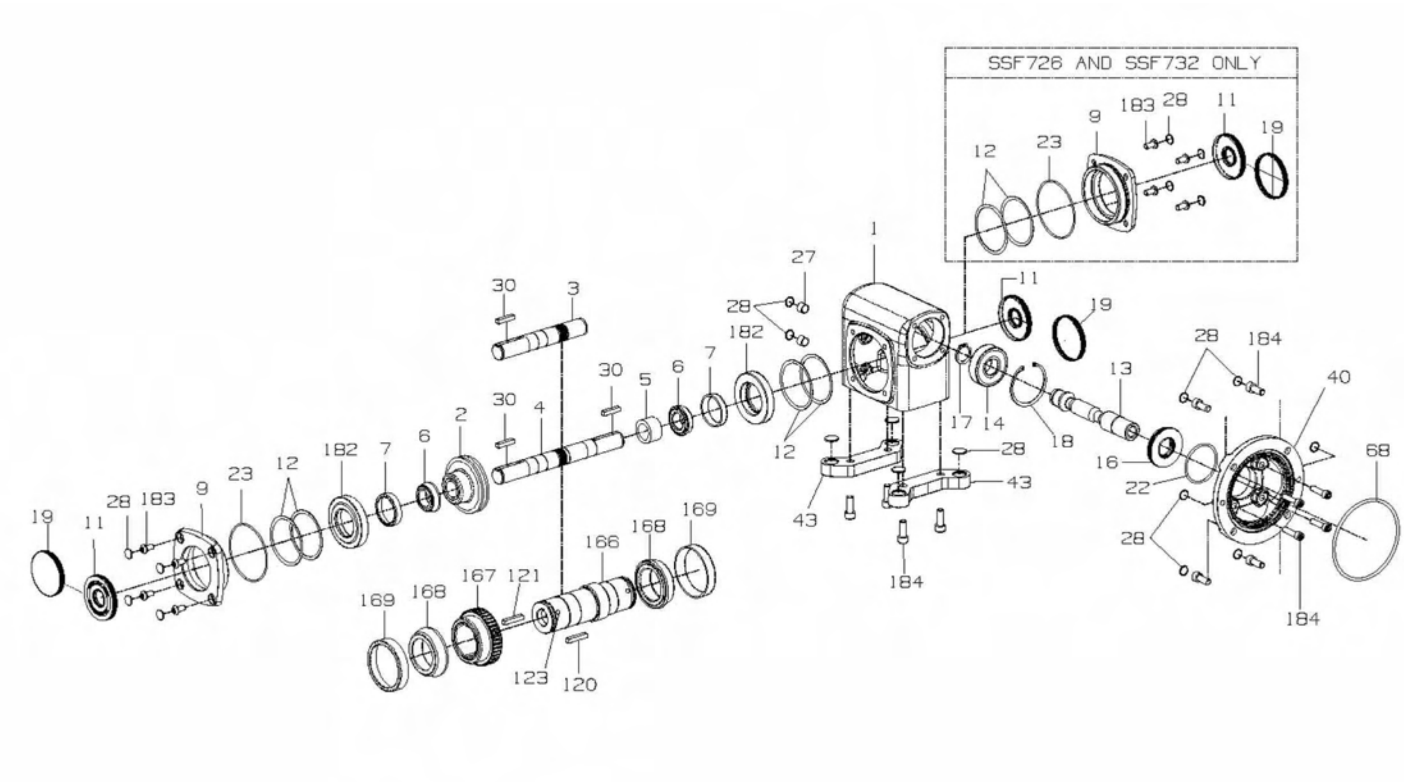
Figure No. 6







PARTS LIST - STAINLESS STEEL MODELS SSF AND SSHF



Part No.	Description
1	HOUSING
2	WORM GEAR
3	SINGLE POJECTING OUTPUT SHAFT
4	DOUBLE PROJECTING OUTPUT SHAFT
5	GEAR SPACER
6	OUTPUT BEARING (CONE)
7	OUTPUT BEARING (CUP)
9	BEARING CARRIER (OPEN)
11	OUTPUT OIL SEAL
12	ADJUSTMENT SHIMS
13	INPUT WORM SHAFT
14	INPUT BEARING
16	INPUT OIL SEAL
17	RETAINING RING (EXTERNAL)
18	RETAINING RING (INTERNAL)
19	BORE PLUG
22	INPUT “O” RING
23	OUTPUT “O” RING
27	PIPE PLUG

Part No.	Description
28	PROTECTIVE CAP PLUG
30	OUTPUT KEY
40	MOTOR FLANGE
43	HORIZONTAL BASE
68	MOTOR FLANGE “O” RING
120	KEY (EXTERNAL)
121	KEY (INTERNAL)
123	SOCKET SETSCREW
166	HOLLOW OUTPUT SHAFT (H) VERSION ONLY
167	WORM GEAR
168	OUTPUT BEARING (CONE)
169	OUTPUT BEARING (CUP)
182	REDUCER BUSHING (MODELS 718 & 721 SOLID SHAFT ONLY)
183	HSBHCS
184	SHCS

Oil Capacity	
Unit Size	**Fluid Oz.**
SSF718	11.2
SSHF718	10.5
SSF721	16.3
SSHF721	15.2
SSF726	32.9
SSHF726	31.1
SSF732	71.4
SSHF732	67.4

700 Series Single Reduction Catalog Number Explanation

HQC 7 32 R - 30 Z F P - B7 H 3 - P20

HQC – STYLE
- Blank - Projecting I/P & O/P shafts (No flange)
- H - Projecting I/P, BostMount hollow O/P shaft, (No flange)
- S - Projecting I/P, hollow O/P shaft (No flange)
- F - Flanged reducer (Quill type), projecting O/P shaft
- HF - Flanged reducer (Quill type), BostMount hollow O/P shaft
- SF - Flanged reducer (Quill type), hollow O/P shaft
- QC - Flanged reducer (Coupling type), projecting O/P shaft
- HQC - Flanged reducer (Coupling type), BostMount hollow O/P shaft
- RF - Flanged reducer (Coupling type), Projecting O/P shaft, 752-760 sizes only
- SSF*- Stainless Flanged reducer (Quill type), Projecting O/P shaft
- SSHF* - Stainless flanged reducer (Quill type) BostMount hollow O/P shaft
- C Prefix - Cast iron flange and base
- (* SSF/SSHF sizes 718, 721, 726, 732 only)

7 – 700 SERIES

32 – CENTER DISTANCE (inches)
10 - 1.00
13 - 1.33
15 - 1.54
18 - 1.75
21 - 2.06
24 - 2.38
26 - 2.62
30 - 3.00
32 - 3.25
38 - 3.75
52 - 5.16
60 - 6.00

R – BASE
- Blank - No base kit required
- A - Horizontal base – Worm under
- B - Horizontal base – Worm over
- C - Vertical High base – I/P right
- D - Vertical Low base – I/P right
- E - Vertical High base – I/P left
- F - Vertical Low base – I/P left
- R/L - BostMount Output Bracket
- X - Input Vertical Up
- Y - Input Vertical Down
- V - Hollow O/P with base – I/P left
- W - Hollow O/P with base – I/P right
- M - Hollow O/P with CFA-I/P left
- N - Hollow O/P with CFA-I/P right

30 – REDUCTION RATIO TO 1
5, 10, 15, 20, 25, 30, 40, 50, 60

Z – LUBRICATION
- Blank - No Lube
- Z - PosiVent® (factory filled with Klubersynth UH1 6-460)
- K - Klubersynth UH1 6-460

F – FAN (732-760 sizes only)
- Blank - No fan
- F - Fan kit

P – P-Pressure Relief

B7 – NEMA MOUNTING

BORE CODE	NEMA MOUNTING	INPUT BORE	KEYWAY
B4	42CZ	.500"	1/8 x 1/16
B5	56C	.625	3/16 x 3/32
B7	140TC/180C	.875	3/16 x 3/32
B9	180TC/210C	1.125	1/4 x 1/8
B11	210TC/250UC	1.375	5/16 x 5/32
B13	250TC	1.625	3/8 x 3/16

H – OUTPUT SHAFT ASSEMBLY
- G* - O/P Projection – Left
- H* - Double O/P Projection
- J* - O/P Projection – Right

* Add "S" after letter for Stainless Steel Shaft (ex. GS, HS, JS)

3 – MOUNTING POSITIONS
- Blank -No Lubrication Supplied

For Factory Prelubrication Indicate Mounting Position
- 1 -Standard Mounting
- 2-6 -Refer to Mounting Positions in Catalog

P20 – H SERIES OUTPUT BORE CODE
For H700 Series only. Specified in 1/16" increments. See Worm Gear Drives Catalog for complete offering. Example: 1-1/4" = P20

700 Series Double Reduction Catalog Number Explanation

HQCWA 7 26 B 100 Z F P - B5 H 3 - P20

HQCWA – STYLE
- WA - Parallel shafts, projecting I/P & O/P
- HWA - Parallel shafts, projecting I/P, BostMount Hollow O/P
- SWA - Parallel shafts, projecting I/P, hollow O/P
- WC - Right Angle shafts, projecting I/P & O/P
- HWC - Right Angle shafts, projecting input, BostMount hollow output shaft.
- SWC - Right Angle shafts, projecting I/P, hollow O/P
- FWA - Parallel shafts, Quill type I/P, projecting O/P
- HFWA - Parallel shafts, Quill type I/P, BostMount Hollow O/P
- SFWA - Parallel shafts, Quill type I/P, hollow O/P
- FWC - Right Angle shafts Quill type I/P, projecting O/P
- HFWC - Right Angle shafts, Quill type input, BostMount hollow output shaft
- SFWC - Right Angle shafts, Quill type I/P, hollow O/P
- QCWA - Parallel shafts, Coupling type I/P, projecting O/P
- HQCWA - Parallel shafts, Coupling type I/P, BostMount hollow O/P
- QCWC - Right Angle shafts, Coupling type I/P, projecting O/P
- HQCWC - Right Angle shafts, Coupling type input, BostMount hollow output shaft
- C Prefix - Cast iron flange and base

7 – 700 SERIES

26 – CENTER DISTANCE (inches)
13 - 1.33
18 - 1.75
21 - 2.06
26 - 2.62
30 - 3.00
32 - 3.25
38 - 3.75
52 - 5.16
60 - 6.00

B – BASE
- Blank - No base kit required
- A & B - Horizontal bases
- C & E - Vertical High bases
- D & F - Vertical Low bases
- R/L - BostMount Output Bracket
- X - Input Vertical Up
- Y - Input Vertical Down
- V/W - Hollow O/P with base
- M/N - Hollow O/P with CFA

100 – REDUCTION RATIO TO 1

100	1200
150	1800
200	2000
300	2400
400	3000
600	3600
900	

Z – LUBRICATION
- Blank - No Lube
- Z - PosiVent® (factory filled with Klubersynth UH1 6-460)
- K - Klubersynth UH1 6-460

F – FAN (732 - 760 only)
- Blank - No Fan
- F - Fan Kit
- E - End Cap available 732 and 738 only, when no fan is used

P – P - Pressure Relief

B5 – NEMA MOUNTING

BORE CODE	NEMA MOUNTING	INPUT BORE	KEYWAY
B4	42CZ	.500"	1/8 x 1/16
B5	56C	.625	3/16 x 3/32
B7	140TC/180C	.875	3/16 x 3/32
B9	180TC/210C	1.125	1/4 x 1/8

H – OUTPUT SHAFT ASSEMBLY (When facing Input)
- G* - O/P Projection – Left
- H* - Double O/P Projection
- J* - O/P Projection – Right

* Add "S" after letter for Stainless Steel Shaft (ex. GS, HS, JS)

3 – MOUNTING POSITIONS
- Blank - No Lubrication Supplied

For Factory Prelubrication Indicate Mounting Position
- 1 - Standard Mounting
- 2 - 6 - Refer to Mounting Positions in Catalog

P20 – H SERIES OUTPUT BORE CODE*
For H700 Series only. Specified in 1/16" increments See Worm Gear Drives Catalog for complete offering. Example: 1-1/4" - P20

*H Series Only

Note: Contact factory for other model numbers.

Stainless Steel 700 Series Single Reduction Catalog Number Explanation

SSF 7 32 B - 30 K - B7 HS - P20

STYLE
SSF - Stainless Flanged reducer (Quill type), Projecting O/P shaft
SSHF - Stainless flanged reducer (Quill type) BostMount hollow O/P shaft

700 SERIES

CENTER DISTANCE (inches)
18 - 1.75
21 - 2.06
26 - 2.62
32 - 3.25

BASE
Blank - No base kit required
B - Horizontal base – Worm over

REDUCTION RATIO TO 1
5
10
15
20
25
30
40
50
60

LUBRICATION
Blank - No Lube
K - Klubersynth UH1 6-460

NEMA MOUNTING

BORE CODE	NEMA MOUNTING	INPUT BORE	KEYWAY
B5	56C	.625	3/16 x 3/32
B7	140TC/180C	.875	3/16 x 3/32

OUTPUT SHAFT ASSEMBLY
(When facing Input)
GS - O/P Projection – Left
HS - Double O/P Projection
JS - O/P Projection – Right

H SERIES OUTPUT BORE CODE
SSHF 700 Series only. Specified in 1/16" increments. See Worm Gear Drives Catalog for complete offering.
Example:
1-1/4" = P20

Warranty

The Company warrants that all 700 Series speed reducers will be free from defects in material and workmanship over the lifetime of the product.

Oil seals are considered to be replaceable maintenance items.

Any products which shall be proved to the Company's satisfaction to have been defective at the time of delivery in these respects will be replaced or repaired by the Company at its option. Freight is the responsibility of the customer. The Company's liability under this warranty is limited to such replacement or repair and it shall not be held liable in any form of action for direct or consequential damages to property or person. THE FOREGOING WARRANTY IS EXPRESSLY MADE IN LIEU OF ALL OTHER WARRANTIES WHATSOEVER, EXPRESS, IMPLIED AND STATUTORY AND INCLUDING WITHOUT LIMITATION THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS. No employee, agent, distributor, or other person is authorized to give additional warranties on behalf of Boston Gear, nor to assume for Boston Gear any other liability in connection with any of its products, except an officer of Boston Gear by a signed writing.

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For remote speed control you will need to set this parameter to either AI1 or AI3 depending on the speed control voltage being used (4-20 DCMA or 0-10VDC)

Each line highlighted in yellow is a individual menu in the VFD

Each menu has a group of parameters under the menu header

Each parameter is defined with the correct setting listerd to the right

The list of menus and parameters to the right are for setting the drive up to run in manual speed control. You are setting up the drive for remote speed control as noted in the block in the upper right hand corner of this document. All of the parameters stay the same with the exception of the parameter circled in red. You will need to check each parameter in each menu listed to the right and set the parameter to the value shown. If the parameter in the drive matches the parameter on this sheet there is no need to change it.

I've attached the manufacturers manual for the drive that explains how to access the drive, menus, and parameters.

Manual Speed Control

- SEt – Settings Menu
  - ACC – Accel Ramp Time – 1 sec
  - dEC – Decel Ramp Time – 0.5 sec
  - LSP – Minimum Speed – 20 Hz
  - HSP – Maximum Speed – 60 Hz
- CtL – Control Menu
  - LAC – Function Access Level – L2 **
  - Fr1 – Speed Reference (Ref.1 channel) – A1P (A1U1)****
- drC – Drive Control Menu
  - (FROM MOTOR NAME PLATE)
  - bFr – Motor Frequency – 60 Hz
  - UnS – Nominal Motor Volts – 460 V
  - FrS – Nominal Motor Frequency – 60Hz
  - nCr – Nominal Motor Current – 0.7 A
  - nSP – Nominal Motor Speed – 1740 RPM
  - TUn – Auto Tuning – YES
  - SCS – Save Configuartion – Str1 * **
- I–O – I/O Menu
  - tCC – Terminal Strip Configuration – 2C (2–Wire) **
  - tCt – 2–wire Control Type – trn (run command transition)
  - rrS – Reverse Operation – nO (disable)
  - r1 – Relay 1 Operation – Flt (Fault)
- FUn – Application Function Menu
  - StC – Sub Menu
    - Stt – Stop Type – rNP (Ramp)
  - JOG – Sub Menu
    - Jog – Jog Operation – L13 (terminal Ref.)
    - JGF – Jog Frequency – 10 Hz

Power up drive,
Press "ENT"
to access parameter menu(s)

Pressing ESC will back out of parameters and Sub Menus.

Notes:
* – Complete Last
** – Must hold for 3 sec to complete change
*** – Change DIP switch from source to CLI (middle)
**** – Model 312 settings

REV.	DRAWN	DATE	DESCRIPTION

Material: VFD PARAMETERS	PRIME EQUIPMENT GROUP Columbus, Ohio 43206 USA	
Quantity: –		
Tolerances (Unless noted): XX = ±0.01 X = .5 XXX = ±0.005 Angular = ±0.5° X/X = 1/16 Angular = 1°	WIRING DIAGRAM CHINCKEN NECK SKINNER	
THIS DRAWING AND THE CONTENTS THEREOFF ARE THE PROPRIETARY PROPERTY OF PRIME EQUIPMENT GROUP. REPRODUCTION OR USE OF THIS INFORMATION WITHOUT PRIOR WRITTEN AUTHORIZATION FROM PRIME EQUIPMENT GROUP IS PROHIBITED.	Drawing By: TC / Scale: NONE / Application: CSNS–1	Approved By: – / Date: 02/2009 / Drawing Number: 14210